

AMENDMENT OF SOLICITATION/MODIFICATION OF CONTRACT				1. CONTRACT ID CODE		PAGE 1		OF PAGES 2	
2. AMENDMENT/MODIFICATION NO. 0001		3. EFFECTIVE DATE 03/02/2026		4. REQUISITION/PURCHASE REQ. NO. 7507874		5. PROJECT NO. (If applicable)			
6. ISSUED BY FEAD CHERRY POINT PSC BOX 8006 CHERRY POINT, NC 28533		Code N40085		7. ADMINISTERED BY (If other than item 6.)			Code		
8. NAME AND ADDRESS OF CONTRACTOR (No., street, county, State and ZIP Code) AMENDMENT MUST BE ACKNOWLEDGED WITH YOUR PROPOSAL				☒	9A. AMENDMENT OF SOLICITATION N400857507874 Replace Remaining Water Source Heat Pumps, B4414				
					9B. DATED (SEE ITEM 11)				
				☐	10A. MODIFICATION OF CONTRACT/ORDER NO.				
10B. DATED (SEE ITEM 13)									
CODE		FACILITY CODE							
11. THIS ITEM ONLY APPLIES TO AMENDMENTS OF SOLICITATIONS									
☒ The above numbered solicitation is amended as set forth in item 14. The hour and date specified for receipt of Offers ☐ is extended ☒ is not extended. Offers must acknowledge receipt of this amendment prior to the hour and date specified in the solicitation or as amended, by one of the following methods: (a) By completing items 8 and 15, and returning 1 copy of the amendment; (b) By acknowledging receipt of this amendment on each copy of the offer submitted; or (c) By separate letter or telegram which includes a reference to the solicitation and amendment numbers. FAILURE OF YOUR ACKNOWLEDGMENT TO BE RECEIVED AT THE PLACE DESIGNATED FOR THE RECEIPT OF OFFERS PRIOR TO THE HOUR AND DATE SPECIFIED MAY RESULT IN REJECTION OF YOUR OFFER. If by virtue of this amendment you desire to change an offer already submitted, such change may be made by telegram or letter, provided each telegram or letter makes reference to the solicitation and this amendment, and is received prior to the opening hour and date specified.									

12. ACCOUNTING AND APPROPRIATION DATA (if required)	
13. THIS ITEM APPLIES ONLY TO MODIFICATIONS OF CONTRACTS/ORDERS, IT MODIFIES THE CONTRACT/ORDER NO. AS DESCRIBED IN ITEM 14.	
☐	A. THIS CHANGE ORDER IS ISSUED PURSUANT TO: (Specify authority) THE CHANGES SET FORTH IN ITEM 14. ARE MADE IN THE CONTRACT ORDER NO. IN ITEM 10A.
☐	B. THE ABOVE NUMBERED CONTRACT/ORDER IS MODIFIED TO REFLECT THE ADMINISTRATION CHANGES (such as changes in paying office, appropriation date, etc.) SET FORTH IN ITEM 14, PURSUANT TO THE AUTHORITY OF FAR 43.103 (b).
☐	C. THIS SUPPLEMENTAL AGREEMENT IS ENTERED INTO PURSUANT TO AUTHORITY OF:
☐	D. OTHER: (specify type of modification and authority)
E. IMPORTANT: Contractor ☐ is not ☐ is required to sign this document and return original to the issuing office.	
14. DESCRIPTION OF AMENDMENT/MODIFICATION (Organized by UCF section headings, including solicitation/contract subject matter where feasible.)	

7507874 Replace Remaining Water Source Heat Pumps, B4414, Marine Corps Air Station Cherry Point, NC

Amendment 0001 is being issued to answer Pre-Proposal Inquiries (PPIs).

See Continuation Sheet.

15A. NAME AND TITLE OF SIGNER (Type or print)		16A. NAME AND TITLE OF CONTRACTING OFFICER (Type or print)	
15B. CONTRACTOR/OFFEROR (Same as Item 8)	15C. DATE SIGNED	16B. UNITED STATES OF AMERICA BY _____ (Signature of Contracting Officer)	16C. DATE SIGNED

PPI Responses:

1. Can Reference drawings be provided to show the location and layout of the existing WSHP's?

Response: Yes. Please see attached PDF of reference DWGs. Drawing No. 4555267-4555269

2. Can a Reference WSHP Schedule be provided?

Response: Yes. Please see attached PDF of reference DWGs. Drawing No. 4555276

3. Can we get Make, Model, and Serial Numbers of the existing Water Source Heat Pumps that are to be replaced?

Response: I do not have the exact make, model, and serial for each unit. However, we would prefer Trane models that are closest in spec to the units identified in the Heat Pump Schedule show on Drawing No. 4555276.

4. Can we get a Schedule for GPM'S of water for each WSHPs to ensure that we have proper water flow to each?

Response: Yes. Please see attached PDF of reference DWGs. Drawing No. 4555276

5. Can a Building Plan be made available to locate the WSHPs in the ceiling?

Response: Please see the attached PDF to reference where the units are on each floor.

6. Can we get an equipment schedule for the WSHPs that are to be replaced?

Response: Yes. Please see attached PDF of reference DWGs. Drawing No. 4555276

7. Can we get as built drawing to show duct connections?

Response: I do not have As-Built drawings. Please refer to the attached PDF for information on ductwork connections.

8. In your SOW it says to "remove refrigerant from the entire system" in WSHPs the refrigerant is self-contained in the unit is this necessary?

Response: Removing refrigerant will not be necessary.

9. In your SOW it says to "Perform a refrigerant leak test" again WSHPs refrigerant is self-contained is this still necessary?

Response: A leak test will not be necessary.

10. In your SOW it says to "Evacuate the system to the proper microns" WSHPs come pre-charged

Response: The contractor is correct. The systems will not need to be evacuated.

11. Can a reflective ceiling plan be provided? During the site visit we noticed that there are some lights that will need to be removed and reinstalled as well as the ACT ceilings.

Response: What I have is lighting and power plans for each floor. Please see the attached PDF titled, "Lighting and Power Plans".

12. Will there be a requirement for TAB? It isn't mentioned in the scope of work. If so, we will need to know the TAB requirements.

Response: A TAB will not be required as this is a like-in-kind replacement of existing units.

D

D

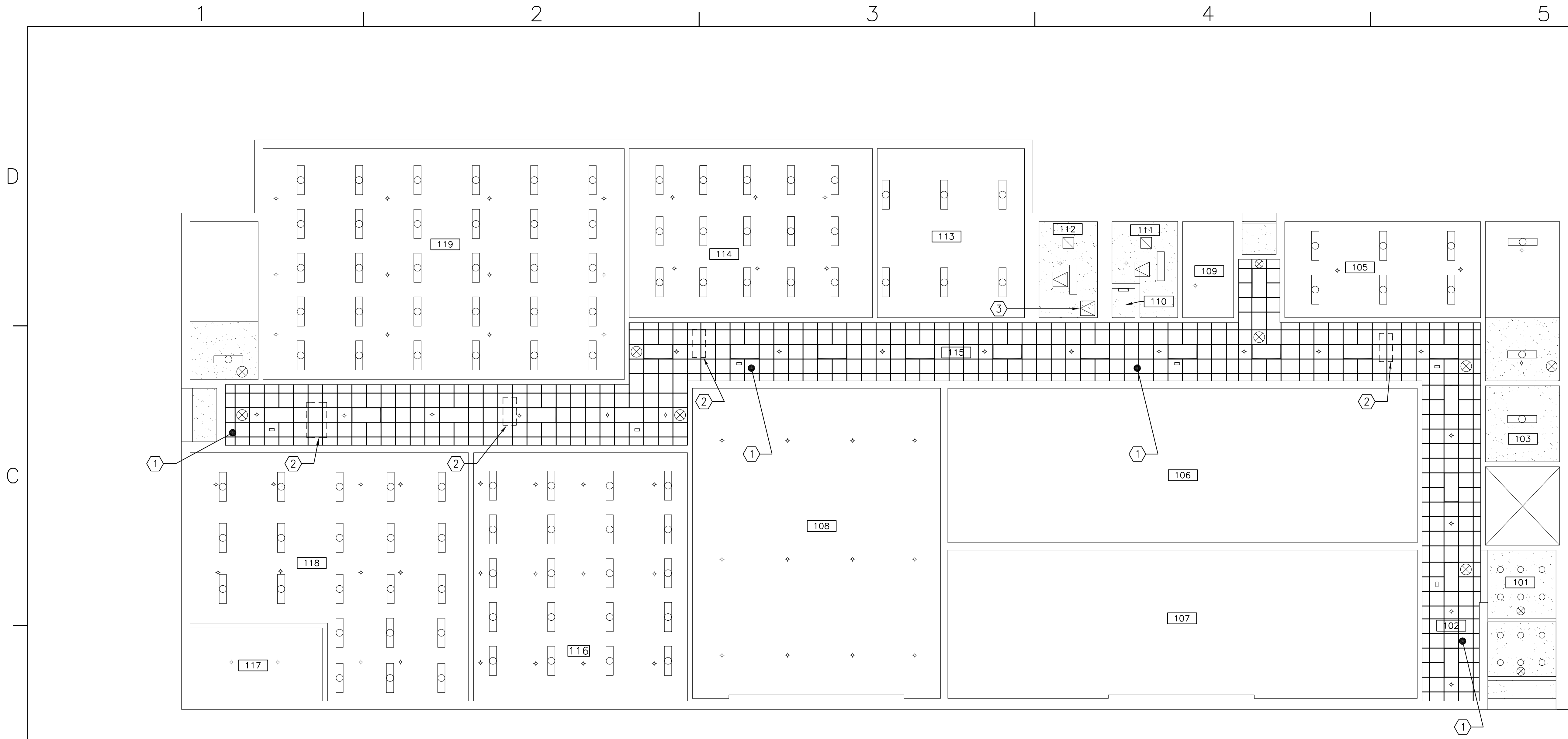


B



B

ADRAWFORM REVISION JULY 2003



 FIRST FLOOR CEILING PLAN – NEW WORK
SCALE: 1/8" = 1'-0"

CONSTRUCTION NOTES (THIS SHEET ONLY)

- ① PROVIDE NEW 2 X 2 SUSPENDED ACOUSTICAL TILE CEILING AND GRID ON COMPLETION OF MECHANICAL WORK.
- ② NEW AND EXISTING MECHANICAL EQUIPMENT ABOVE CEILING. PROVIDE REMOVABLE CEILING GRID AND TILES FOR ACCESS TO MECHANICAL EQUIPMENT.
- ③ PROVIDE NEW CEILING ACCESS DOOR 36" x 36" IN EXISTING GWB CEILING. REFER TO "CEILING ACCESS PANEL", F/A-102/A-501. FOR ADDITIONAL INFORMATION.

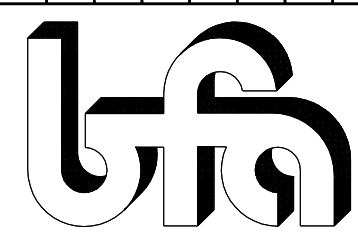
WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

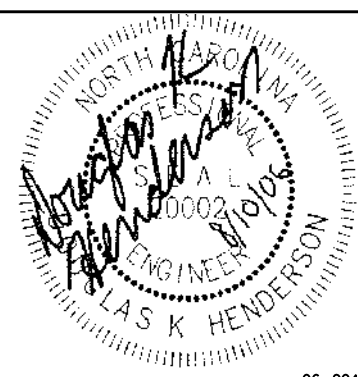
NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.



APPROVED		DATE	APPR
ACTIVITY – SATISFACTORY TO			
DATE			
APPROVED			
FOR EFD FOR COMMANDER NAVFAC			
DATE			
A/E	DESIGN	EFD	
BJW	DRAWN		
DKH	REVIEW		
SGR	QC		
SGR	CHIEF ARCH/ ENGR.		
PROJECT MANAGER		JS	
FIRE PROTECTION			
BRANCH MANAGER		RM	
DESIGN DIRECTOR		CC	



Bowman, Foster & Associates, P.C.
Consulting Engineers
#4 Interstate Corporate Center, Suite 1
Norfolk, VA 23502-4118

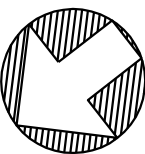


06-004C

NAVAL FACILITIES ENGINEERING COMMAND
MARINE CORPS AIR STATION, CHERRY POINT, N.C.
REPLACE HVAC SYSTEM BLDG 4414
FIRST FLOOR CEILING PLAN – NEW WORK

CODE ID. NO. 80091	SIZE 0
SCALE: AS SHOWN	
FED. NO. 2827328	
STA. PROJ. NO. CP0641M	
SPEC. NO. -	
CONSTR. CONTR. NO. -	
NAVFAC DRAWING NO. 4555250	
SHEET 6 OF 49	
A-102	

DRAWFORM REVISION JULY 2003



SECOND FLOOR CEILING PLAN – NEW WORK

SCALE: 1/8" = 1'-0"

CONSTRUCTION NOTES (THIS SHEET ONLY)

- 1 PROVIDE NEW 2 X 2 SUSPENDED ACOUSTICAL TILE CEILING AND GRID ON COMPLETION OF MECHANICAL WORK.
- 2 PROVIDE 2 X 2 SUSPENDED ACOUSTICAL TILE CEILING AND GRID IN AREA EFFECTED BY MECHANICAL WORK. CONTRACTOR SHALL USE TILES REMOVED FROM CORRIDOR THAT WERE UNDAMAGED.
- 3 BLANK OFF EXISTING LOUVER. REFER TO "TYPICAL LOUVER INFILL DETAIL" D/A-103/501 FOR ADDITIONAL INFORMATION.
- 4 NEW AND EXISTING MECHANICAL EQUIPMENT ABOVE CEILING. PROVIDE REMOVABLE CEILING GRID AND TILES FOR ACCESS TO MECHANICAL EQUIPMENT.

WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:

1/8" = 1'-0"

APPROVED	DATE	DESCRIPTION	SYMBOL
Bowman, Foster & Associates, P.C. Consulting Engineers #4 Interstate Corporate Center, Suite 1 Norfolk, VA 23502-4118			
APPROVED			
ACTIVITY - SATISFACTORY TO			
DATE			
APPROVED			
FOR EFD FOR COMMANDER NAVFAC			
DATE			
A/E	DESIGN	EFD	
BJW	DRAWN		
DKH	REVIEW		
SGR	QC		
SGR	CHIEF ARCH/ ENGR.		
PROJECT MANAGER			
FIRE PROTECTION			
BRANCH MANAGER			
DESIGN DIRECTOR			
NAVAL FACILITIES ENGINEERING COMMAND			
MARINE CORPS AIR STATION, CHERRY POINT, N.C.			
REPLACE HVAC SYSTEM BLDG 4414			
SECOND FLOOR CEILING PLAN – NEW WORK			
CODE ID. NO. 80091			
SCALE: AS SHOWN			
FED. NO. 2827328			
STA. PROJ. NO. CP0641M			
SPEC. NO. -			
CONSTR. CONTR. NO. -			
NAVFAC DRAWING NO. 4555251			
SHEET 7 OF 49			
A-103			
DRAWING REVISION JULY 2003			



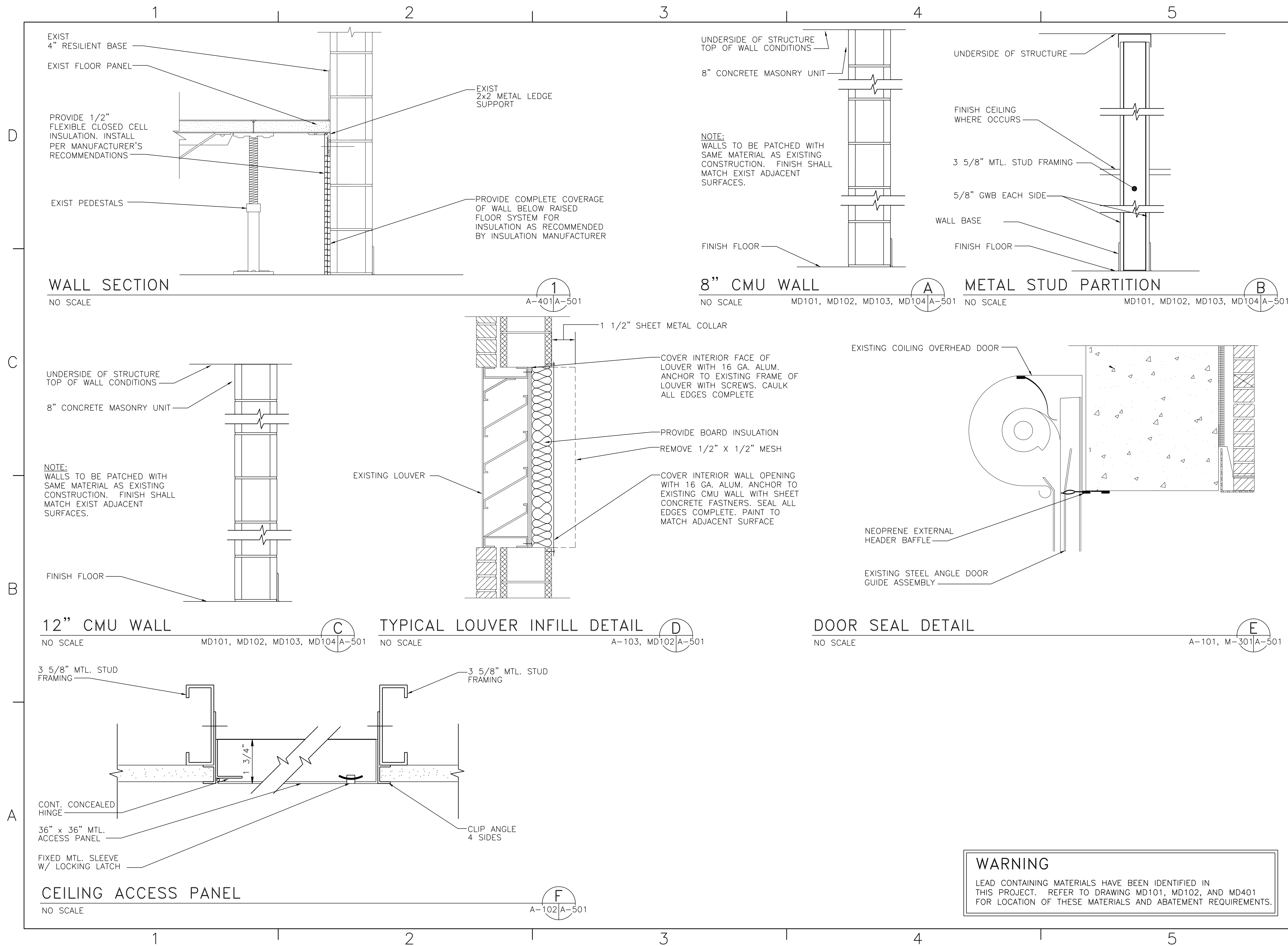
SCALE: $1/8" = 1'-0"$

- ① PROVIDE NEW 2 X 2 SUSPENDED ACOUSTICAL TILE CEILING AND GRID ON COMPLETION OF MECHANICAL WORK.
- ② PROVIDE 2 X 2 SUSPENDED ACOUSTICAL TILE CEILING AND GRID IN AREA EFFECTED BY MECHANICAL WORK. CONTRACTOR SHALL USE TILES REMOVED FROM CORRIDOR THAT WERE UNDAMAGED.
- ③ NEW AND EXISTING MECHANICAL EQUIPMENT ABOVE CEILING. PROVIDE REMOVABLE CEILING GRID AND TILES FOR ACCESS TO MECHANICAL EQUIPMENT.

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

$\frac{1}{8}'' = 1'-0''$

A



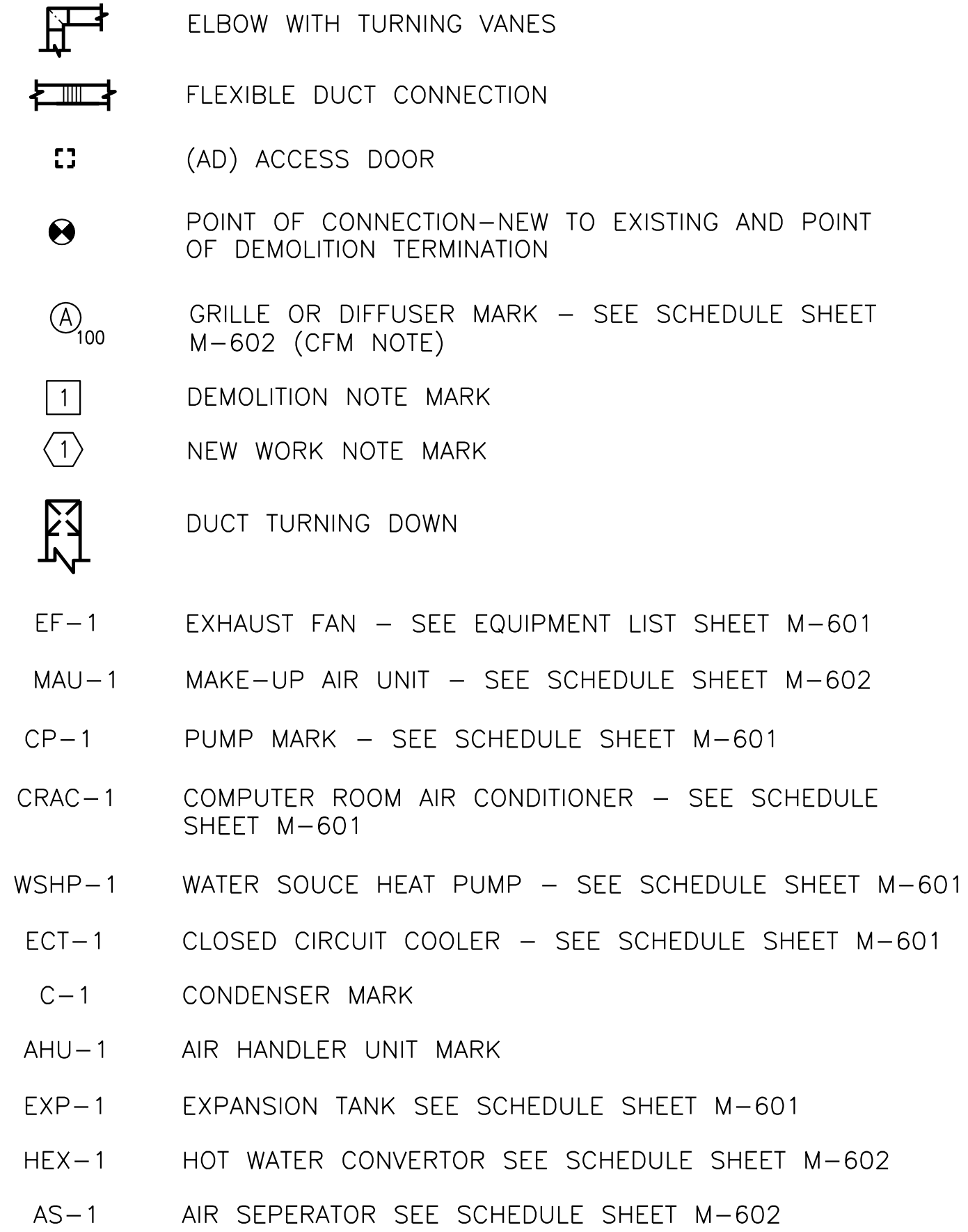
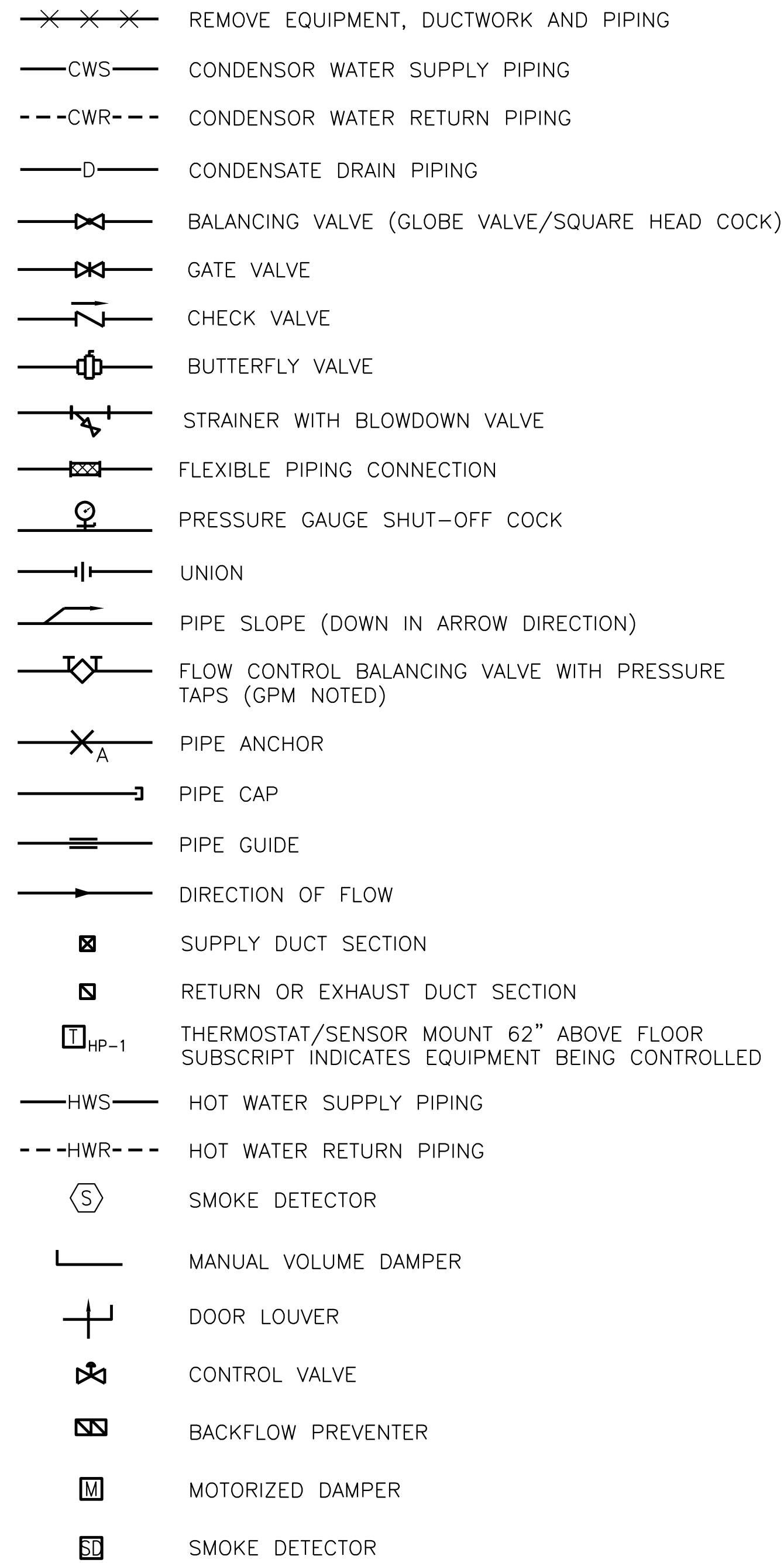
WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

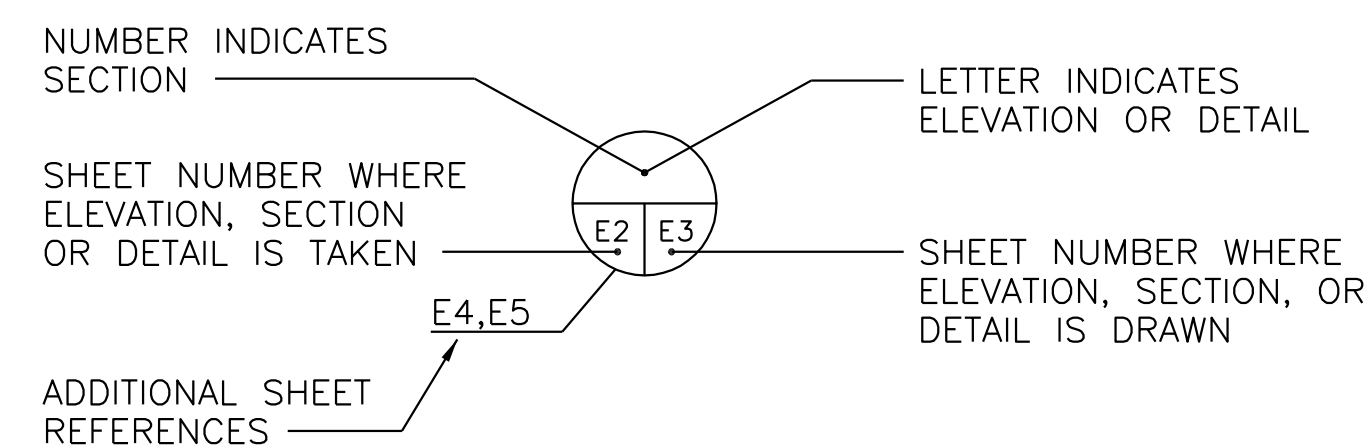
DATE		DESCRIPTION	SYN.
APPROVED	DATE		
APPROVED			
ACTIVITY - SATISFACTORY TO			
DATE			
APPROVED			
FOR EFD FOR COMMANDER NAVFAC			
DATE			
A/E	DESIGN	EFD	
BJW	DRAWN		
DKH	REVIEW		
SGR	QC		
SGR	CHIEF ARCH/ ENGR.	JS	
PROJECT MANAGER		ARCH/ ENGR.	
FIRE PROTECTION		RM	
BRANCH MANAGER		CC	
DESIGN DIRECTOR			
NAVAL FACILITIES ENGINEERING COMMAND			
MARINE CORPS AIR STATION, CHERRY POINT, N.C.			
REPLACE HVAC SYSTEM BLDG 4414			
DETAILS			
DEPARTMENT OF THE NAVY			
CODE ID. NO. 80091			
SCALE: AS SHOWN			
FED. NO. 2827328			
STA. PROJ. NO. CP0641M			
SPEC. NO. -			
CONSTR. CONTR. NO. -			
NAVFAC DRAWING NO. 4555254			
SHEET 10 OF 49			
A-501			
DRAWING REVISION JULY 2003			

AD	ACCESS DOOR	OA	OUTSIDE AIR
AFF	ABOVE FINISHED FLOOR	OD	OUTSIDE DIAMETER
AFG	ABOVE FINISHED GRADE	PD	PRESSURE DROP
APD	AIR PRESSURE DROP	%RH	PERCENT RELATIVE HUMIDITY
APPROX	APPROXIMATELY	PRV	PRESSURE RELIEF VALVE
BD	BALANCING DAMPER	RA	RETURN AIR
BDD	BACK DRAFT DAMPER	RBJ	RUN BETWEEN JOIST
BTUH	BRITISH THERMAL UNIT PER HOUR	RLA	RATED LOAD AMPS
CFH	CUBIC FEET PER HOUR	RPM	REVOLUTIONS PER MINUTE
CI	LITERS PER SECOND	RTJ	RUN THRU JOIST
CLS	CUBIC LITERS PER SECOND	SA	SUPPLY AIR
CM	CENIMETERS	SCT	SATURATED CONDENSING TEMPERATURE
COP	COEFFICIENT OF PERFORMANCE	SEER	SEASONAL ENERGY EFFICIENCY RATING
CWB	DEGREE WET BULB	SM	SQUARE METER
DN	DOWN	SP	STATIC PRESSURE
EAT	ENTERING AIR TEMPERATURE	SST	SATURATED SUCTION TEMPERATURE
EDB	ENTERING DRY BULB	TS	TIP SPEED
EMD	END OF MAIN DRIP	T'STAT	THERMOSTAT
EWB	ENTERING WET BULB	TYP	TYPICAL
EWT	ENTERING WATER TEMPERATURE	"WC	INCHES WATER COLUMN
ESP	EXTERNAL STATIC PRESSURE	WPD	WATER PRESSURE DROP
EXIST	EXISTING	PH	PHASE
FD	FIRE DAMPER WITH ACCESS DOOR	(E)	EXISTING
FDB	DEGREE FAHRENHEIT DRY BULB	FD	FLOOR DRAIN
FF	FINISHED FLOOR	CW	COLD WATER
GA	GAUGE	LPS	LOW PRESSURE STEAM
ID	INSIDE DIAMETER	HORZ	HORIZONTAL
kW	KILOWATT	CFM	CUBIC FEET PER MINUTE
L	ANGLE	HP	HORSE POWER
LAT	LEAVING AIR TEMPERATURE	MBH	EXISTING
L/S	LITERS PER SECOND	FWB	*F WETBULB
Kpa	KILO PASCALS	PSIG	POUND PER SQUARE INCH GRAVITY
LDB	LEAVING DRY BULB	MVD	MANUAL VOLUME DAMPER
LWB	LEAVING WET BULB	ø	DIAMETER
LWT	LEAVING WATER TEMPERATURE	RS	REFRIGERANT SUCTION
m	METER	RL	REFRIGERANT LIQUID
MAX	MAXIMUM	CO	CLEAN OUT
MIN	MINIMUM	EFF	EFFICIENCY
KW-hr	KILOWATTS-HOURS	MOCp	MAXIMUM OVERLOAD PROTECTOR
MCA	MINIMUM CIRCUIT AMPS	GPM	GALLONS PER MINUTE
MFS	MAXIMUM FUSE SIZE	PH	PHASE
NOM	NOMINAL	FLA	FULL LOAD AMPS
NO	NUMBER	16/12ø	OVAL DUCT
NC	NOISE COEFIENT		
MBH	MEGA BRITISH THERMAL UNIT PER HOUR		

NOTE: HEAVY LINE WEIGHT INDICATES NEW EQUIPMENT OR EQUIPMENT TO BE REMOVED, LIGHT LINE WEIGHT INDICATES EQUIPMENT WHICH IS EXISTING TO REMAIN

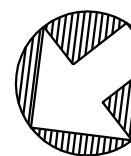
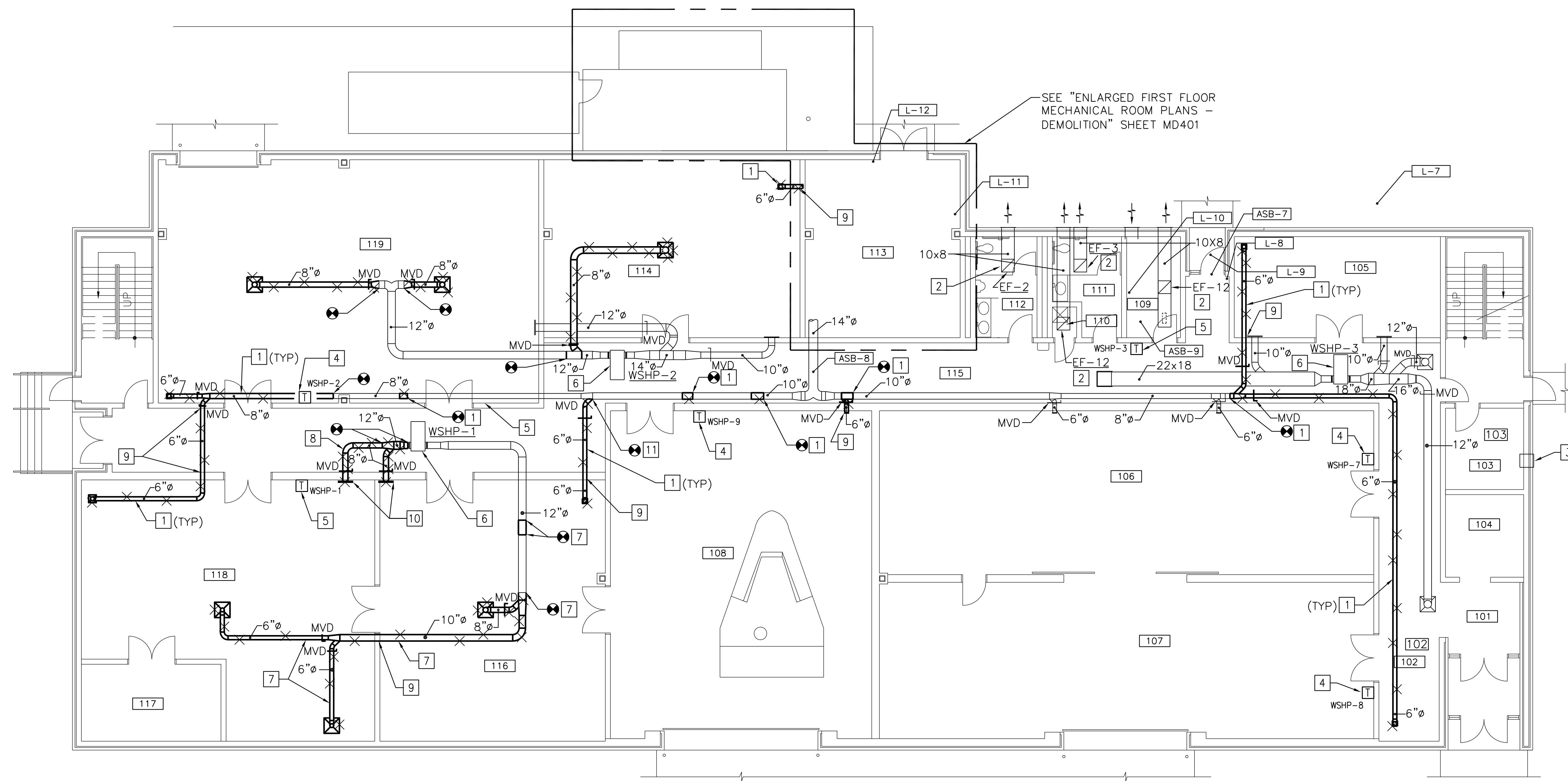


LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.



ELEVATION, SECTION OR DETAIL SYMBOL

[illegible]



FIRST FLOOR MECHANICAL PLAN – DUCTWORK – DEMOLITION

SCALE: 1/8" = 1'-0"

DEMOLITION NOTES (THIS SHEET ONLY)

- 1 REMOVE OUTSIDE AIR SUPPLY DUCT AND DIFFUSERS AS INDICATED ON FLOOR PLAN.
- 2 EXISTING EXHAUST FAN TO REMAIN.
- 3 EXISTING PACKAGED TERMINAL AIR CONDITIONER TO REMAIN.
- 4 EXISTING SENSOR TO REMAIN.
- 5 REMOVE EXISTING SENSOR CLEAN AND STORE FOR REUSE.
- 6 EXISTING WATER SOURCE HEAT PUMP TO REMAIN.
- 7 REMOVE EXISTING SUPPLY AIR DUCTWORK AND DIFFUSERS TO POINT INDICATED ON PLAN.
- 8 REMOVE RETURN AIR DUCTWORK AND WALL REGISTER TO POINT INDICATED ON PLAN.
- 9 PATCH WALL TO MATCH EXISTING SEE ARCHITECTURAL DRAWING SHEET A-501 DETAILS A, B, AND C. REFER TO ARCHITECTURAL CEILING PLAN SHEET A-101 FOR WALL SIZES.
- 10 MODIFY WALL PENETRATION TO FACILITATE INSTALLATION OF NEW DUCTWORK. SEE SHEET M-101
- 11 CAP END OF DUCT. SEAL AND INSULATE TOP MATCH EXISTING.

WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:



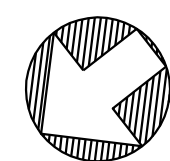
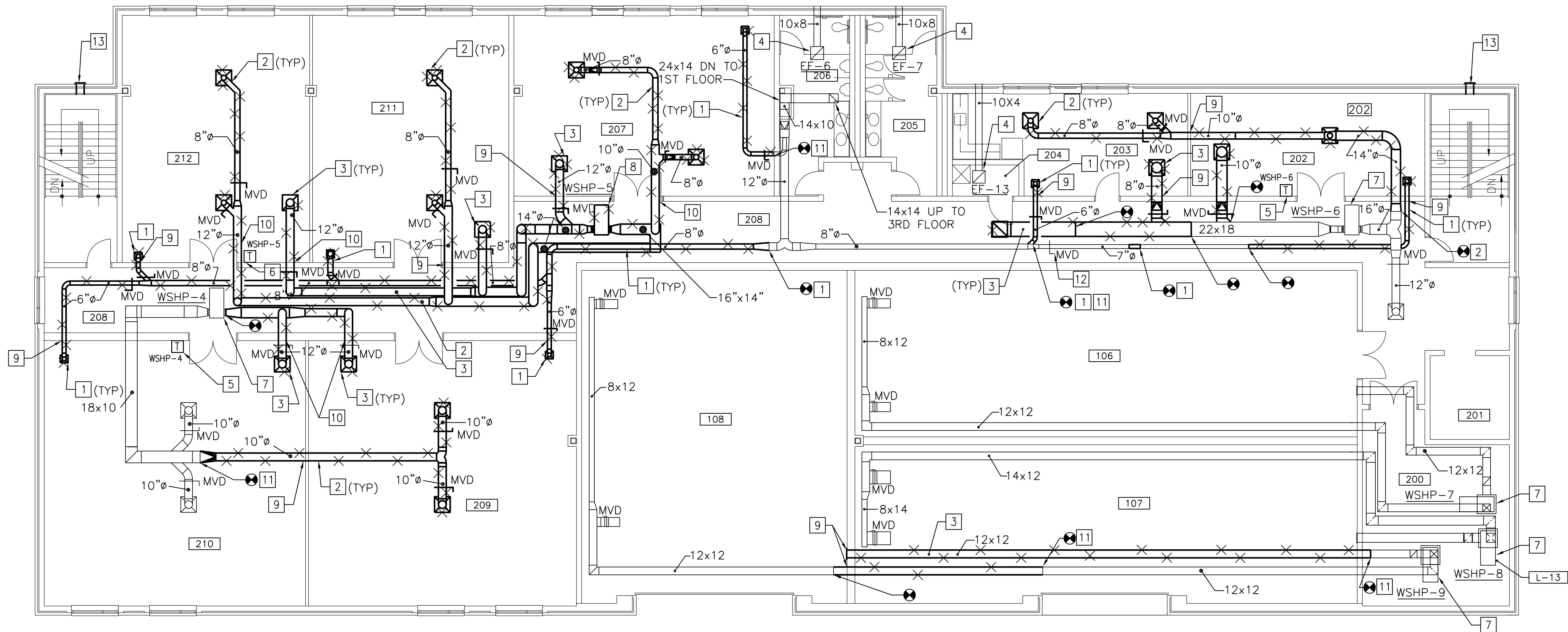
DATE	APPR
DESCRIPTION	
SYN	

Bowmar, Foster & Associates, P.C.
Consulting Engineers
#4 Interstate Corporate Center, Suite 1
Norfolk, VA 23502-4118

APPROVED	06-004C	
ACTIVITY - SATISFACTORY TO		
DATE		
APPROVED		
FOR EFD FOR COMMANDER NAVFAC		
DATE		
A/E	DESIGN	EFD
BJW	DRAWN	
DKH	REVIEW	
SGR	QC	
SGR	CHIEF ARCH/ ENGR.	
PROJECT MANAGER	JS	
FIRE PROTECTION		
BRANCH MANAGER	RM	
DESIGN DIRECTOR	CC	

NAVAL FACILITIES ENGINEERING COMMAND
MARINE CORPS AIR STATION, CHERRY POINT, N.C.
REPLACE HVAC SYSTEM BLDG 4414
FIRST FLOOR MECHANICAL PLAN - DUCTWORK - DEMOLITION

CODE ID. NO. 80091	SIZE 0
SCALE: AS SHOWN	
FED. NO. 2827328	
STA. PROJ. NO. CP0641M	
SPEC. NO. -	
CONSTR. CONTR. NO. -	
NAVFAC DRAWING NO. 4555256	
SHEET 12 OF 49	
MD101	
DRAWN REVISION JULY 2003	



SECOND FLOOR MECHANICAL PLAN – DUCTWORK – DEMOLITION

SCALE: 1/8" = 1'-0"

DEMOLITION NOTES (THIS SHEET ONLY)

- 1 REMOVE OUTSIDE AIR SUPPLY DUCT AND DIFFUSERS AS INDICATED ON FLOOR PLAN.
- 2 REMOVE SUPPLY AIR DUCTWORK AS INDICATED ON FLOOR PLAN.
- 3 REMOVE RETURN AIR DUCTWORK AS INDICATED ON FLOOR PLAN.
- 4 EXISTING EXHAUST FAN TO REMAIN.
- 5 EXISTING SENSOR TO REMAIN.
- 6 REMOVE EXISTING SENSOR CLEAN AND STORE FOR REUSE.
- 7 EXISTING WATER SOURCE HEAT PUMP TO REMAIN.
- 8 REMOVE EXISTING WATER SOURCE HEAT PUMP CLEAN AND STORE FOR REUSE.
- 9 PATCH WALL TO MATCH EXISTING SEE ARCHITECTURAL DRAWING SHEET A-501 DETAILS A, B, AND C. REFER TO ARCHITECTURAL CEILING PLAN SHEET A-103 FOR WALL SIZES.
- 10 MODIFY WALL PENETRATION TO FACILITATE INSTALLATION OF NEW DUCTWORK. SEE SHEET M-101.
- 11 SEAL AND INSULATE TO MATCH EXISTING.
- 12 LOCK EXISTING MVD IN FULL-OPEN POSITION.
- 13 CAP WALL LOUVER SEE ARCHITECTURAL DRAWING SHEET M-501 DETAIL D.

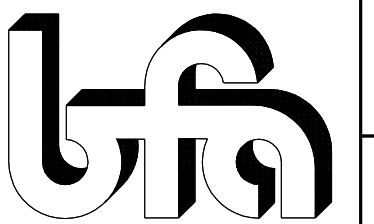
WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

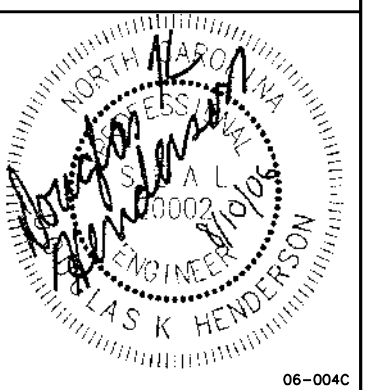
NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:

1/8" = 1'-0" 



Bowmar, Foster & Associates, P.C.
Consulting Engineers
#4 Interstate Corporate Center, Suite 1
Norfolk, VA 23502-4118

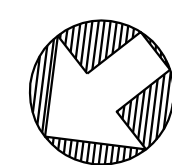
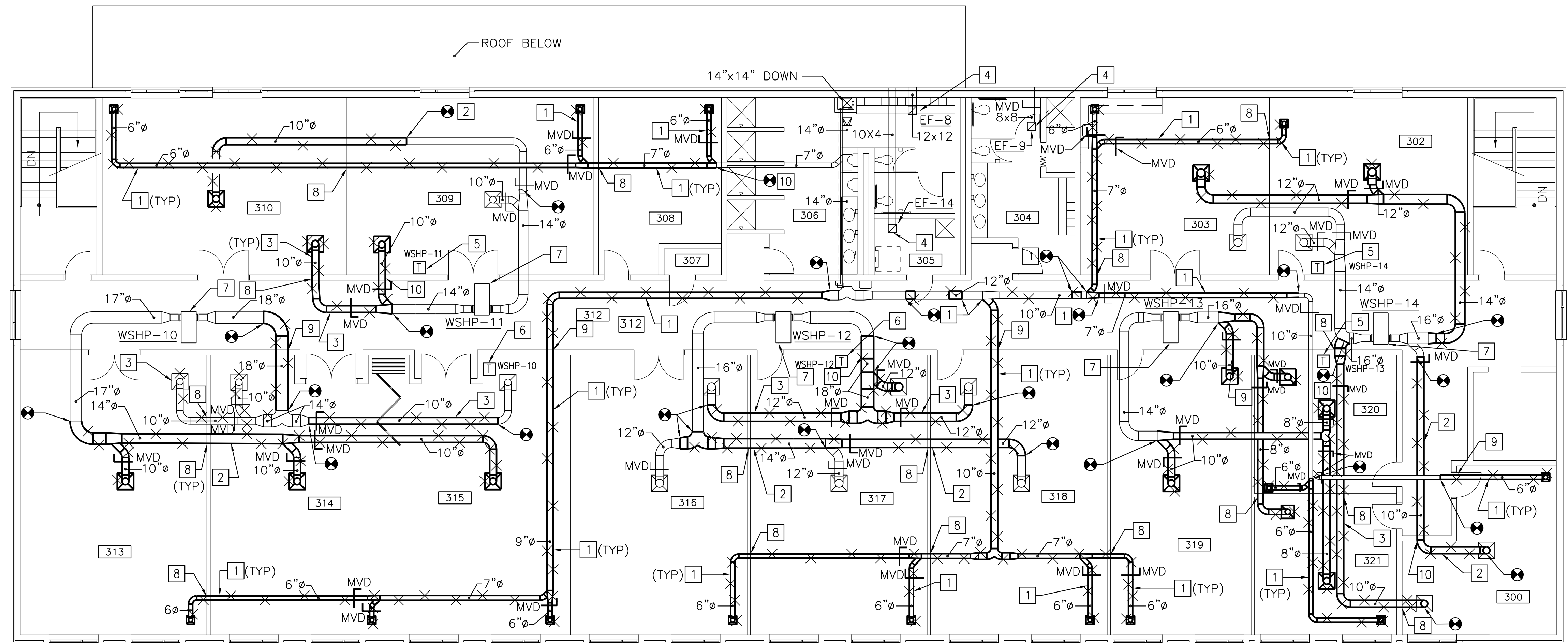


APPROVED	DATE
ACTIVITY – SATISFACTORY TO	DATE
APPROVED	DATE
FOR EFD FOR COMMANDER NAVFAC	DATE
A/E	DESIGN
BJW	DESIGN
BJW	DRAWN
DKH	REVIEW
SGR	QC
SGR	CHIEF ARCH/ ENGR.
PROJECT MANAGER	JS
FIRE PROTECTION	
BRANCH MANAGER	RM
DESIGN DIRECTOR	CC

NAVAL FACILITIES ENGINEERING COMMAND
MARINE CORPS AIR STATION, CHERRY POINT, N.C.
REPLACE HVAC SYSTEM BLDG 4414
SECOND FLOOR MECHANICAL PLAN – DUCTWORK – DEMOLITION

CODE ID. NO. 80091	SIZE 0
SCALE: AS SHOWN	
FED. NO. 2827328	
STA. PROJ. NO. CP0641M	
SPEC. NO. –	
CONSTR. CONTR. NO. –	
NAVFAC DRAWING NO. 4555257	
SHEET 13 OF 49	
MD102	

DRAWING REVISION JULY 2003



THIRD FLOOR MECHANICAL PLAN - DUCTWORK - DEMOLITION

SCALE: 1/8" = 1'-0"

DEMOLITION NOTES (THIS SHEET ONLY)

- [1] REMOVE OUTSIDE AIR SUPPLY DUCT AND DIFFUSERS AS INDICATED ON FLOOR PLAN.
- [2] REMOVE SUPPLY AIR DUCTWORK AS INDICATED ON FLOOR PLAN.
- [3] REMOVE RETURN AIR DUCTWORK AS INDICATED ON FLOOR PLAN.
- [4] EXISTING EXHAUST FAN TO REMAIN.
- [5] EXISTING SENSOR TO REMAIN.
- [6] REMOVE EXISTING SENSOR CLEAN AND STORE FOR REUSE.
- [7] EXISTING WATER SOURCE HEAT PUMP TO REMAIN.
- [8] PATCH WALL TO MATCH EXISTING SEE ARCHITECTURAL DRAWING SHEET A-501 DETAILS A, B, AND C. REFER TO ARCHITECTURAL REFLECTED CEILING PLAN SHEET A-104 FOR WALL SIZES.
- [9] ENLARGE OPENING IN WALL TO ROUTE NEW DUCTWORK AND SEAL ANY OPENINGS.
- [10] CAP END OF DUCT. SEAL AND INSULATE TO MATCH EXISTING.

WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

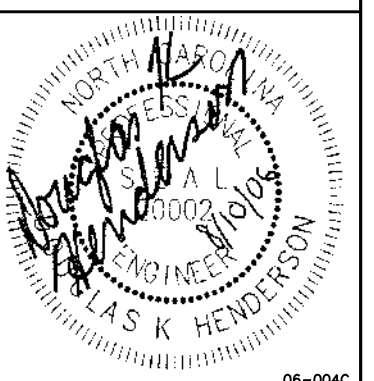
NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:

1/8" = 1'-0" 



Bowmar, Foster & Associates, P.C.
Consulting Engineers
#4 Interstate Corporate Center, Suite 1
Norfolk, VA 23502-4118



06-004C

APPROVED

ACTIVITY - SATISFACTORY TO

DATE

APPROVED

FOR EFD FOR COMMANDER NAVFAC

DATE

A/E	DESIGN	EFD
BJW	DRAWN	
DKH	REVIEW	
SGR	QC	
SGR	CHIEF ARCH/ ENGR.	

PROJECT MANAGER JS

FIRE PROTECTION

BRANCH MANAGER RM

DESIGN DIRECTOR CC

DEPARTMENT OF THE NAVY
MARINE CORPS AIR STATION, CHERRY POINT, N.C.

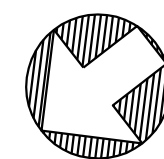
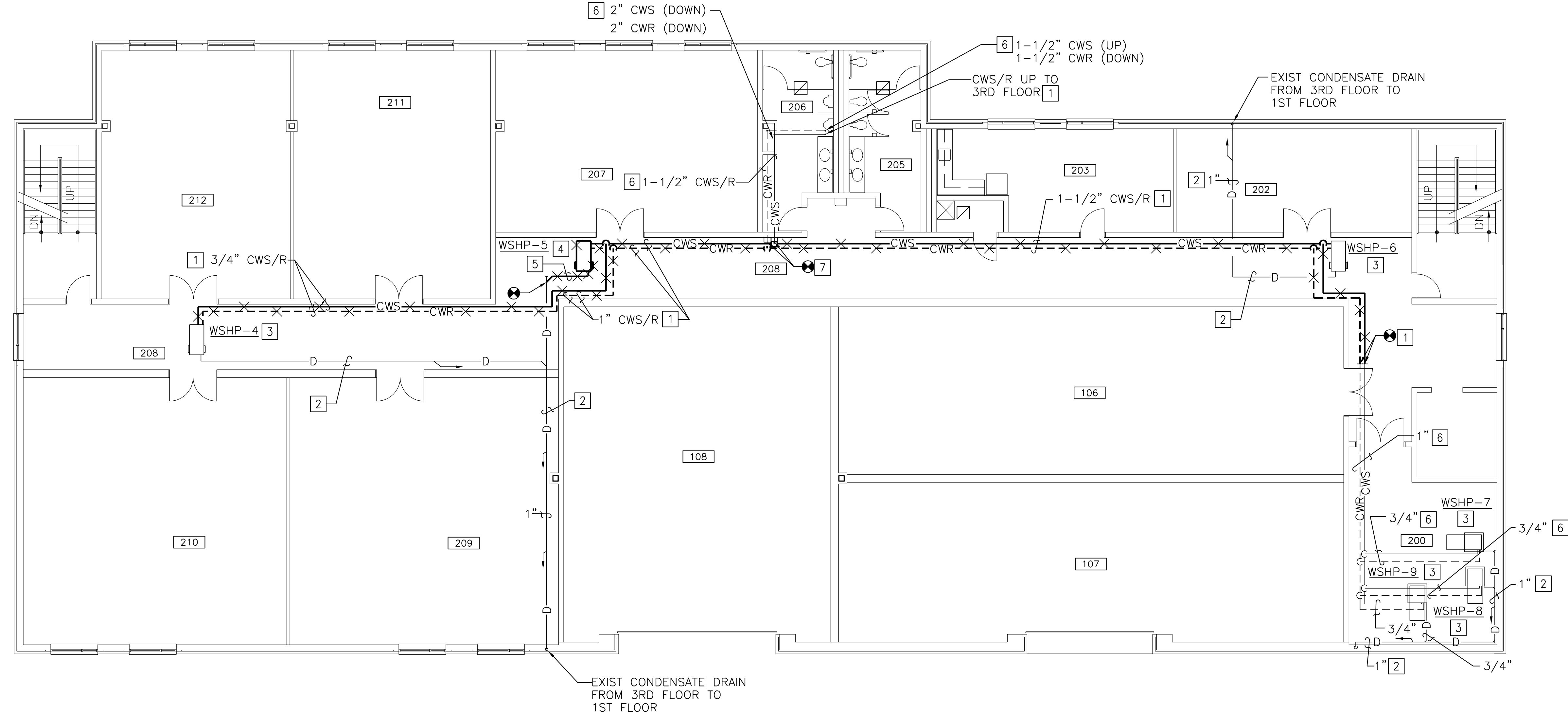
REPLACE HVAC SYSTEM BLDG 4414

THIRD FLOOR MECHANICAL PLAN - DUCTWORK - DEMOLITION

CODE ID. NO. 80091	SIZE 0
SCALE: AS SHOWN	
FED. NO. 2827328	
STA. PROJ. NO. CP0641M	
SPEC. NO. -	
CONSTR. CONTR. NO. -	

NAVFAC DRAWING NO. 4555258
SHEET 14 OF 49

MD103
DRAWING REVISION JULY 2003



SECOND FLOOR MECHANICAL PLAN - PIPING - DEMOLITION

SCALE: 1/8" = 1'-0"

DEMOLITION NOTES (THIS SHEET ONLY)

- 1 REMOVE CWS AND CWR PIPING INCLUDING HANGERS AND SUPPORTS UP TO POINT INDICATED ON FLOOR PLAN.
- 2 EXISTING DRAIN PIPING TO REMAIN.
- 3 EXISTING WATER SOURCE HEAT PUMP TO REMAIN.
- 4 REMOVE EXISTING WATER SOURCE HEAT PUMP (WSHP-5) COMPLETE CLEAN AND STORE FOR REUSE.
- 5 REMOVE DRAIN PIPING BACK TO POINT INDICATED ON PLANS AND CAP.
- 6 EXISTING CWS AND CWR PIPING SHALL REMAIN.
- 7 CAP EXISTING CWS & CWR ABOVE DROP CEILING IN CORRIDOR.

WARNING

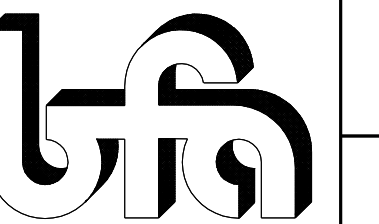
LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.

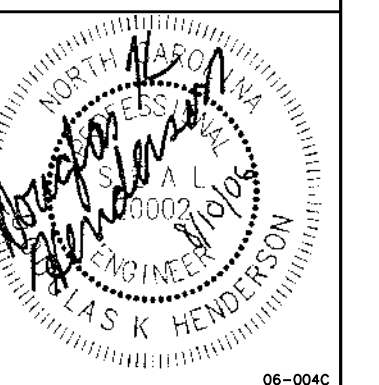
GRAPHIC SCALE:

1/8" = 1'-0"

DATE	APPR
DESCRIPTION	SYN



Bowman, Foster & Associates, P.C.
Consulting Engineers
#4 Interstate Corporate Center, Suite 1
Norfolk, VA 23502-4118

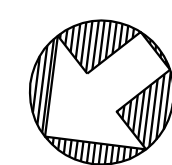


APPROVED	06-004C
ACTIVITY - SATISFACTORY TO	
DATE	
APPROVED	
FOR EFD FOR COMMANDER NAVFAC	
DATE	
A/E	EFD
DESIGN	
DRAWN	
REVIEW	
QC	
ENGR.	
PROJECT MANAGER	JS
FIRE PROTECTION	
BRANCH MANAGER	RM
DESIGN DIRECTOR	CC

NAVAL FACILITIES ENGINEERING COMMAND	REPLACE HVAC SYSTEM BLDG 4414
MARINE CORPS AIR STATION, CHERRY POINT, N.C.	SECOND FLOOR MECHANICAL PLAN - PIPING - DEMOLITION

CODE ID. NO. 80091	SIZE 0
SCALE: AS SHOWN	
FED. NO. 2827328	
STA. PROJ. NO. CP0641M	
SPEC. NO. -	
CONSTR. CONTR. NO. -	
NAVFAC DRAWING NO. 4555260	
SHEET 16 OF 49	
MD105	

DRAWING REVISION JULY 2003

[illegible]

DEMOLITION NOTES (THIS SHEET ONLY)

- 1 REMOVE AIR SEPARATOR COMPLETE.

2 REMOVE EXPANSION TANK COMPLETE. RETAIN CONCRETE PAD FOR REUSE.

3 REMOVE HEAT EXCHANGER COMPLETE.

4 REMOVE BASE MOUNTED CENTRIFUGAL PUMP COMPLETE. RETAIN CONCRETE PAD FOR REUSE.

5 REMOVE CWS AND CWR PIPING COMPLETE INCLUDING HANGERS.

6 REMOVE LPS UP TO POINT INDICATED.

7 REMOVE CLOSED CIRCUIT COOLER COMPLETE.

8 REMOVE HEAT TRACE TAPE COMPLETE.
- 9 DISCONNECT CW TO EXIST ECT-1.

10 REMOVE STEAM CONTROL VALVE.

11 CAP PIPING TO UPPER FLOORS AT CEILING AND ABANDON IN PLACE.

12 REMOVE HWS AND HWR PIPING FOR AHU-1.

13 REMOVE ST-2 COMPLETE AND CONDENSATE RETURN PIPING TO POINT INDICATED ON DIAGRAM.

FOR CONTINUATION SEE SHEET MD104

FOR CONTINUATION SEE SHEET MD104

FOR CONTINUATION SEE BELOW

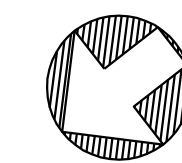
WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

MECHANICAL ROOM PIPING SCHEMATIC – DEMOLITION

NO SCALE

APPROVED	DATE	DESCRIPTION	SYMBOL
			
Bowman, Foster & Associates, P.C. Consulting Engineers #4 Interstate Corporate Center, Suite 1 Norfolk, VA 23502-4118			
			
APPROVED			
ACTIVITY – SATISFACTORY TO			
DATE			
APPROVED			
FOR EFD FOR COMMANDER NAVFAC			
DATE			
A/E	DESIGN	EFD	
BJW	DRAWN		
DKH	REVIEW		
SGR	QC		
SGR	CHIEF ARCH/ ENGR.	JS	
PROJECT MANAGER			
FIRE PROTECTION			
BRANCH MANAGER		RM	
DESIGN DIRECTOR		CC	
NAVAL FACILITIES ENGINEERING COMMAND			
MARINE CORPS AIR STATION, CHERRY POINT, N.C.			
REPLACE HVAC SYSTEM BLDG 4414			
MECHANICAL ROOM PIPING SCHEMATIC – DEMOLITION			
CODE ID. NO. 80091			
SCALE: AS SHOWN			
FED. NO. 2827328			
STA. PROJ. NO. CP0641M			
SPEC. NO. -			
CONSTR. CONTR. NO. -			
NAVFAC DRAWING NO. 4555263			
SHEET 19 OF 49			
MD501			
DRAWING REVISION JULY 2003			



FIRST FLOOR MECHANICAL PLAN – DUCTWORK – NEW WORK

SCALE: 1/8" = 1'-0"

CONSTRUCTION NOTES (THIS SHEET ONLY)

- EXTEND FLEXIBLE DUCT INTO ESCAPE HATCH OF SIMULATOR (ACTIVITY SHOULD VERIFY IF THIS IS ACCEPTABLE).
- PROVIDE HORIZONTAL WATER SOURCE HEAT PUMP. REFER TO "WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-101/M-504 FOR ADDITIONAL INFORMATION.
- EXISTING WATER SOURCE HEAT PUMP TO REMAIN.
- PROVIDE NEW RETURN AIR DUCTWORK. REFER TO "TYPICAL BRANCH CONNECTION" A/M-101/M-501 FOR ADDITIONAL INFORMATION.
- PROVIDE OUTSIDE AIR DUCTWORK CONNECT TO RETURN AIR DUCTWORK OF THE WATER SOURCE HEAT PUMPS.
- PROVIDE SUPPLY AIR DUCTWORK. REFER TO "SUPPLY AIR DIFFUSERS DUCTWORK DETAIL" B/M-101/M-502 FOR ADDITIONAL INFORMATION.
- PROVIDE VERTICAL WATER SOURCE HEAT PUMP. LOCATE ON FLOOR COORDINATE EXACT LOCATION WITH ROICC. REFER TO "WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-101/M-504 FOR ADDITIONAL INFORMATION.
- EXISTING DUCTWORK TO REMAIN.

- EXISTING DIFFUSERS TO REMAIN BALANCE TO INDICATE AIRFLOW.
- PROVIDE NEW WALL MOUNTED SENSOR.
- EXISTING SENSOR TO REMAIN.
- RELOCATE EXISTING SENSOR TO LOCATION INDICATED.
- EXISTING RETURN GRILLES TO REMAIN BALANCE TO INDICATED AIR FLOWS.
- PROVIDE DUCT SLEEVE. REFER TO "DUCTWORK WALL SLEEVE DETAIL" B/M-101/M-501 AND "INSULATED ROUND DUCT PENETRATION DETAIL" C/M-101/M-501 FOR ADDITIONAL INFORMATION.
- CONNECT EXIST SENSOR TO NEW WATER SOURCE HEAT PUMP.
- PROVIDE SPEED CONTROLLER AND REBALANCE AIR FLOW AS INDICATED IN EQUIPMENT LIST ON SHEET M-601.
- PROVIDE SMOKE DETECTION IN SUPPLY DUCTWORK OF WSHP-15 ONLY.

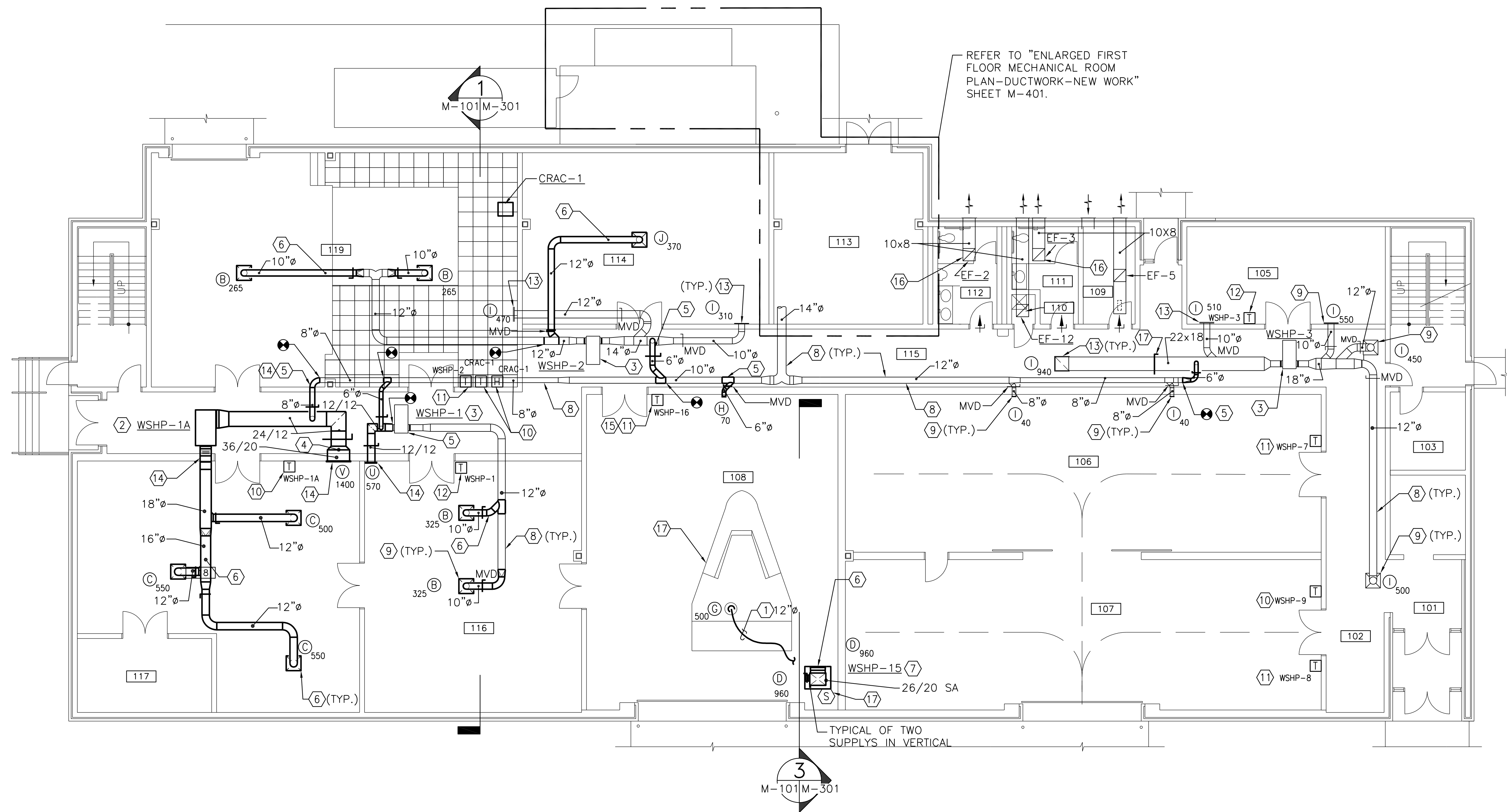
WARNING

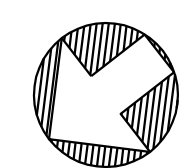
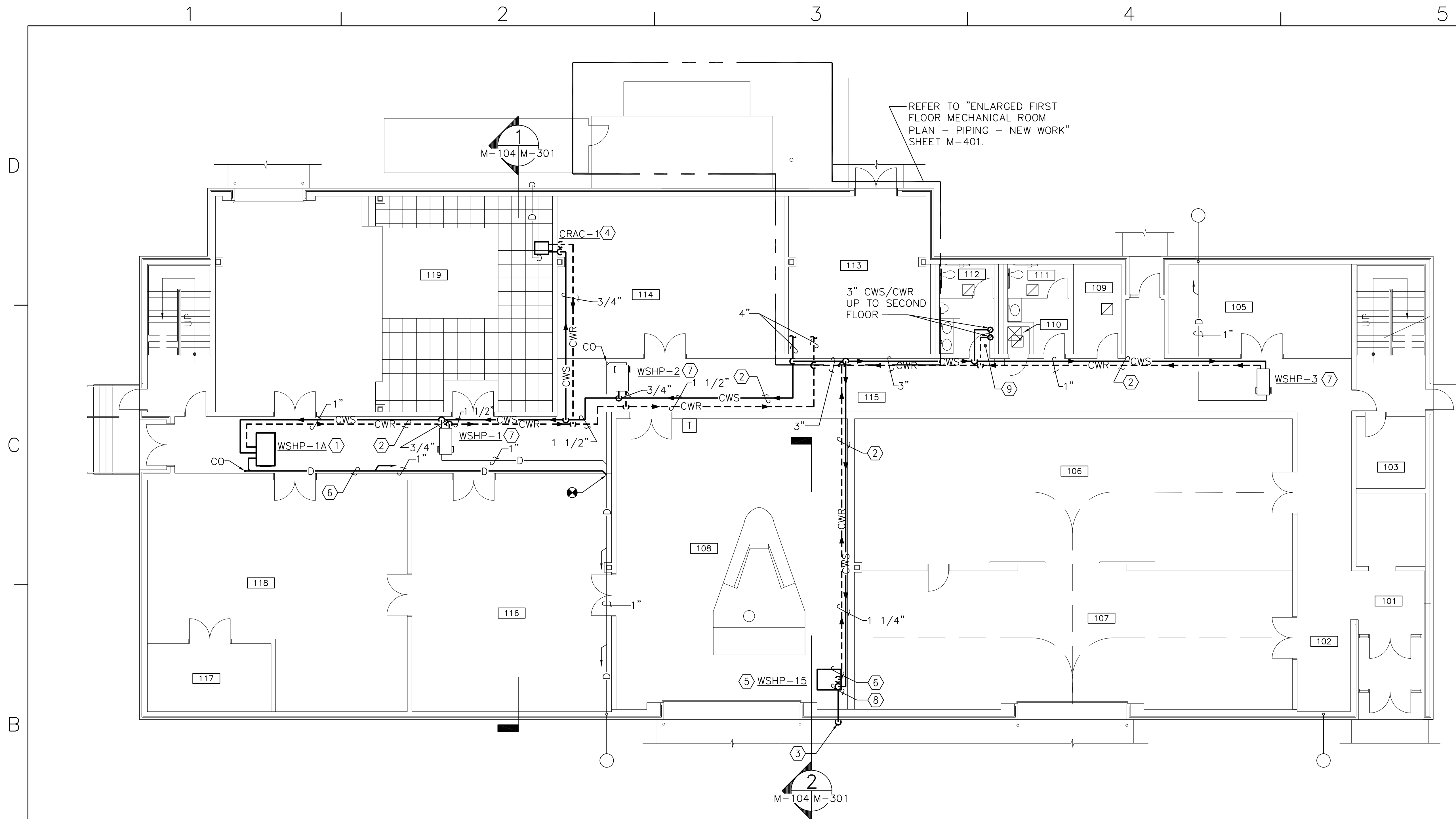
LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:

1/8" = 1'-0" 





FIRST FLOOR MECHANICAL PLAN - PIPING - NEW WORK

SCALE: 1/8" = 1'-0"

CONSTRUCTION NOTES (THIS SHEET ONLY)

- ① PROVIDE WATER SOURCE HEAT PUMP. REFER TO "WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-104/M-504 FOR ADDITIONAL INFORMATION.
- ② PROVIDE CONDENSER SUPPLY AND RETURN PIPING SUPPORT FROM STRUCTURE ABOVE.
- ③ EXTEND DRAIN PIPING THROUGH WALL AND TURN DOWN.
- ④ PROVIDE COMPUTER ROOM AIR CONDITIONING UNIT. RECONNECT EXISTING CONDENSATE DRAIN.
- ⑤ PROVIDE VERTICAL WATER SOURCE HEAT PUMP. REFER TO "WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-104/M-504 FOR ADDITIONAL INFORMATION.
- ⑥ PROVIDE CONDENSATE DRAIN PIPING CONNECT TO EXISTING DRAIN PIPING. REFER TO "TYPICAL CONDENSATE DRAW TRAP DETAIL" A/M-104/M-502 FOR ADDITIONAL INFORMATION.
- ⑦ EXISTING WATER SOURCE HEAT PUMP TO REMAIN. PROVIDE NEW CWS AND CWR PIPING. WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-104/M-504 FOR ADDITIONAL INFORMATION.
- ⑧ RUN TIGHT TO WATER.
- ⑨ PROVIDE ACCESS DOOR IN EXISTING GWB CEILING. REFER TO ARCHITECTURAL DRAWING SHEET A-102 FOR ADDITIONAL INFORMATION.

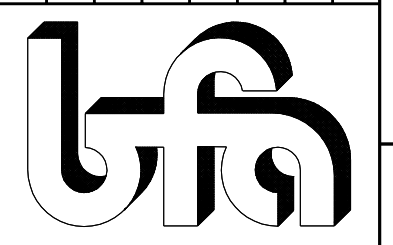
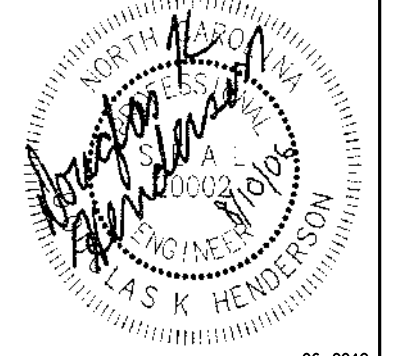
WARNING

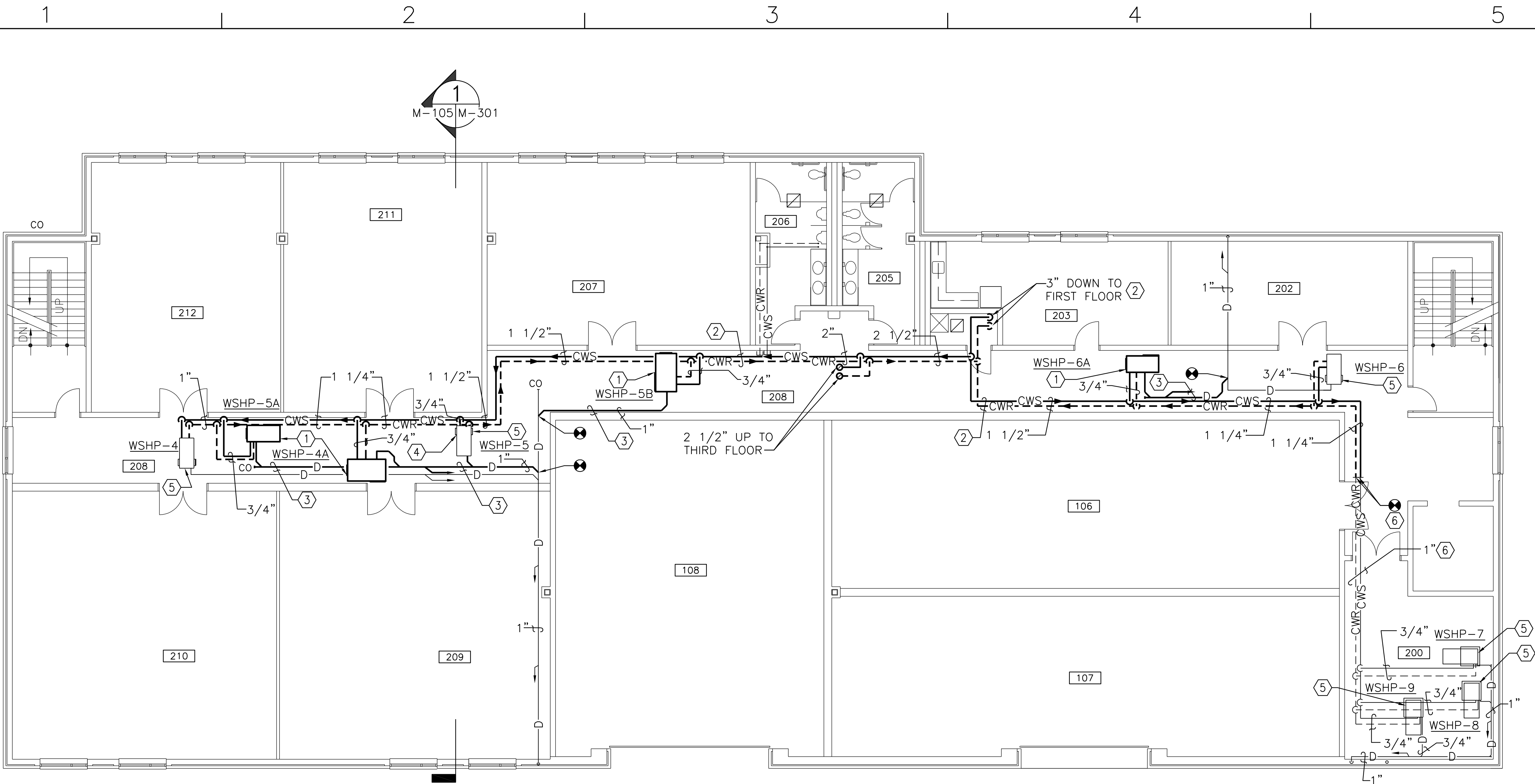
LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:

1/8" = 1'-0" 

DATE		DESCRIPTION	SYMBOL
DATE	SYMBOL		
			
Bowman, Foster & Associates, P.C. Consulting Engineers 44 Interstate Corporate Center, Suite 1 Norfolk, VA 23502-4118			06-004C
APPROVED			
ACTIVITY - SATISFACTORY TO			
DATE			
APPROVED			
FOR EFD FOR COMMANDER NAVFAC			
DATE			
A/E	DESIGN	EFD	
BJW	DRAWN		
DKH	REVIEW		
SGR	QC		
SGR	CHIEF ARCH/ ENGR.		
PROJECT MANAGER		JS	
FIRE PROTECTION		RM	
BRANCH MANAGER		CC	
DESIGN DIRECTOR			
NAVAL FACILITIES ENGINEERING COMMAND			
MARINE CORPS AIR STATION, CHERRY POINT, N.C.			
REPLACE HVAC SYSTEM BLDG 4414			
FIRST FLOOR MECHANICAL PLAN - PIPING - NEW WORK			
CODE ID. NO. 80091		SIZE 0	
SCALE: AS SHOWN			
FED. NO. 2827328			
STA. PROJ. NO. CP0641M			
SPEC. NO. -			
CONSTR. CONTR. NO. -			
NAVFAC DRAWING NO. 4555267			
SHEET 23 OF 49			
M-104			
DRAWN/REVISION JULY 2003			



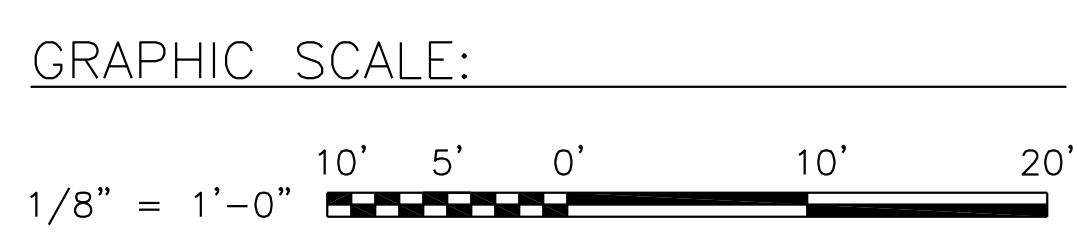
SECOND FLOOR MECHANICAL PLAN – PIPING – NEW WORK
SCALE: 1/8" = 1'-0"

CONSTRUCTION NOTES (THIS SHEET ONLY)

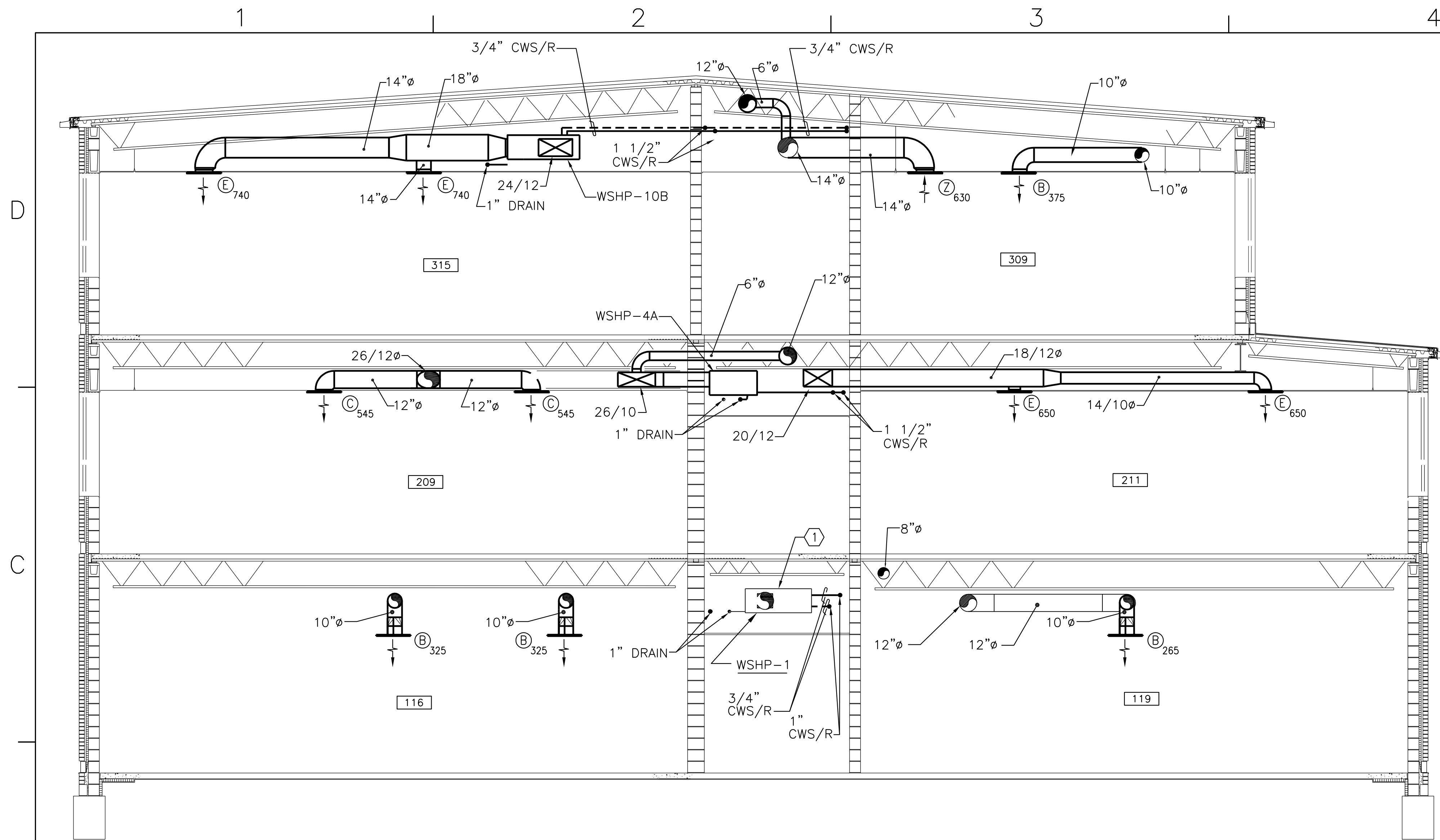
- ① PROVIDE WATER SOURCE HEAT PUMP. REFER TO "WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-105/M-504 FOR ADDITIONAL INFORMATION.
- ② PROVIDE CONDENSER SUPPLY AND RETURN PIPING SUPPORT FROM STRUCTURE ABOVE.
- ③ PROVIDE CONDENSATE DRAIN PIPING CONNECT TO EXISTING DRAIN PIPING. REFER TO "TYPICAL CONDENSATE DRAIN TRAP DETAIL" A/M-104/M-502 FOR ADDITIONAL INFORMATION.
- ④ RELOCATE EXISTING WATER SOURCE HEAT PUMP (WSHP-5). REFER TO "WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-104/M-504 FOR ADDITIONAL INFORMATION.
- ⑤ EXISTING WATER SOURCE HEAT PUMP SHALL REMAIN. PROVIDE CWS AND CWR PIPING REFER TO "WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-105/M-502 FOR ADDITIONAL INFORMATION.
- ⑥ EXISTING CWS AND CWR PIPING TO CONNECT TO NEW CWS AND CWR PIPING.

WARNING
LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.



APPROVED		DATE		APPR	
ACTIVITY - SATISFACTORY TO		DATE		SYN	
DATE		APPROVED		DESCRIPTION	
FOR EFD FOR COMMANDER NAVFAC		DATE		SYN	
A/E		EFD		SYN	
DESIGN		EFD		SYN	
DRAWN		EFD		SYN	
REVIEW		EFD		SYN	
QC		EFD		SYN	
ARCH/ ENGR.		EFD		SYN	
PROJECT MANAGER		EFD		SYN	
FIRE PROTECTION		EFD		SYN	
BRANCH MANAGER		EFD		SYN	
DESIGN DIRECTOR		EFD		SYN	
NAVAL FACILITIES ENGINEERING COMMAND		EFD		SYN	
MARINE CORPS AIR STATION, CHERRY POINT, N.C.		EFD		SYN	
REPLACE HVAC SYSTEM BLDG 4414		EFD		SYN	
SECOND FLOOR MECHANICAL PLAN - PIPING - NEW WORK		EFD		SYN	
CODE ID. NO. 80091		EFD		SYN	
SCALE: AS SHOWN		EFD		SYN	
FED. NO. 2827328		EFD		SYN	
STA. PROJ. NO. CP0641M		EFD		SYN	
SPEC. NO.		EFD		SYN	
CONSTR. CONTR. NO.		EFD		SYN	
NAVFAC DRAWING NO. 4555268		EFD		SYN	
SHEET 24 OF 49		EFD		SYN	
M-105		EFD		SYN	
DRAWING REVISION JULY 2003		EFD		SYN	

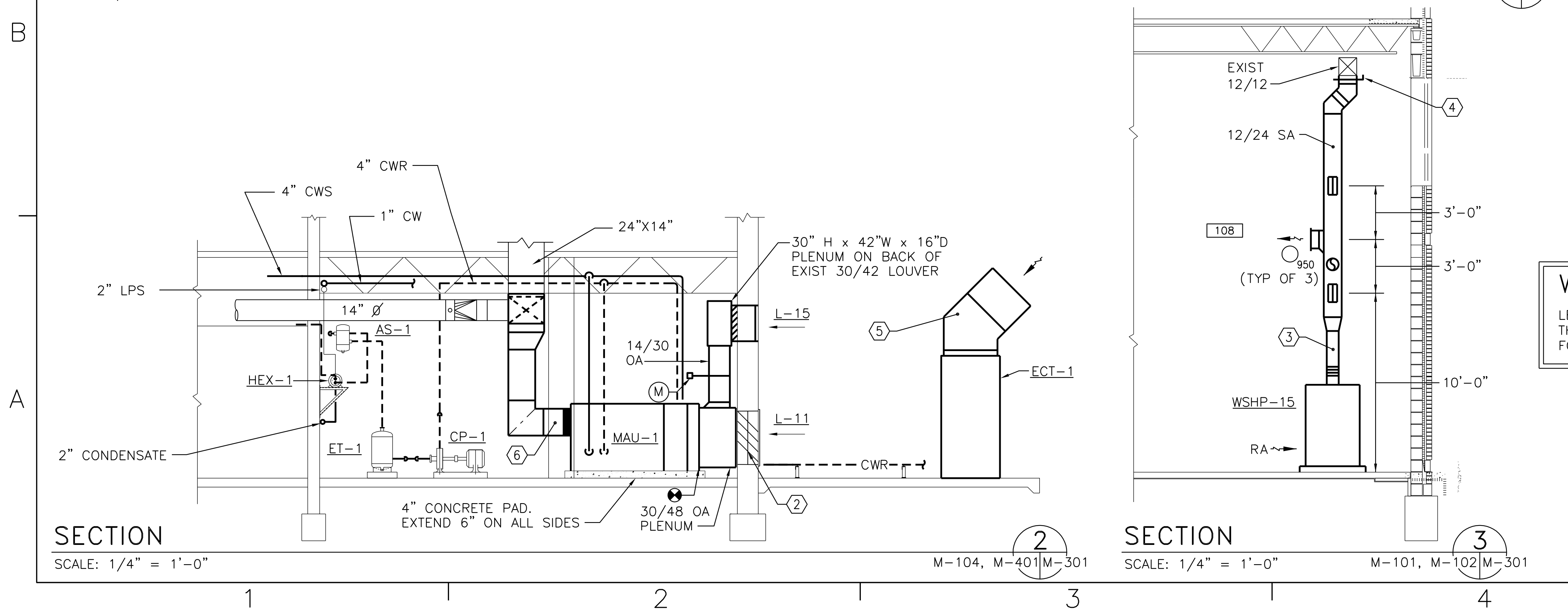


CONSTRUCTION NOTES (THIS SHEET ONLY)

- EXISTING WATER SOURCE HEAT PUMP. REFER TO "WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-301/M-504 FOR ADDITIONAL INFORMATION.
- EXISTING LOUVER TO REMAIN REFER TO "TYPICAL LOUVER INFILL DETAIL" D/M-301/A-501 FOR ADDITIONAL INFORMATION.
- SUPPLY AIR DUCT FULL SIZE OF UNIT CONNECTION RUN 2'-0" OF STRAIGHT DUCT.
- BALANCE TO 650 CFM TO EXIST REGISTERS.
- PROVIDE ALUMINUM DISCHARGE HOOD ON ECT-1 TO DISCHARGE AIR AWAY FROM BUILDING.
- 18/18 FULL SIZE OF UNIT CONNECTION RUN 18" OF STRAIGHT DUCT.

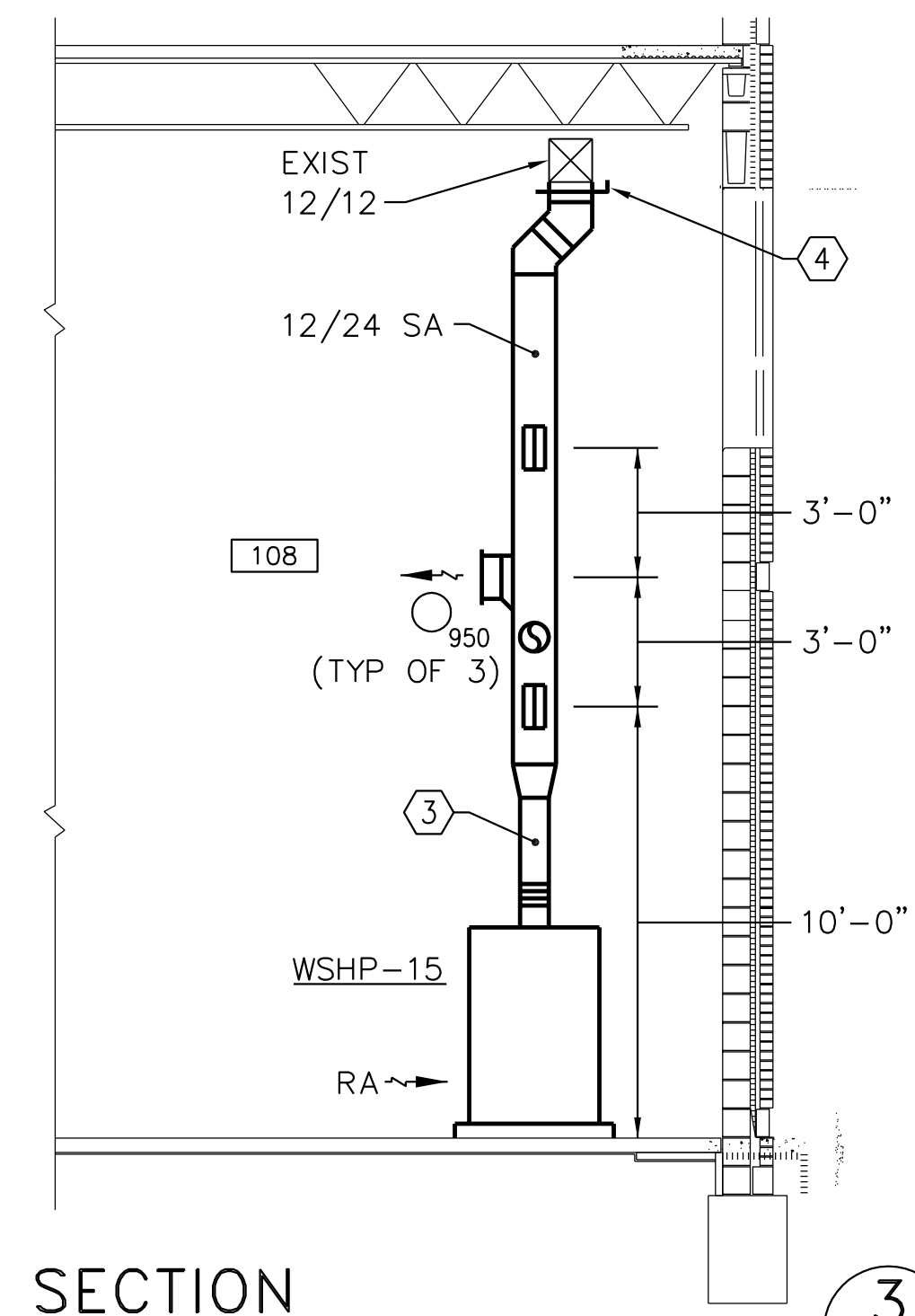
SECTION

SCALE: 1/4" = 1'-0"



SECTION

SCALE: 1/4" = 1'-0"



SECTION

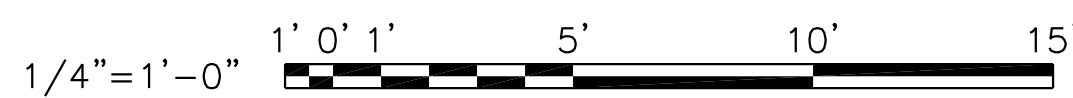
SCALE: 1/4" = 1'-0"

WARNING

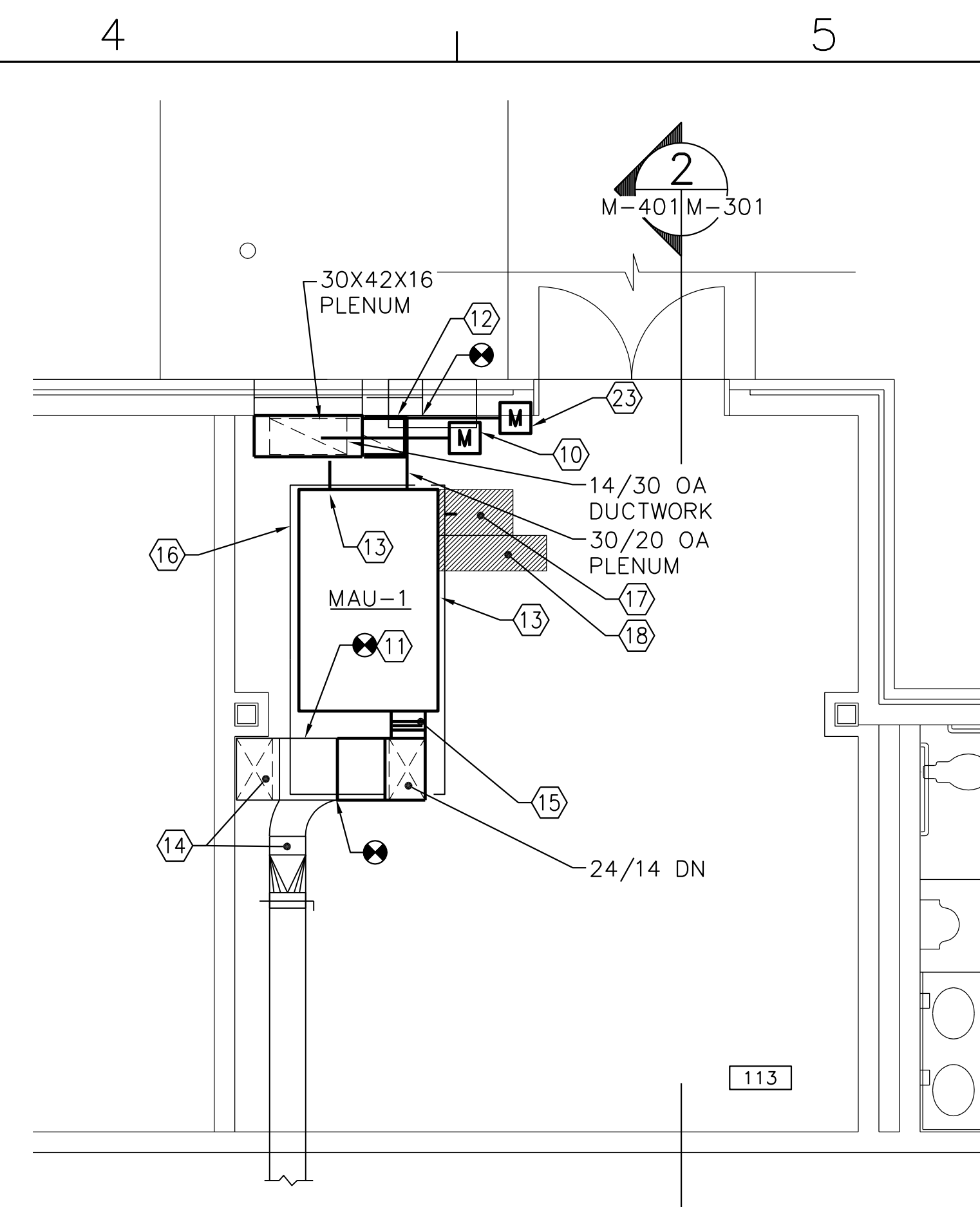
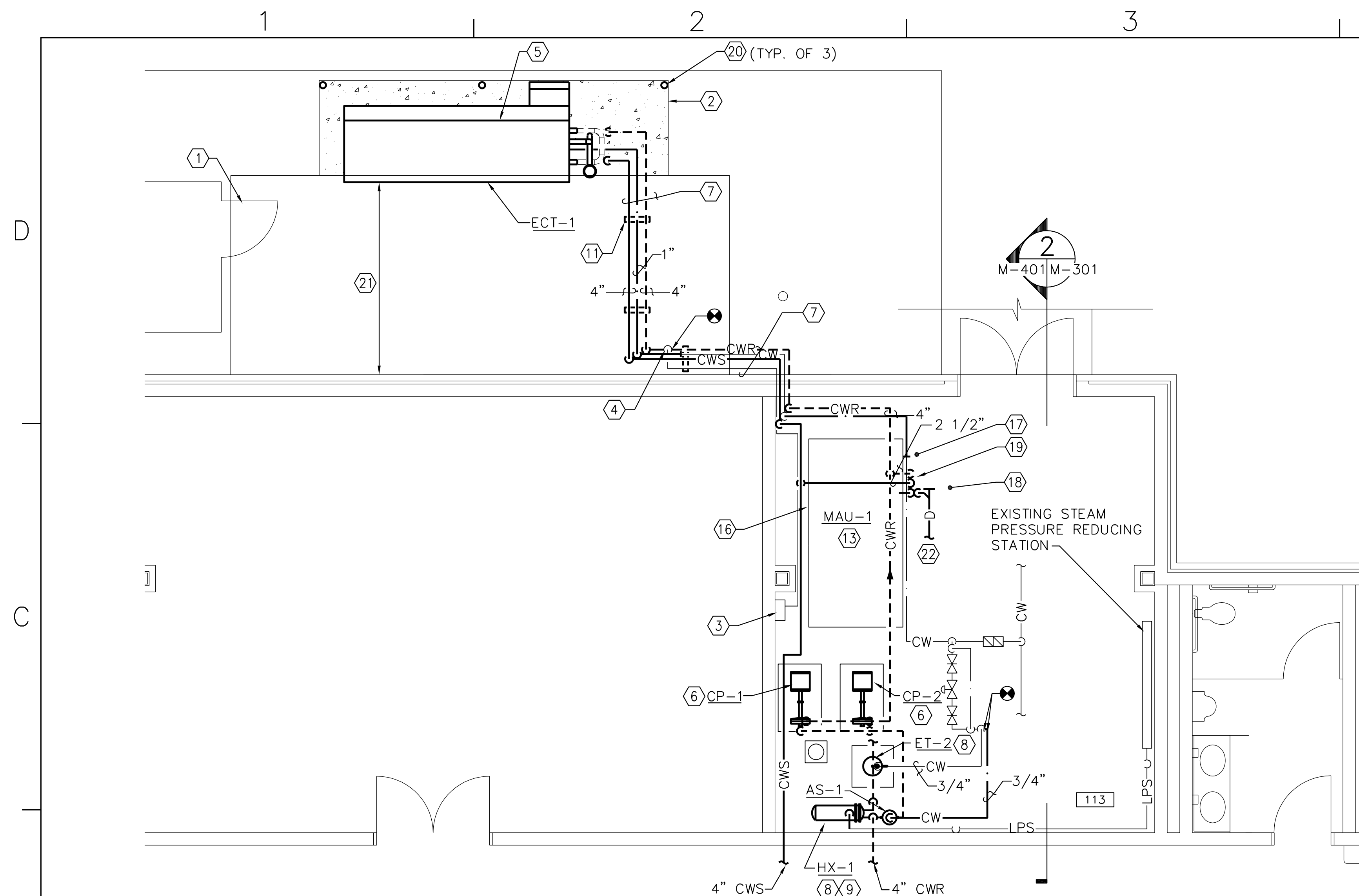
LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:

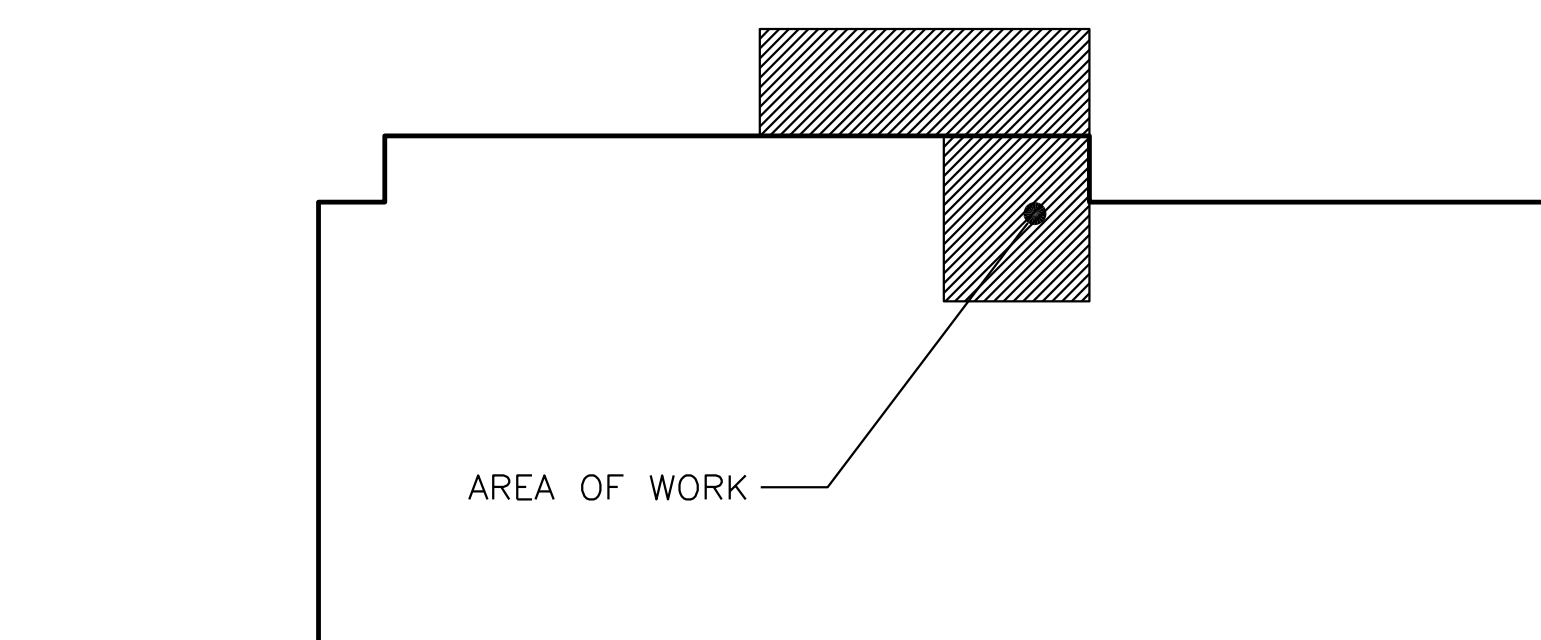


APPROVED	DATE	DESCRIPTION	SYMBOL
Bowman, Foster & Associates, P.C. Consulting Engineers #4 Interstate Corporate Center, Suite 1 Norfolk, VA 23502-4118			
APPROVED			
ACTIVITY - SATISFACTORY TO			
DATE			
APPROVED			
FOR EFD FOR COMMANDER NAVFAC			
DATE			
A/E	DESIGN	EFD	
BJW			
BJW	DRAWN		
DKH	REVIEW		
SGR	QC		
SGR	CHIEF ARCH/ ENGR.		
PROJECT MANAGER		JS	
FIRE PROTECTION		RM	
BRANCH MANAGER		CC	
DESIGN DIRECTOR			
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND MARINE CORPS AIR STATION, CHERRY POINT, N.C. REPLACE HVAC SYSTEM BLDG 4414 SECTIONS			
CODE ID. NO. 80091		SIZE 0	
SCALE: AS SHOWN			
FED. NO. 2827328			
STA. PROJ. NO. CP0641M			
SPEC. NO.			
CONSTR. CONTR. NO.			
NAVFAC DRAWING NO. 4555270			
SHEET 26 OF 49			
M-301			
DRAWING REVISION JULY 2003			



ENLARGED FIRST FLOOR MECHANICAL ROOM
PLAN — DUCTWORK — NEW WORK

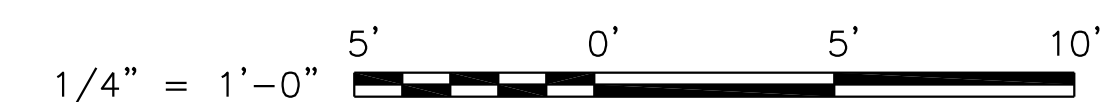
SCALE: 1/4" = 1'-0"



 **KEY PLAN**
NO SCALE

NOTE: IF THIS DRAWING IS A REDUCTION,
GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:



 ENLARGED FIRST FLOOR MECHANICAL ROOM PLAN – PIPING – NEW WORK
SCALE: 1/4" = 1'-0"

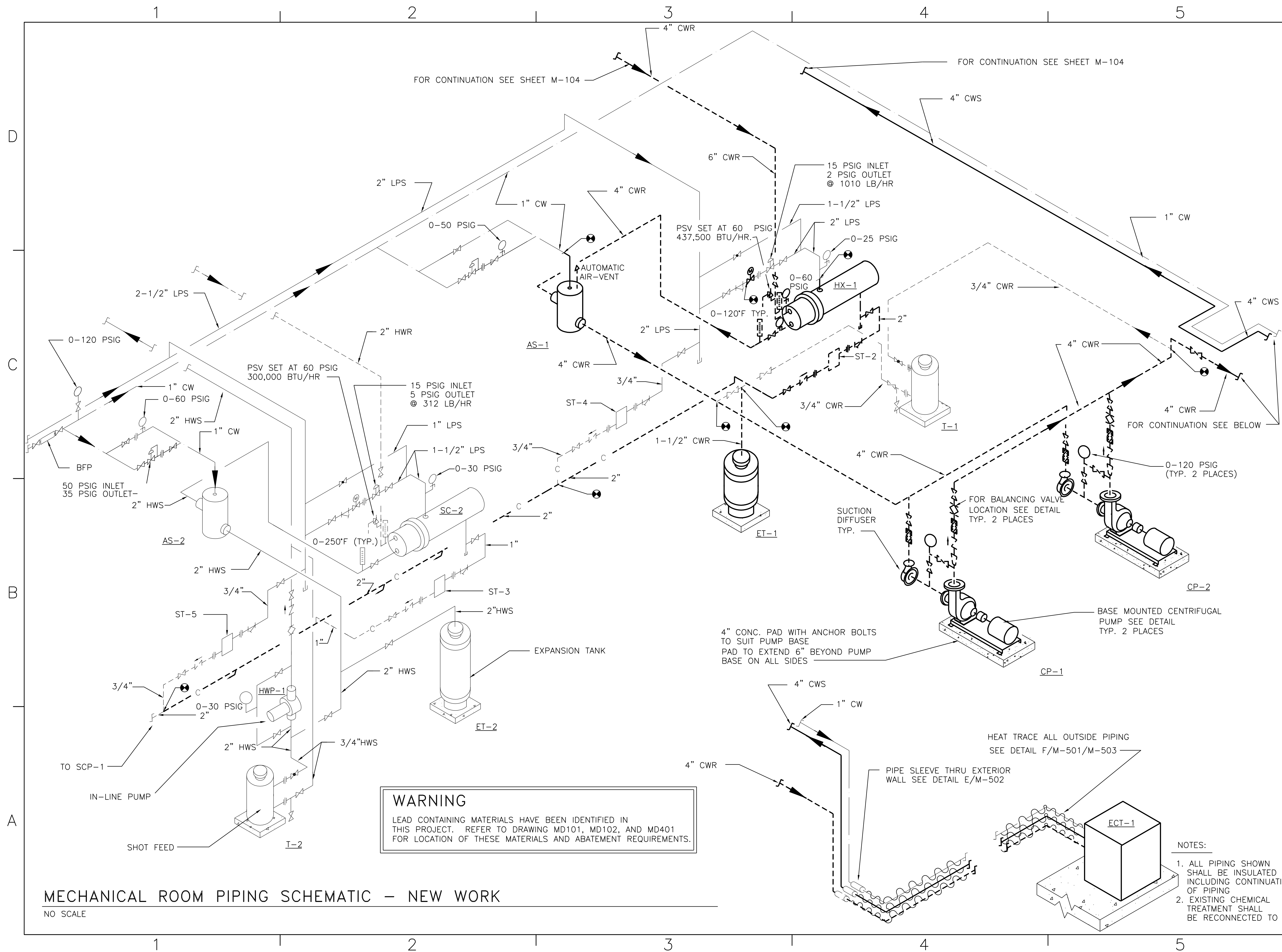
CONSTRUCTION NOTES (THIS SHEET ONLY)

- | | |
|---|---|
| <ul style="list-style-type: none">① MAINTAIN ACCESS TO TRAILER.② PROVIDE CONCRETE PAD TO EDGE OF SIDEWALK FOR NEW ECT-1. REFER TO "CONCRETE PAD DETAIL" G/M-401/M-502 FOR ADDITIONAL INFORMATION.③ EXIST WATER TREATMENT.④ EXTEND WATER TREATMENT PIPING TO ECT-1.⑤ ECT-1 SHALL BE LOCATED TO DISCHARGE AWAY FROM BUILDING.⑥ REFER TO "BASE MOUNTED CENTRIFUGAL PUMP DETAIL" E/M-401/M-501 FOR ADDITIONAL INFORMATION.⑦ PROVIDE HEAT TRACING ON ALL EXTERIOR PIPING. SEE "HEAT TRACING DETAILS" F/M-401/M-501.⑧ SEE MECHANICAL ROOM PIPING SCHEMATIC NEW WORK SHEET M-503.⑨ REFER TO "CONVERTOR SUPPORT DETAIL" D/M-401/M-502 FOR ADDITIONAL INFORMATION.⑩ PROVIDE NEW MOTORIZED DAMPER IN NEW OUTSIDE AIR DUCTWORK.⑪ PROVIDE PIPE SUPPORT. REFER TO "PIPE SUPPORT DETAIL" D/M-401/M-501 FOR ADDITIONAL INFORMATION.⑫ CONNECT OUTSIDE AIR DUCTWORK TO EXISTING LOUVER.⑬ PROVIDE MAKE-UP AIR UNIT. CONNECT NEW OUTSIDE AIR DUCTWORK TO UNIT. | <ul style="list-style-type: none">⑭ EXISTING SUPPLY AIR DUCTWORK SHALL REMAIN.⑮ 18/18 SA FULL SIZE OF UNIT CONNECTION MIN. 18" OF STRAIGHT DUCT.⑯ EXTEND 4" CONCRETE PAD AS REQUIRED FOR NEW MAU-1.⑰ FILTER ACCESS SPACE.⑱ COIL PULL AREA.⑲ REFER TO "WATER SOURCE HEAT PUMP PIPING DETAIL" A/M-401/M-504 FOR ADDITIONAL INFORMATION.⑳ PROVIDE GUARD POST (TYPICAL OF 3). REFER TO "GUARD POST DETAIL" H/M-401/M-502 FOR ADDITIONAL INFORMATION.㉑ MAXIMIZE DISTANCE AWAY FROM BUILDING.㉒ 1 1/4" DRAIN TO FLOOR DRAIN. REFER TO "TYPICAL CONDENSATE DRAIN TRAP DETAIL" A/M-401/M-502 FOR ADDITIONAL INFORMATION.㉓ PROVIDE MOTORIZED DAMPER IN INTAKE LOUVER RECONNECT TO DIRECT DIGITAL CONTROL SYSTEM. DAMPER TO OPEN WHEN EXISTING EXHAUST FAN IN MECHANICAL ROOM 113 IS ENERGIZED. |
|---|---|

WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

[illegible]



MECHANICAL ROOM PIPING SCHEMATIC – NEW WORK

NO SCALE

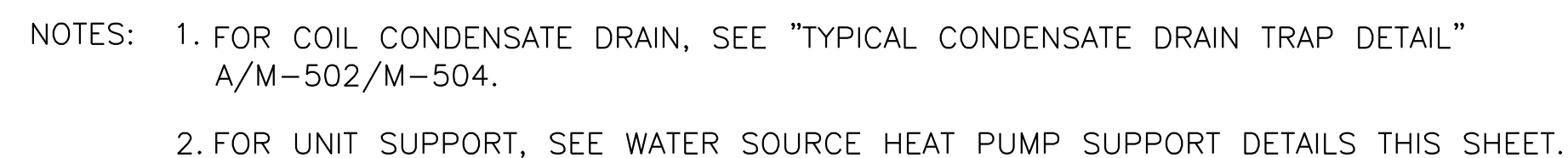
WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

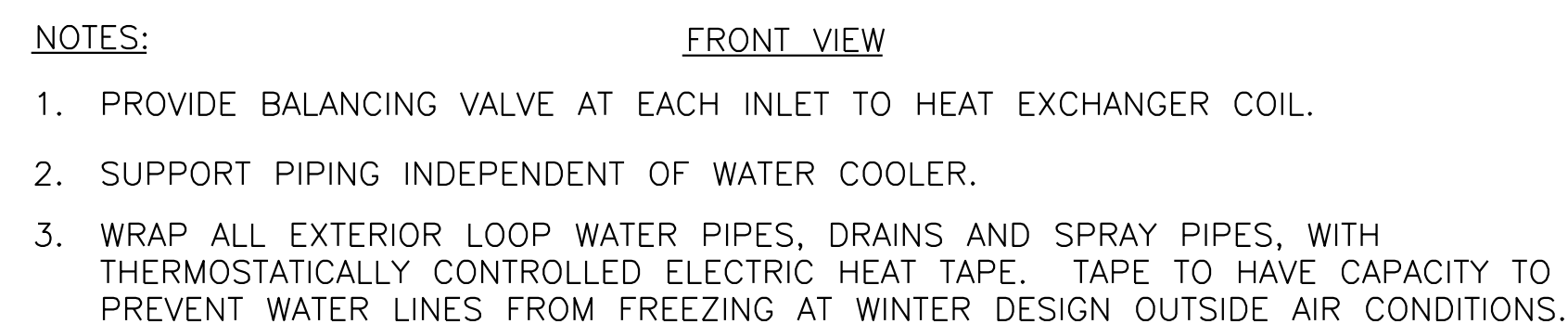
NOTES:

- ALL PIPING SHOWN SHALL BE INSULATED INCLUDING CONTINUATION OF PIPING
- EXISTING CHEMICAL TREATMENT SHALL BE RECONNECTED TO

APPROVED	DATE	DESCRIPTION	SYMBOL
FOR CONTINUATION SEE SHEET M-104			
FOR CONTINUATION SEE SHEET M-105			
FOR CONTINUATION SEE BELOW			
FOR BALANCING VALVE LOCATION SEE DETAIL TYP. 2 PLACES			
BASE MOUNTED CENTRIFUGAL PUMP SEE DETAIL TYP. 2 PLACES			
HEAT TRACE ALL OUTSIDE PIPING SEE DETAIL F/M-501/M-503			
PIPE SLEEVE THRU EXTERIOR WALL SEE DETAIL E/M-502			
SHOT FEED			
IN-LINE PUMP			
TO SCP-1			
HWP-1			
EXPANSION TANK			
ST-5			
ST-4			
ST-3			
ST-2			
T-1			
T-2			
ET-1			
ET-2			
SC-2			
AS-1			
AS-2			
BFP			
0-120 PSIG			
0-60 PSIG			
0-50 PSIG			
0-30 PSIG			
0-25 PSIG			
0-120°F TYP.			
0-250°F (TYP.)			
15 PSIG INLET 5 PSIG OUTLET @ 312 LB/HR			
15 PSIG INLET 2 PSIG OUTLET @ 1010 LB/HR			
PSV SET AT 60 PSIG 300,000 BTU/HR			
PSV SET AT 60 PSIG 437,500 BTU/HR			
AUTOMATIC AIR-VENT			
HUX-1			
SUCTION DIFFUSER TYP.			
4" CONC. PAD WITH ANCHOR BOLTS TO SUIT PUMP BASE PAD TO EXTEND 6" BEYOND PUMP BASE ON ALL SIDES			
4" CWR			
4" CWS			
1" CW			
2" LPS			
2" HWS			
3/4" CWR			
3/4" HWS			
2" HWR			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" C			
2" C			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			
2" HWS			
3/4" HWS			
2" HWS			
1" CW			
1" LPS			
1-1/2" LPS			
1-1/2" CWR			
3/4" CWR			
2" LPS			
4" CWR			
4" CWS			
1" CW			



NO SCALE (HORIZONTAL UNIT SHOW VERTICAL UNIT SIMILAR.) M-101, M-102, M-103, M-104, M-105, M-106, M-301, M-401, M-504



NO SCALE



NO SCALE



NO SCALE



NO SCALE



NO SCALE

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.



DETAIL

NO SCALE



NO SCALE

[illegible]

MARK	LOCATION	FAN DATA			DIRECT EXPANSION WATER COIL (SEE NOTE 2)								ELECTRIC REHEAT COIL				HUMIDIFIER			ELECTRICAL DATA					REMARKS
		SUPPLY CFM	ESP "WC	MOTOR HP	TOTAL MBH	SENSIBLE MBH	EAT DEG. FDB		LAT DEG. FDB		WPD FT	WATER FLOW GPM	TOTAL MBH	EAT DEG. FDB	LAT DEG. FDB	HEATER KW	TYPE	CAPACITY LBS/HR K _w		VOLTS	PH	FLA	MCA	MOCP	
CRAC-1	ROOM 108	1600	0.50	1	50.6	40.6	75	62.5	57.9	54.9	4.4	12	41.0	70	93.6	12	STEAM GENERATING	10	3.2	460	3	24	30	35	1

NOTES:

- DOWN FLOW UNIT WITH ELECTRIC REHEAT, HUMIDIFIER AND FLOOR STAND.
- CONDENSER WATER 89 DEG. F EWT.

MARK	FAN DATA				WATER FLOW GPM	TOTAL MBH	SENSIBLE MBH	COOLING CAPACITY				HEAT REJEC TION MBH	HEATING CAPACITY		ELECTRICAL DATA				WPD FT	FILTER		UNIT TYPE	REMARKS	
	CFM	OUTSIDE AIR CFM	ESP "WC	MOTOR HP				EAT		LAT			TOTAL MBH	HEAT ABSORPTION MBH	TOTAL FLA	MFS	VOLTS	PH		TYPE	SIZE IN			
								DEG. FDB	DEG. FWB	DEG. FDB	DEG. FWB													
WSHP-1	650	80	--	--	3.00	--	--	--	--	--	--	15.0	--	10.2	--	--	--	--	--	--	--	--	--	1
WSHP-2	900	120	--	--	5.20	--	--	--	--	--	--	26.0	--	22.3	--	--	--	--	--	--	--	--	--	1
WSHP-3	1500	70	--	--	10.00	--	--	--	--	--	--	50.0	--	33.9	--	--	--	--	--	--	--	--	--	1
WSHP-4	1300	100	--	--	6.00	--	--	--	--	--	--	30.0	--	25.3	--	--	--	--	--	--	--	--	--	1
WSHP-5	1300	100	--	--	6.80	--	--	--	--	--	--	34.0	--	24.0	--	--	--	--	--	--	--	--	--	1
WSHP-6	1500	80	--	--	10.00	--	--	--	--	--	--	50.0	--	33.9	--	--	--	--	--	--	--	--	--	1
WSHP-7	650	60	--	--	3.00	--	--	--	--	--	--	15.0	--	13.4	--	--	--	--	--	--	--	--	--	1
WSHP-8	900	40	--	--	4.00	--	--	--	--	--	--	20.0	--	17.6	--	--	--	--	--	--	--	--	--	1
WSHP-9	650	40	--	--	4.40	--	--	--	--	--	--	22.0	--	15.8	--	--	--	--	--	--	--	--	--	1
WSHP-10	1350	70	--	--	6.00	--	--	--	--	--	--	30.0	--	25.3	--	--	--	--	--	--	--	--	--	1
WSHP-11	750	120	--	--	4.50	--	--	--	--	--	--	22.5	--	17.5	--	--	--	--	--	--	--	--	--	1
WSHP-12	1500	90	--	--	7.50	--	--	--	--	--	--	37.5	--	32.3	--	--	--	--	--	--	--	--	--	1
WSHP-13	950	30	--	--	5.00	--	--	--	--	--	--	25.0	--	19.5	--	--	--	--	--	--	--	--	--	1
WSHP-14	1350	180	--	--	12.00	--	--	--	--	--	--	60.0	--	55.5	--	--	--	--	--	--	--	--	--	1
WSHP-1A	1600	200	0.26	0.5	7.00	42.5	33.9	75	62.5	55.7	53.3	57.8	54.2	40.3	7.80	15	460	3	4.66	THROWAWAY	1	HORIZONTAL		
WSHP-4A	1090	100	0.83	0.5	4.80	30.1	23.7	75	62.5	55.1	52.9	40.3	41.8	31.9	16.40	30	265	1	4.91	THROWAWAY	1	HORIZONTAL		
WSHP-5A	1300	100	0.49	0.5	8.00	35.7	28.7	75	62.5	55.0	53.0	47.7	46.3	34.4	6.50	15	460	3	5.80	THROWAWAY	1	HORIZONTAL		
WSHP-5B	2000	100	0.43	1.0	12.00	53.2	42.6	75	62.5	55.5	53.3	70.8	71.4	54.4	10.20	15	460	3	9.70	THROWAWAY	1	HORIZONTAL		
WSHP-6A	640	80	0.28	0.125	3.30	17.1	13.5	75	62.5	55.7	53.2	22.7	21.4	16.4	7.10	15	265	1	5.13	THROWAWAY	1	HORIZONTAL		
WSHP-10A	1000	130	0.02	0.333	5.50	25.9	20.7	75	62.5	56.1	53.6	34.2	32.6	25.0	12.40	25	265	1	9.90	THROWAWAY	1	HORIZONTAL		
WSHP-10B	1480	280	0.42	0.5	7.00	42.1	32.5	75	62.5	55.0	52.6	57.1	53.7	39.6	16.40	30	265	1	4.76	THROWAWAY	1	HORIZONTAL		
WSHP-11A	1490	160	0.41	0.5	7.50	42.2	32.7	75	62.5	55.0	52.6	57.3	53.8	39.8	7.80									

CLOSED-CIRCUIT COOLER SCHEDULE															
MARK	WATER FLOW GPM	COOLING LOAD MBH	EWT F DEG.	LWT F DEG.	WPD FT	EAT (WB) F DEG.	FAN			SPRAY PUMP		SUMP HEATER	ELECTRICAL		REMARKS
							TYPE	AIR FLOW CFM	HP	FLOW GPM	MOTOR HP	KW	VOLTS	PH	
ECT-1	252.4	1438	100.4	89	4.8	78	CENTRIFUGAL	26200	20	245	1.5	3	460	3	
NOTES															
1.MOTOR STARTERS: FAN-NEMA 3R SIZE 2 SPRAY PUMP NEMA 3R SIZE 0.															

PUMP SCHEDULE													
MARK	LOCATION	TYPE	SYSTEM SERVED	WATER FLOW GPM	HEAD FT	MOTOR					MOTOR STARTER		REMARKS
						HP	VOLTS	PHASE	RPM	EFF.	SIZE	TYPE	
CP-1	MECH. ROOM	BASE MOUNTED	CONDENSER WATER	252.4	105	15	460	3	1750	70	2	NEMA 1	
CP-2	MECH. ROOM	BASE MOUNTED	CONDENSER WATER	252.4	105	15	460	3	1750	70	2	NEMA 1	
NOTES:													

EXPANSION TANK SCHEDULE							
MARK	SERVICE	TYPE	SIZE (IN.)		VOLUME (GALLONS)		REMARKS
			DIAMETER	LENGTH	TANK	ACCEPTANCE	
ET-1	CONDENSER WATER	FLOOR MOUNTED	16.25	45.125	33.6	11.3	
NOTES:							

HVAC DESIGN CONDITIONS						
INDOOR			OUTDOOR			
SUMMER		WINTER	SUMMER			WINTER
FDB	% RH	FDB	FDB	MEAN COINCIDENT FWB		FDB
75	50	70	90	78		22
NOTES:						

EF-2

REBALANCE TO 200 CFM @ .15" ESP (EXISTING @ 350 CFM @ .20" ESP)

EF-3

REBALANCE TO 200 CFM @ .15" ESP (EXISTING @ 350 CFM @ .20" ESP)

EF-6

REBALANCE TO 225 CFM @ .15" ESP (EXISTING @ 200 CFM @ .20" ESP)

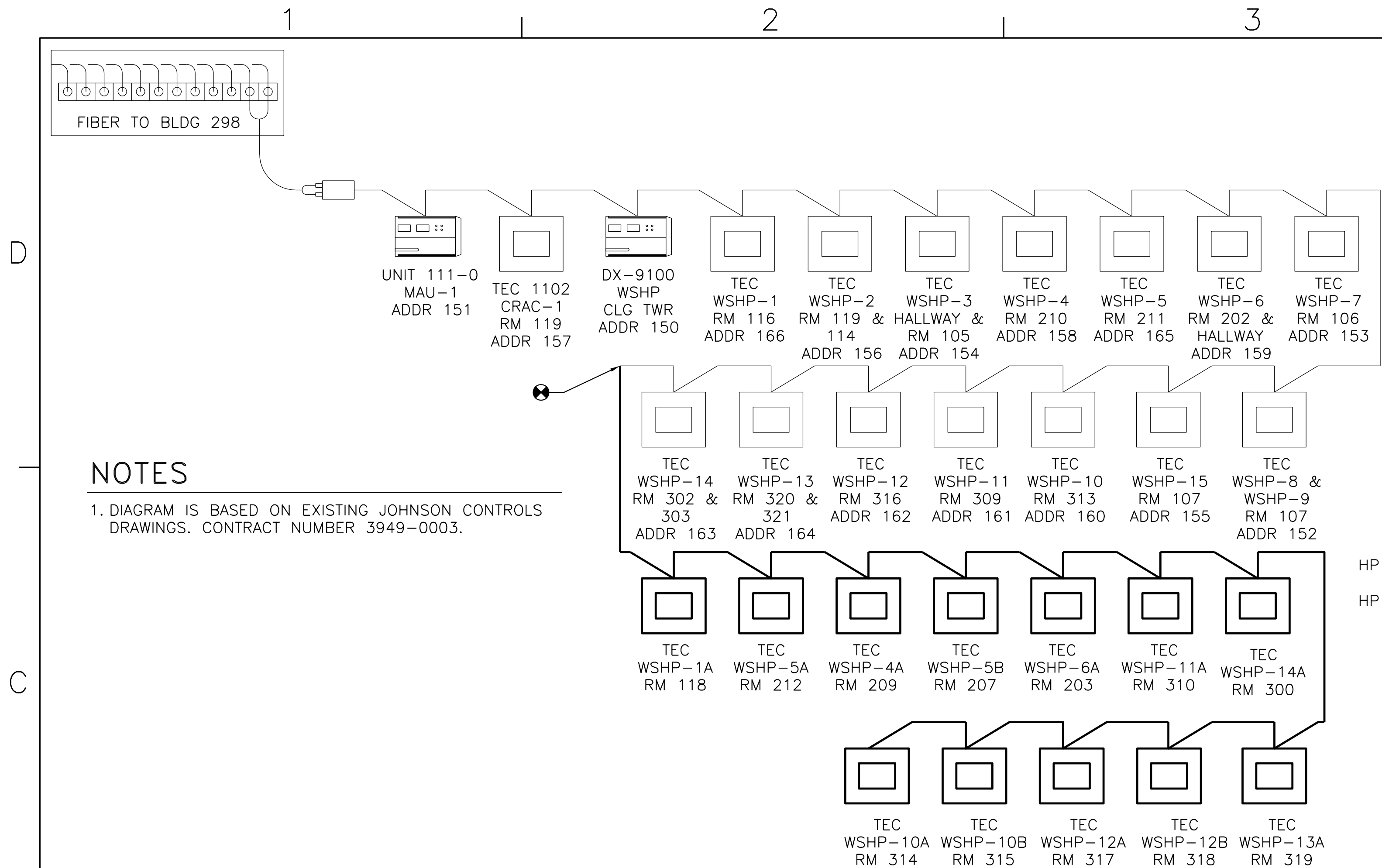
EF-7

REBALANCE TO 225 CFM @ .15" ESP (EXISTING @ 200 CFM @ .20" ESP)

PROVIDE SPEED CONTROLLER ON EF-2, 3, 6, & 7 TO REDUCE AIR FLOW AS INDICATED.

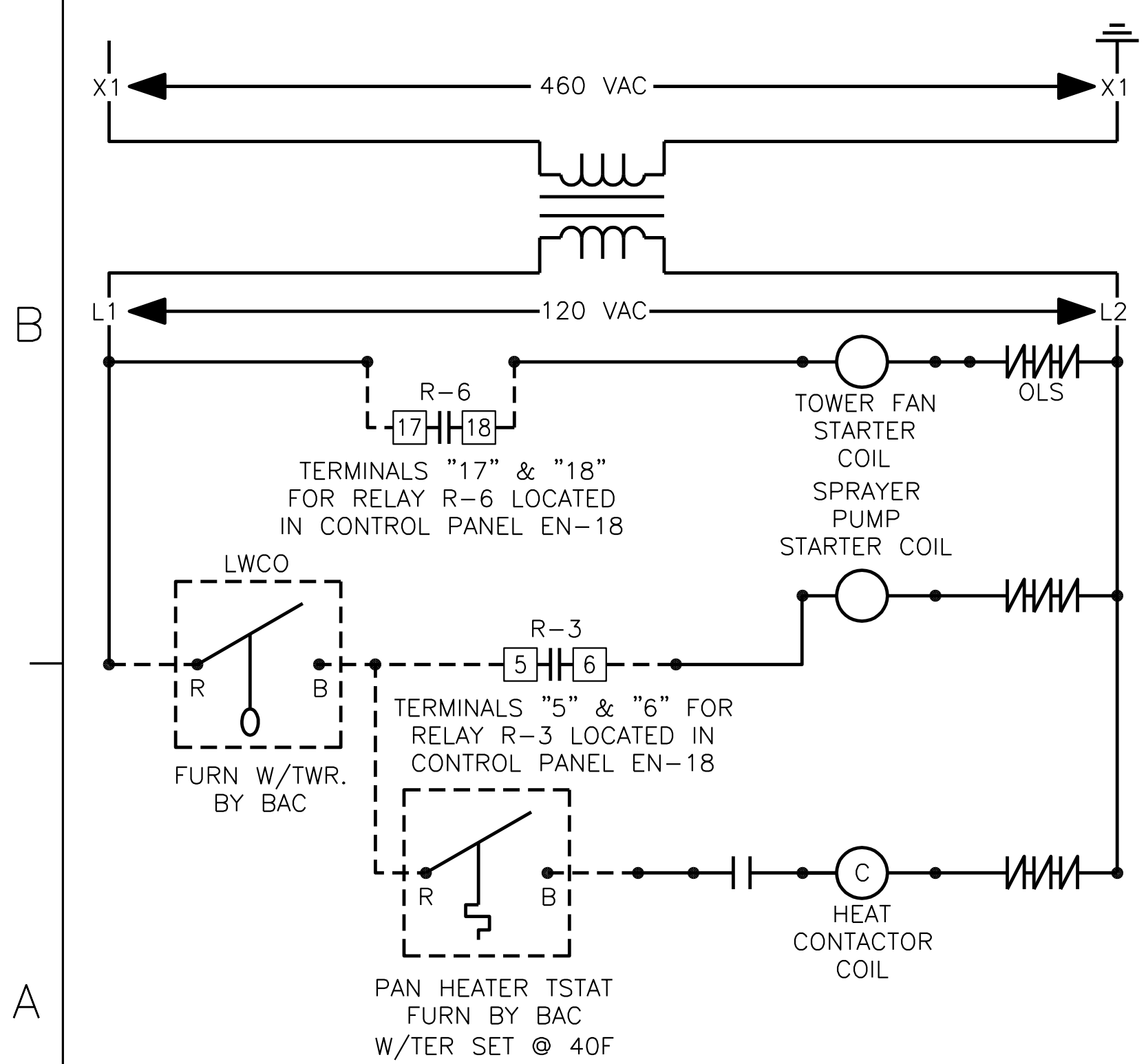
LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

[illegible]



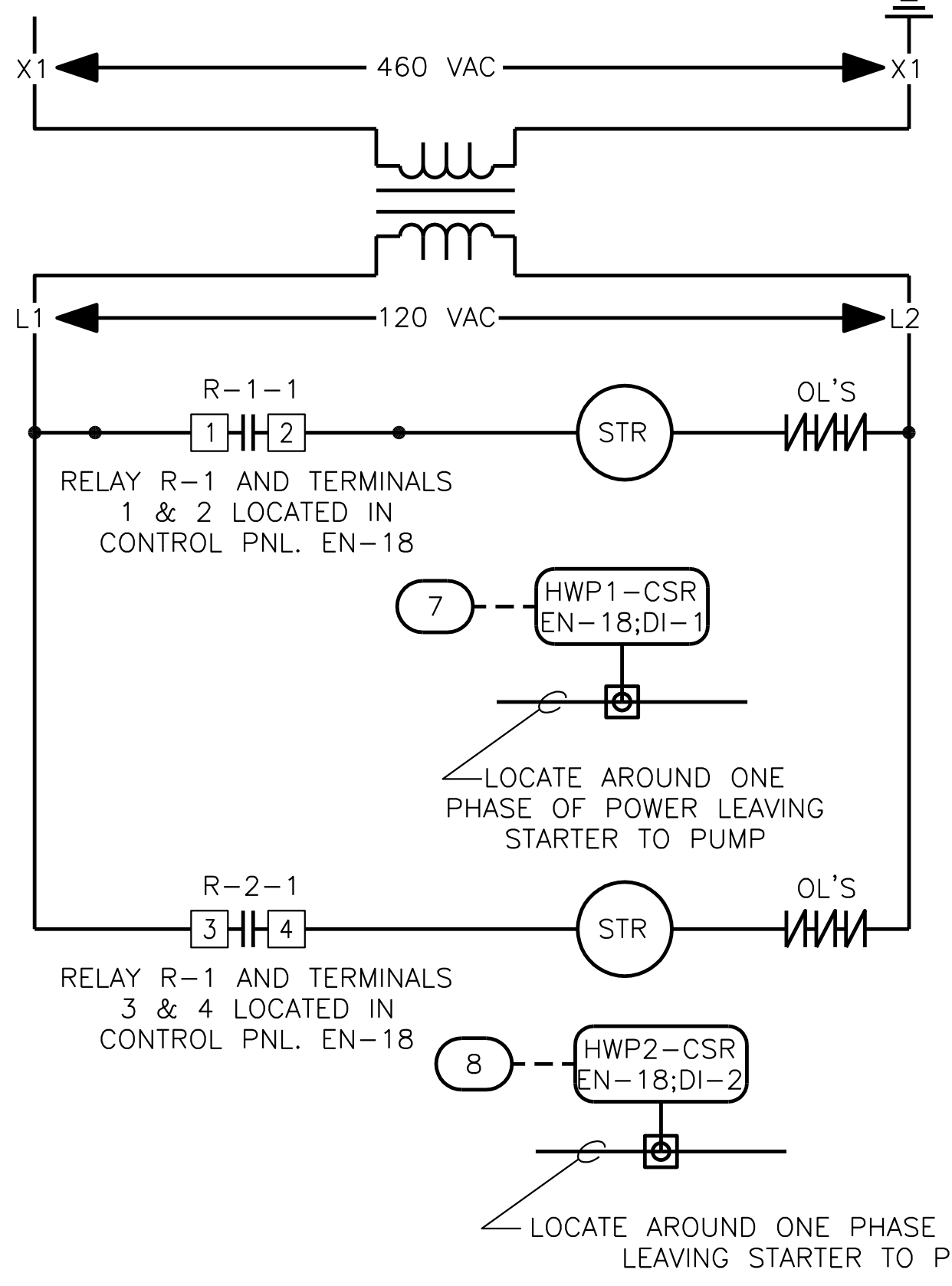
NOTES

1. DIAGRAM IS BASED ON EXISTING JOHNSON CONTROLS DRAWINGS. CONTRACT NUMBER 3949-0003.



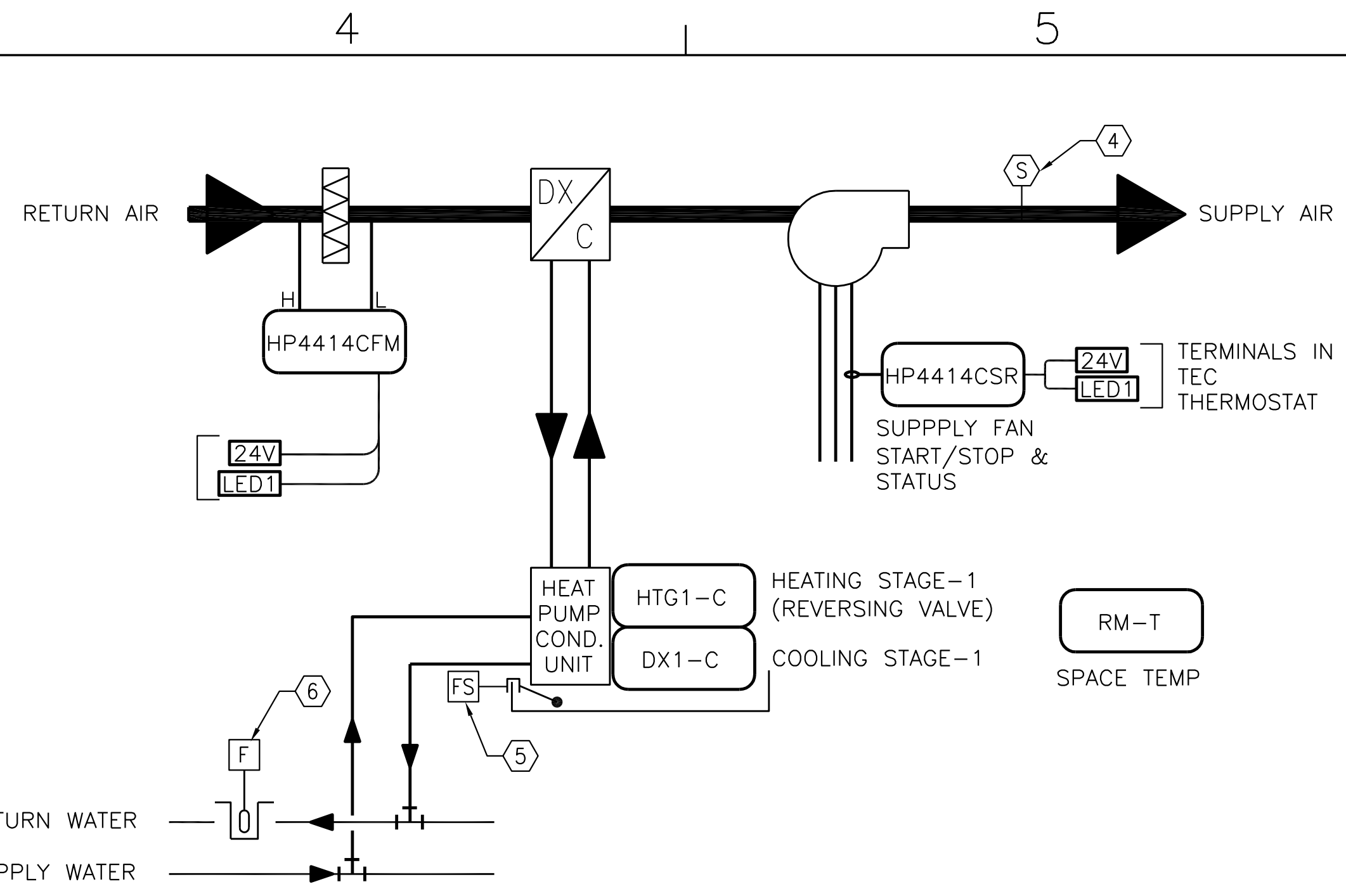
SYSTEM TOWER SYSTEM WIRING DIAGRAM

NO SCALE



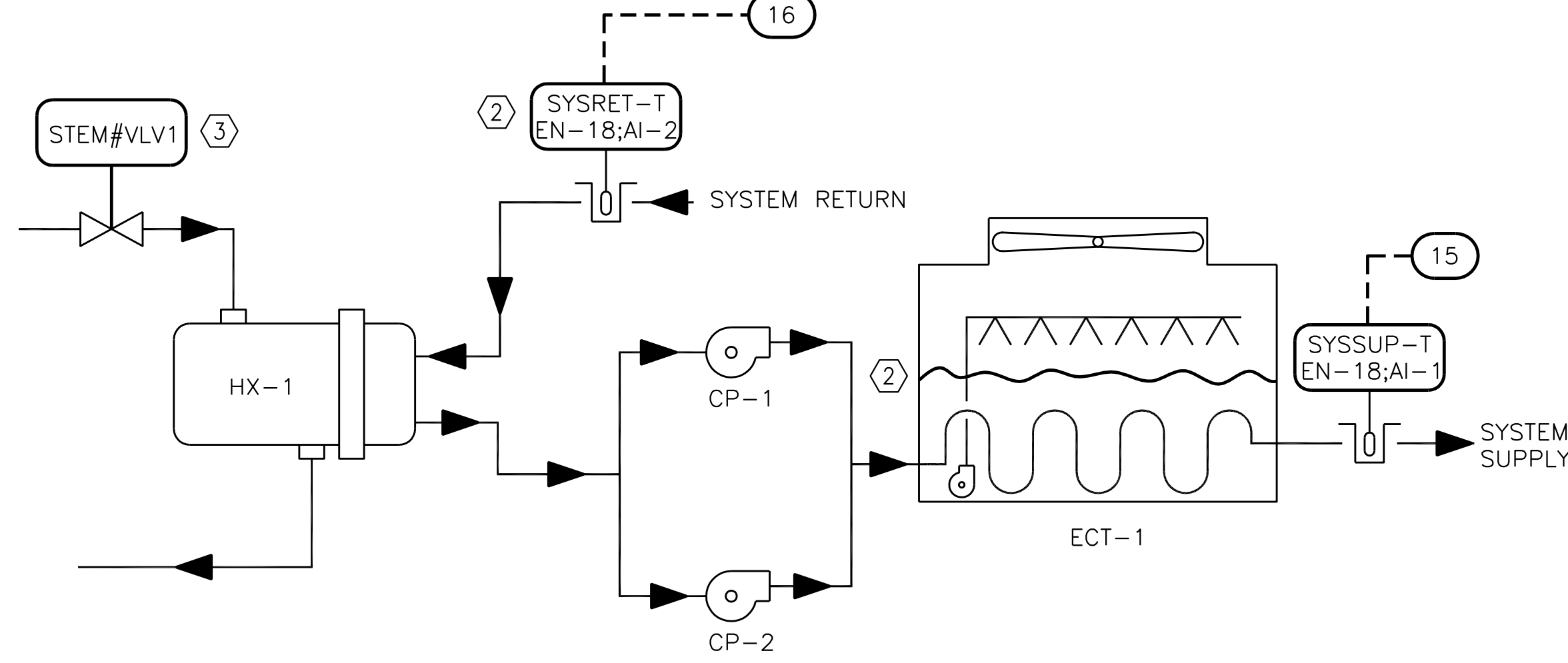
SYSTEM WATER PUMP STARTER CONTROL WIRING

NO SCALE (TYPICAL FOR HOT WATER PUMP HVA4414S AND HVA4414T)



WATER SOURCE HEAT PUMP DIAGRAM ①

NO SCALE



WATER SOURCE HEAT PUMP LOOP SYSTEM ①

NO SCALE

WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

CONSTRUCTION NOTES (THIS SHEET ONLY)

- ① DISCONNECT CONTROLS SHOWN AND RECONNECT TO NEW EQUIPMENT.
- ② PROVIDE NEW SENSOR.
- ③ PROVIDE NEW VALVE.
- ④ PROVIDE SMOKE DUCT DETECTOR TO WSHP-5B AND WSHP-15 (ONLY) WIRE TO SHUT-DOWN UNIT AND SEND ALARM TO FIRE ALARM CONTROL PANEL.
- ⑤ PROVIDE FLOAT SWITCH IN AUX. DRAIN PAN.
- ⑥ PROVIDE FLOW SWITCH.

DATE	APPR
DATE	SYN
DESCRIPTION	
BOWMAN, FOSTER & ASSOCIATES, P.C. Consulting Engineers 44 Interstate Corporate Center, Suite 1 Norfolk, VA 23502-4118	
APPROVED	
ACTIVITY - SATISFACTORY TO	
DATE	
APPROVED	
FOR EFD FOR COMMANDER NAVFAC	
DATE	
A/E	DESIGN
BJW	BJW
DKH	DKH
SGR	SGR
SGR	SGR
PROJECT MANAGER	ARCH/ENGR
FIRE PROTECTION	RM
BRANCH MANAGER	CC
DESIGN DIRECTOR	CC
NAVAL FACILITIES ENGINEERING COMMAND	
MARINE CORPS AIR STATION, CHERRY POINT, N.C.	
REPLACE HVAC SYSTEM BLDG 4414	
CONTROL DIAGRAMS	
CODE ID. NO. 80091	SIZE 0
SCALE: AS SHOWN	
FED. NO. 2827328	
STA. PROJ. NO. CP0641M	
SPEC. NO. -	
CONSTR. CONTR. NO. -	
NAVFAC DRAWING NO. 4555278	
SHEET 34 OF 49	
M-701	
DRAWING REVISION JULY 2003	

DDC SEQUENCE OF OPERATION

MAKE-UP AIR UNIT MAU-1

RUN CONDITIONS – INTERLOCKED:

THE UNIT MAU-1 SHALL BE ENABLED TO RUN BASED ON A TIME SCHEDULE UNLESS SHUTDOWN ON SAFETIES.

EMERGENCY SHUTDOWN:

THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING AN EMERGENCY SHUTDOWN SIGNAL.

FREEZE PROTECTION:

THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING A FREEZESTAT STATUS.

SMOKE DETECTION:

THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM UPON RECEIVING A SMOKE DETECTOR STATUS.

OUTSIDE AIR DAMPER:

THE OUTSIDE AIR DAMPER SHALL OPEN ANYTIME THE UNIT RUNS AND SHALL CLOSE ANYTIME THE UNIT STOPS. THE SUPPLY FAN SHALL START ONLY AFTER THE DAMPER STATUS HAS PROVEN THE DAMPER IS OPEN. THE OUTSIDE AIR DAMPER SHALL CLOSE 4SEC (ADJ.) AFTER THE SUPPLY FAN STOPS.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- OUTSIDE AIR DAMPER FAILURE: COMMANDED OPEN, BUT THE STATUS IS CLOSED.
- OUTSIDE AIR DAMPER IN HAND: COMMANDED CLOSED, BUT THE STATUS IS OPEN.

PREFILTER DIFFERENTIAL PRESSURE MONITOR:

THE CONTROLLER SHALL MONITOR THE DIFFERENTIAL PRESSURE ACROSS THE PREFILTER.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- PREFILTER CHANGE REQUIRED: PREFILTER DIFFERENTIAL PRESSURE EXCEEDS A USER DEFINABLE LIMIT (ADJ.).

SUPPLY AIR TEMPERATURE:

THE CONTROLLER SHALL MONITOR THE SUPPLY AIR TEMPERATURE.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- HIGH SUPPLY AIR TEMP: IF THE SUPPLY AIR TEMPERATURE IS GREATER THAN 120°F (ADJ.).
- LOW SUPPLY AIR TEMP: IF THE SUPPLY AIR TEMPERATURE IS LESS THAN 45°F (ADJ.).

UNIT MOUNTED CONTROLS:

SUPPLY FAN:

THE SUPPLY FAN SHALL RUN ANYTIME THE UNIT IS COMMANDED TO RUN. TO PREVENT SHORT CYCLING, THE SUPPLY FAN SHALL HAVE A USER DEFINABLE (ADJ.) MINIMUM RUNTIME, UNLESS SHUTDOWN ON SAFETIES.

ALARMS SHALL BE PROVIDED AS FOLLOWS:

- SUPPLY FAN FAILURE: COMMANDED ON, BUT THE STATUS IS OFF.
- SUPPLY FAN IN HAND: COMMANDED OFF, BUT THE STATUS IS ON.
- SUPPLY FAN RUNTIME EXCEEDED: STATUS RUNTIME EXCEEDS A USER DEFINABLE LIMIT (ADJ.).

COOLING/DEHUMIDIFICATION MODE:

COOLING COIL LEAVING AIR TEMPERATURE SETPOINT – FIXED:

THE CONTROLLER SHALL MONITOR THE COOLING COIL LEAVING AIR TEMPERATURE AND SHALL MAINTAIN A FIXED SUPPLY AIR DEWPOINT SETPOINT OF 50°F (ADJ.).

SUPPLY AIR TEMPERATURE SETPOINT – FIXED:

THE CONTROLLER SHALL MONITOR THE SUPPLY AIR TEMPERATURE AND SHALL MAINTAIN A FIXED SUPPLY AIR TEMPERATURE SETPOINT OF 72°F (ADJ.).

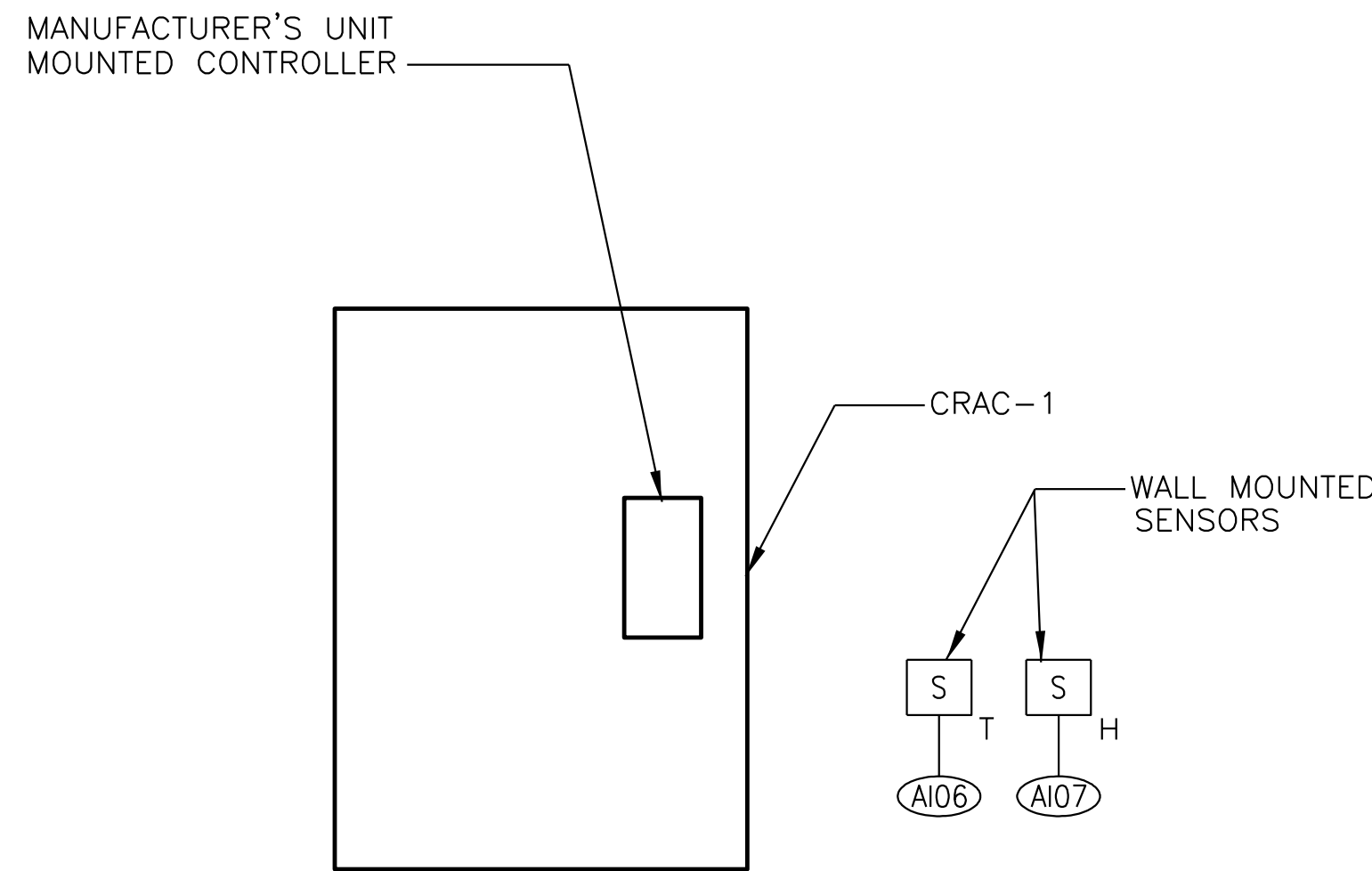
HEATING MODE:

SUPPLY LEAVING AIR TEMPERATURE SETPOINT – FIXED:

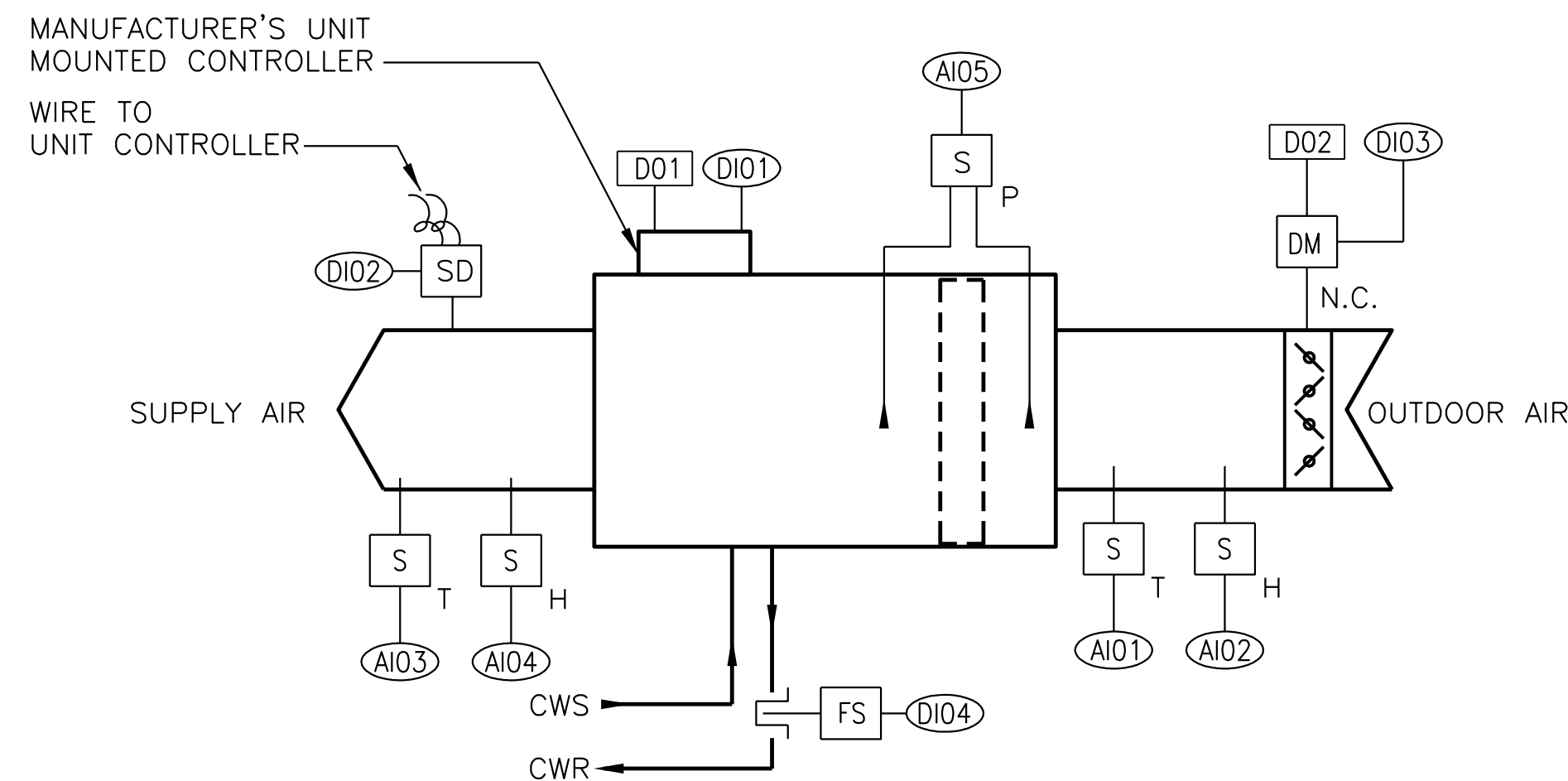
THE CONTROLLER SHALL MONITOR THE SUPPLY AIR TEMPERATURE AND SHALL MAINTAIN A FIXED SUPPLY AIR TEMPERATURE SETPOINT OF 68°F (ADJ.)

THE HEATING SHALL BE ENABLED WHENEVER:

- OUTSIDE AIR TEMPERATURE IS LESS THAN 50°F (ADJ.).
- AND THE FAN STATUS IS ON.



COMPUTER ROOM AIR HANDLING UNIT (CRAC-1) CONTROL DIAGRAM



MAKE-UP AIR HANDLING UNIT (MAU-1) CONTROL DIAGRAM

CONTROLS LEGEND

AI	ANALOG INPUT TO DDC/FMS
DDC	DIRECT DIGITAL CONTROL
NO	NORMALLY OPEN
NC	NORMALLY CLOSED
AI (AI) DI (DI)	DDC INPUT POINT (ANALOG, DIGITAL)
DO (DO)	DDC OUTPUT POINT (DIGITAL)
DM (DM)	DAMPER ACTUATOR 2-10 VAC SIGNAL OR 2 POSITION AS REQUIRED
S (S) P (P)	DIFFERENTIAL PRESSURE TRANSMITTER
SD (SD)	DUCT MOUNTED SMOKE DETECTOR
S (S) T (T)	TEMPERATURE SENSOR
S (S) H (H)	HUMIDITY SENSOR
FS (FS)	FLOW SWITCH

POINTS LIST	
LOCAL DDC CONTROLLER	
INPUTS	
AI01	OUTSIDE AIR TEMPERATURE
AI02	OUTSIDE AIR HUMIDITY
AI03	SUPPLY AIR TEMPERATURE
AI04	SUPPLY AIR HUMIDITY
AI05	FILTER STATIC PRESSURE
AI06	ROOM 119 TEMPERATURE
AI07	ROOM 119 HUMIDITY
DI01	MAU-1 ALARM
DI02	SMOKE DETECTOR
DI03	DAMPER END SWITCH
DI04	FLOW SWITCH
OUTPUTS	
DO01	MAU-1 ENABLE
DO02	OUTSIDE AIR DAMPER

WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

DATE	APR
DESCRIPTION	
SYN	
Bowmar, Foster & Associates, P.C. Consulting Engineers #4 Interstate Corporate Center, Suite 1 Norfolk, VA 23502-4118	
APPROVED	06-004C
ACTIVITY - SATISFACTORY TO	
DATE	
APPROVED	
FOR EFD FOR COMMANDER NAVFAC	
DATE	
A/E	DESIGN
BJW	DRAWN
DKH	REVIEW
SGR	QC
SGR	CHIEF ARCH/ ENGR.
PROJECT MANAGER	JS
FIRE PROTECTION	RM
BRANCH MANAGER	CC
DESIGN DIRECTOR	CC
NAVAL FACILITIES ENGINEERING COMMAND	
DEPARTMENT OF THE NAVY	
MARINE CORPS AIR STATION, CHERRY POINT, N.C.	
REPLACE HVAC SYSTEM BLDG 4414	
CONTROL DIAGRAM POINTS LIST AND SEQUENCE OF OPERATION	
CODE ID. NO. 80091	SIZE 0
SCALE: AS SHOWN	
FED. NO. 2827328	
STA. PROJ. NO. CP0641M	
SPEC. NO. -	
CONSTR. CONTR. NO. -	
NAVFAC DRAWING NO. 4555280	
SHEET 36 OF 49	
M-703	
DRAWING REVISION JULY 2003	

NOTE: DASHED SYMBOLS INDICATE EXISTING DEVICES, EQUIPMENT, CONDUIT, ETC. TO REMAIN, UNLESS OTHERWISE NOTED. AN "X" THRU A DASHED SYMBOL INDICATES REMOVE.

- | | |
|---|--|
|  | 2'x4' FLUORESCENT LIGHTING FIXTURE. |
|  | 2'x4' FLUORESCENT LIGHTING FIXTURE. |
|  | 1'x4' FLUORESCENT LIGHTING FIXTURE. |
|  | 1'x4' FLUORESCENT LIGHTING FIXTURE WITH INTEGRAL BATTERY BACK UP. |
|  | EXIT LIGHTING FIXTURE. ARROW, WHEN USED INDICATES DIRECTION (SHADING INDICATES FACE OF SIGN ORIENTATION). |
|  | S SINGLE POLE SWITCH. 20 A, 120/277 V FLUSH MOUNT 48" AFF UON. |
|  | M MOTOR RATED SWITCH. |
|  | E EQUIPMENT CONNECTION AS NOTED. |
|  | J JUNCTION BOX. |
|  | M MAGNETIC MOTOR CONTROLLER (BY MECHANICAL CONTRACTOR). |
|  | DISCONNECT SWITCH. 600 V UON, IN NEMA 1 ENCLOSURE UON
3P = NO. OF POLES, 60 = SWITCH RATING, 40 = FUSE RATING (NF INDICATES NON-FUSIBLE). |
|  | E ELECTRICAL PANELBOARD (480Y/277 V). |
|  | E ELECTRICAL PANELBOARD (208Y/120 V). |
|  | B BRANCH CIRCUIT OR FEEDER WIRING IN CONDUIT. NO TICK MARKS INDICATE TWO #12 CONDUCTORS & ONE #12 GROUND IN 1/2" CONDUIT U.O.N. TICK MARKS, WHEN SHOWN, INDICATE NO. OF #12 CONDUCTORS IF OTHER THAN THREE; (—) INDICATES GROUND. CONDUIT LARGER THAN 1/2" & WIRE LARGER THAN #12 SHALL BE AS INDICATED. |
|  | H HOMERUNS TO PANEL. PANEL & CIRCUIT DESIGNATIONS AS INDICATED. |
|  | I INDICATES CONDUIT RUN. |
|  | E EXISTING UNDERGROUND SECONDARY, CONCRETE ENCASED UON. |
|  | U UNDERGROUND SECONDARY. |
|  | U CONDUIT TURNED UP TOWARD POWER SOURCE. |
|  | U CONDUIT TURNED DOWN TOWARD POWER SOURCE. |
|  | E EXISTING PAD MOUNTED TRANSFORMER. |
|  | T TRANSFORMER. |
|  | C CABLE TERMINATIONS. |
|  | C CIRCUIT BREAKER. |
|  | G GROUND. |
|  | N NEW TO EXISTING CONNECTION. |
|  | D DUCT SMOKE DETECTOR. |
|  | F FIRE ALARM CONTROL PANEL. |
|  | D DEMOLITION NOTE SYMBOL. |
|  | C CONSTRUCTION NOTE SYMBOL. |

NOTE: SEE SHEET M-001 FOR MECHANICAL EQUIPMENT ABBREVIATIONS.

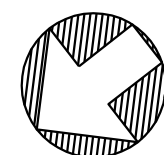
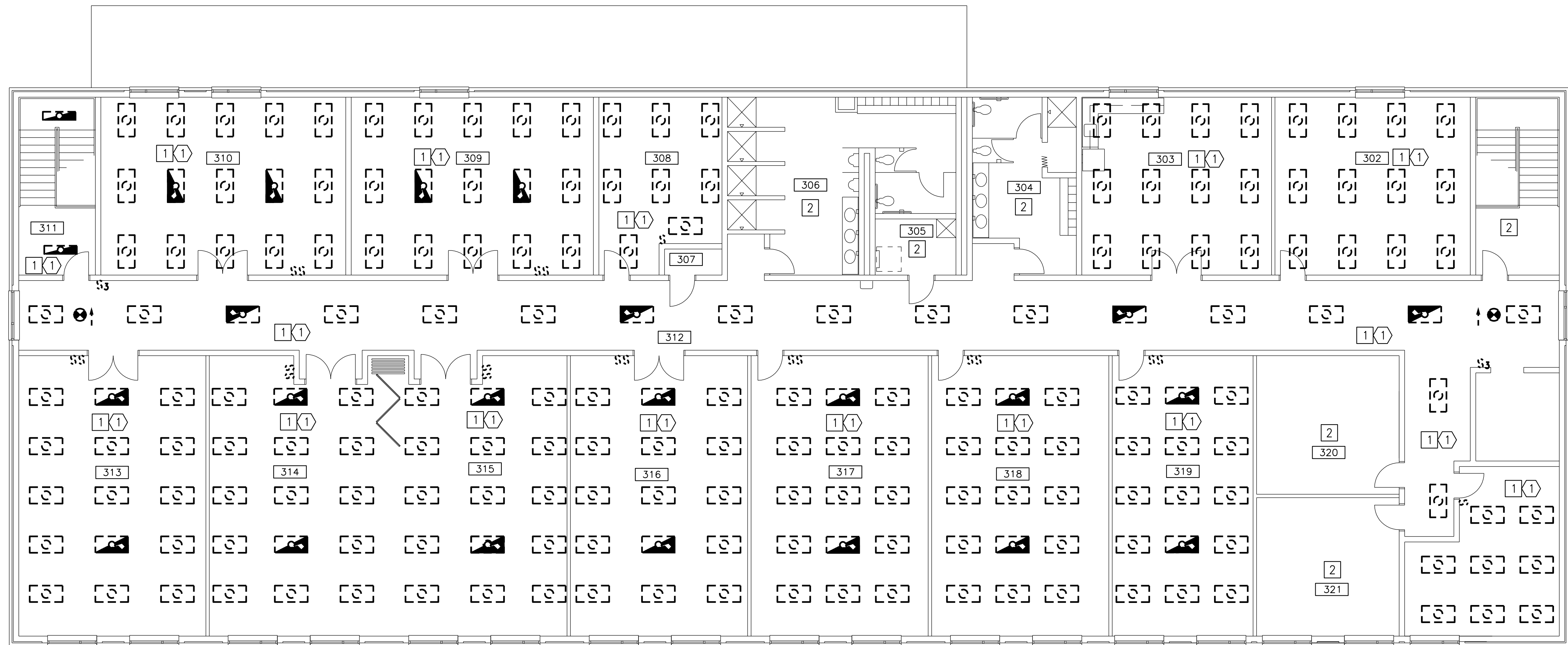
- | | |
|--------|---|
| Ø, PH | PHASE |
| A, AMP | AMPERE |
| AFF | ABOVE FINISHED FLOOR |
| AFG | ABOVE FINISHED GRADE |
| AIC | AMPERE INTERRUPTING CAPACITY
(RMS SYMMETRICAL) |
| BLDG | BUILDING |
| C | CONDUIT |
| EC | EMPTY CONDUIT |
| EF | EXHAUST FAN |
| EMT | ELECTRICAL METALLIC TUBING |
| EXIST | EXISTING |
| FACP | FIRE ALARM CONTROL PANEL |
| GFEP | GROUND—FAULT EQUIPMENT PROTECTION |
| GND | GROUND |
| HID | HIGH INTENSITY DISCHARGE |
| HP | HORSE POWER |
| kAIC | KILOAMPERE INTERRUPTING CAPACITY |
| kW | KILOWATTS |
| kVA | KILOVOLT AMPERES |
| MCB | MAIN CIRCUIT BREAKER |
| MH | MANHOLE |
| N | NEUTRAL |
| NF | NONFUSIBLE |
| N.I.C. | NOT IN CONTRACT |
| P | POLE |
| RCPT | RECEPTACLE |
| PNL | PANELBOARD |
| RSC | RIGID STEEL CONDUIT |
| TYP | TYPICAL |
| UON | UNLESS OTHERWISE NOTED |
| V | VOLT, VOLTAGE |
| VAC | VOLTS ALTERNATING CURRENT |
| WP | WEATHERPROOF |
| XFMR | TRANSFORMER |

1. THE CONTRACTOR SHALL PROVIDE THE COMPLETE PANEL SCHEDULES, INDICATING CIRCUIT USED, AREA SERVED, LOADS, AND SPARES AVAILABLE AFTER COMPLETION OF ELECTRICAL WORK FOR ALL PANELBOARDS.
2. THE CONTRACTOR SHALL REPLACE IN KIND ANY LIGHTING FIXTURE DAMAGED DURING CONSTRUCTION.
3. UPON COMPLETION OF THE WORK, REMOVE ALL MATERIALS, SCRAP AND DEBRIS RELATIVE TO THE INSTALLATION AND LEAVE THE PREMISES IN AN ORDERLY CONDITION.
4. ELECTRICAL POWER INTERRUPTION: SHUTDOWN OF ANY SYSTEM SHALL BE COORDINATED AND APPROVED BY THE GOVERNMENT REPRESENTATIVE.
5. THE CONTRACTOR SHALL STUDY IN DETAIL ALL AREAS AND EQUIPMENT BEING AFFECTED. VERIFY EXISTING CONDITIONS AND TAKE ANY AND ALL MEASURES NECESSARY TO ASSURE THAT ALL SERVICES ARE MAINTAINED OR RESTORED AS REQUIRED IN THE TIME PERIOD OUTLINED.
6. ALL WORK AND MATERIAL NOT SPECIFICALLY NOTED AS "EXISTING" OR TO BE "FURNISHED BY THE GOVERNMENT" IS NEW AND SHALL BE PROVIDED BY THE CONTRACTOR.

1. DRAWINGS ARE BASED ON EXISTING BUILDING PLANS AND FIELD INVESTIGATION WITHOUT DEMOLITION. RENOVATION HAS OCCURRED IN VARIOUS STAGES AND RECORD DRAWINGS ARE NOT AVAILABLE FOR ALL STAGES OF PAST CONSTRUCTION.
2. EXISTING CONDUITS THAT WILL NOT BE REUSED SHALL BE REMOVED WHERE THEY WILL BE EXPOSED AFTER COMPLETION OF CONSTRUCTION. OTHER CONDUITS MAY BE ABANDONED WHEN CONCEALED UNDERGROUND. DISCONNECT AND REMOVE WIRING FROM CONDUITS TO BE ABANDONED. REMOVAL OF ALL ASSOCIATED CONDUIT AND CONDUCTORS SHALL INCLUDE REMOVAL OF ALL CONDUIT, SUPPORT SYSTEMS, JUNCTION BOXES. CONDUCTORS SHALL BE REMOVED BACK TO POWER SOURCE. RETAIN EXISTING CIRCUIT BREAKERS AND LABEL AS SPARE.
3. EXISTING CONDUIT AND CONDUCTOR SIZES INDICATED ON THESE DRAWINGS ARE BASED ON INFORMATION OBTAINED FROM EXISTING DRAWINGS AND VISUAL INSPECTION FROM GROUND (FLOOR) LEVEL AND SHALL BE CONSIDERED AS APPROXIMATE.
4. PRIOR TO DEMOLITION CONTRACTOR SHALL VERIFY WITH GOVERNMENT REPRESENTATIVE PHASE ROTATION OF EXISTING SYSTEM. CONTRACTOR SHALL COMPLETE PROJECT WITH SAME PHASE AS ROTATION PRIOR TO DEMOLITION AND VERIFY ALL EQUIPMENT IS IN PROPER ROTATION.

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

[illegible]



THIRD FLOOR LIGHTING PLAN – DEMOLITION AND NEW WORK

SCALE: 1/8" = 1'-0"

DEMOLITION NOTES (THIS SHEET ONLY)

- 1 REMOVE EXISTING LIGHTING FIXTURES AND STORE FOR REINSTALLATION IN SAME LOCATION AFTER COMPLETION OF MECHANICAL WORK. CONTRACTOR SHALL PERMANENTLY MARK EACH FIXTURE WITH ROOM NUMBER. CONTRACTOR SHALL VERIFY NO. & TYPE OF LAMPS. SEE GENERAL DEMOLITION NOTE 4 ON SHEET E-001. LIGHTING FIXTURES SHALL BE DISCONNECTED FROM THE EXISTING JUNCTION BOX ABOVE THE CEILING. IF JUNCTION BOX DOES NOT EXIST THEN DISCONNECT FIXTURE WHIP AT THE LIGHTING FIXTURE AND SECURE WHIP FROM STRUCTURE ABOVE.
- 2 IT IS NOT NECESSARY TO REMOVE FIXTURES IN THIS AREA.

CONSTRUCTION NOTES (THIS SHEET ONLY)

- 1 REINSTALL EXISTING FIXTURES. SEE THIS SHEET DEMOLITION NOTE 1. CONTRACTOR SHALL CLEAN EXISTING LIGHTING FIXTURES LENS PRIOR TO INSTALLATION BY DIPPING THEM IN A CLEANING SOLUTION AS RECOMMENDED BY A FIXTURE MANUFACTURER, AND RINSE TWICE IN CLEAN WATER AND SET TO AIR DRY PRIOR TO RE-INSTALLATION. CLEAN REFLECTOR LENS FRAME AND INTERIOR OF FIXTURE WITH A CLEANSER AS RECOMMENDED BY A FIXTURE MANUFACTURER. REPLACE EXISTING LAMPS WITH NEW LAMPS.

WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

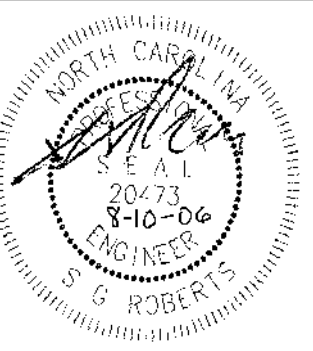
NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:

1/8" = 1'-0" 



Bowmar, Foster & Associates, P.C.
Consulting Engineers
#4 Interstate Corporate Center, Suite 1
Norfolk, VA 23502-4118



APPROVED 06-004C

ACTIVITY – SATISFACTORY TO

DATE

APPROVED

FOR EFD FOR COMMANDER NAVFAC

DATE

A/E EFD

BAS DESIGN

LAW DRAWN

MED REVIEW

SGR QC

SGR CHIEF ARCH/ ENGR.

PROJECT MANAGER JS

FIRE PROTECTION

BRANCH MANAGER RM

DESIGN DIRECTOR CC

NAVAL FACILITIES ENGINEERING COMMAND

MARINE CORPS AIR STATION, CHERRY POINT, N.C.

REPLACE HVAC SYSTEM BLDG 4414

THIRD FLOOR LIGHTING PLAN – DEMOLITION & NEW WORK

CODE ID. NO. 80091 SIZE 0

SCALE: AS SHOWN

FED. NO. 2827328

STA. PROJ. NO. CP0641M

SPEC. NO. -

CONSTR. CONTR. NO. -

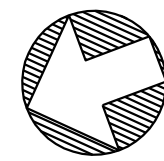
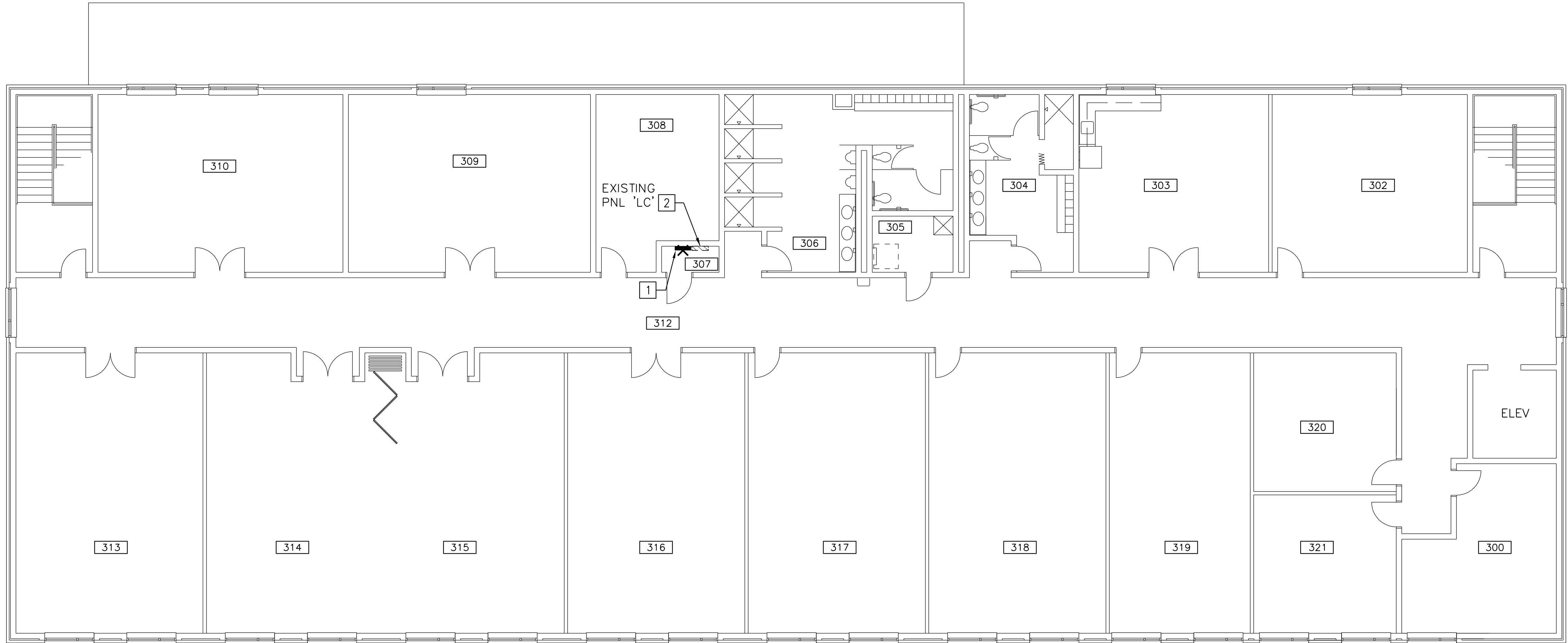
NAVFAC DRAWING NO.

4555284

SHEET 40 OF 49

ED103

DRAWFORM REVISION JULY 2003



THIRD FLOOR POWER PLAN – DEMOLITION

SCALE: 1/8" = 1'-0"

DEMOLITION NOTES

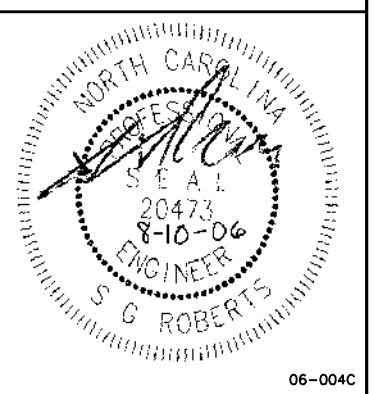
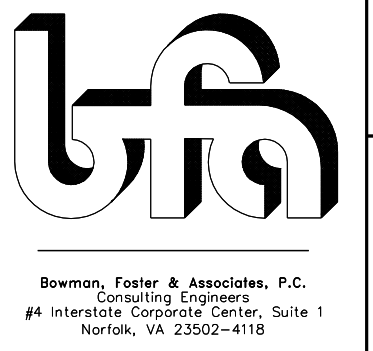
- 1 DISCONNECT AND REMOVE EXISTING PANELBOARD "HC" COMPLETE. EXISTING FEEDER AND BRANCH CIRCUIT CONDUITS AND WIRES SHALL REMAIN FOR REUSE.
- 2 EXISTING PANEL 'LC' SHALL REMAIN FOR REUSE.

NOTE: IF THIS DRAWING IS A REDUCTION,
GRAPHIC SCALE MUST BE USED.

GRAPHIC SCALE:

1/8" = 1'-0" 

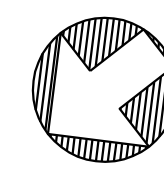
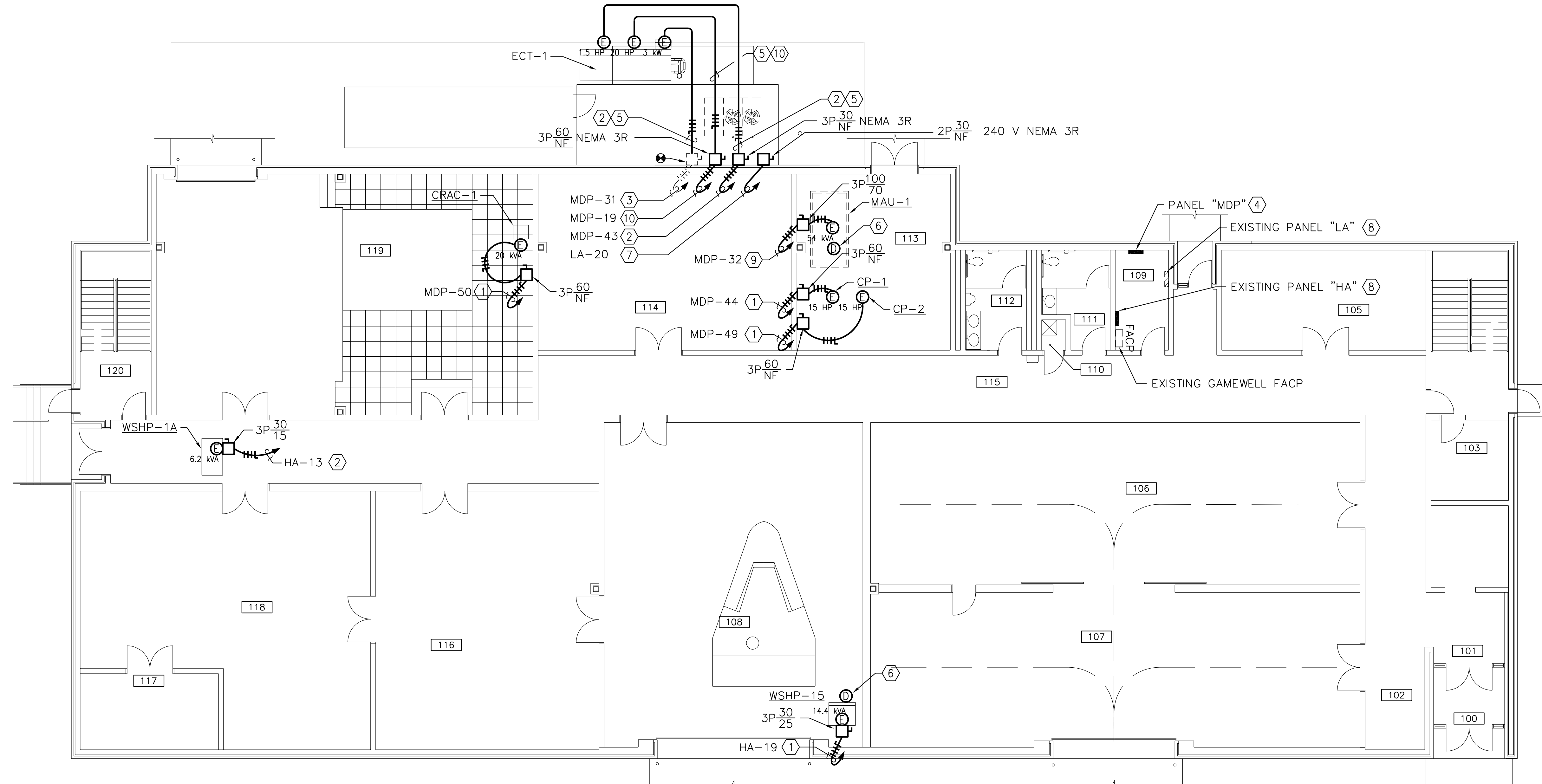
DATE	APPR
DESCRIPTION	SYN



APPROVED		
ACTIVITY - SATISFACTORY TO		
DATE		
APPROVED		
FOR EFD FOR COMMANDER NAVFAC		
DATE		
A/E		EFD
BAS	DESIGN	
LAW	DRAWN	
MED	REVIEW	
SGR	QC	
SGR	CHIEF ARCH./ ENGR.	
PROJECT MANAGER	JS	
FIRE PROTECTION		
BRANCH MANAGER	RM	
DESIGN DIRECTOR	CC	

DEPARTMENT OF THE NAVY	NAVAL FACILITIES ENGINEERING COMMAND
MARINE CORPS AIR STATION, CHERRY POINT, N.C.	REPLACE HVAC SYSTEM BLDG 4414
THIRD FLOOR POWER PLAN – DEMOLITION	

CODE ID. NO. 80091	SIZE 0
SCALE: AS SHOWN	
FED. NO. 2827328	
STA. PROJ. NO. CP0641M	
SPEC. NO. –	
CONSTR. CONTR. NO. –	
NAVFAC DRAWING NO. 4555287	
SHEET 43 OF 49	
ED106	
DRAWFORM REVISION JULY 2003	



FIRST FLOOR POWER PLAN – NEW WORK

SCALE: 1/8" = 1'-0"

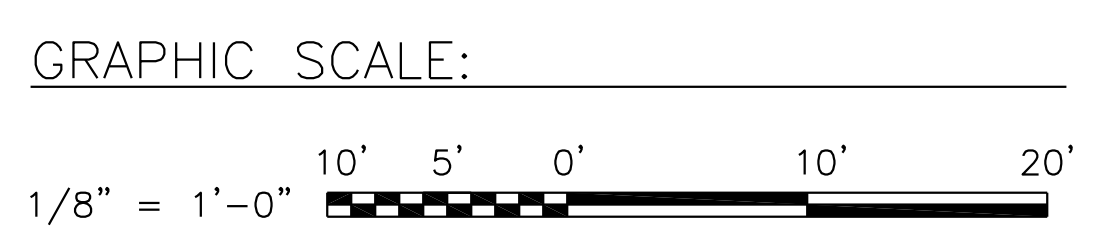
CONSTRUCTION NOTES (THIS SHEET ONLY)

- | | |
|---|--|
| <p>(1) THREE #10 AND ONE #10 GND IN 1/2" C.</p> <p>(2) THREE #12 AND ONE #12 GND IN 1/2" C.</p> <p>(3) SEE SHEET ED104 DEMOLITION NOTE (6).</p> <p>(4) PROVIDE 800A, 480Y/277V PANEL "MDP". SEE ONE-LINE DIAGRAM ON SHEET E-601 AND PANELBOARD SCHEDULE ON SHEET E-603.</p> <p>(5) CONDUITS SHALL RUN UNDERGROUND PARALLEL TO MECHANICAL PIPING.</p> <p>(6) SEE FIRE ALARM RISER DIAGRAM NOTE 3 ON SHEET E-603.</p> | <p>(7) REMOVE EXISTING SPARE SINGLE POLE, 20 A CB, AND PROVIDE GFEP TYPE SINGLE POLE, 20 A, 240 V CB TO MATCH EXISTING SIEMENS TYPE S3 PANELBOARD.</p> <p>(8) SEE PANELBOARD SCHEDULE ON SHEET E-602.</p> <p>(9) THREE #4 AND ONE #8 GND IN 1" C.</p> <p>(10) THREE #8 AND ONE #10 GND IN 3/4"C.</p> |
|---|--|

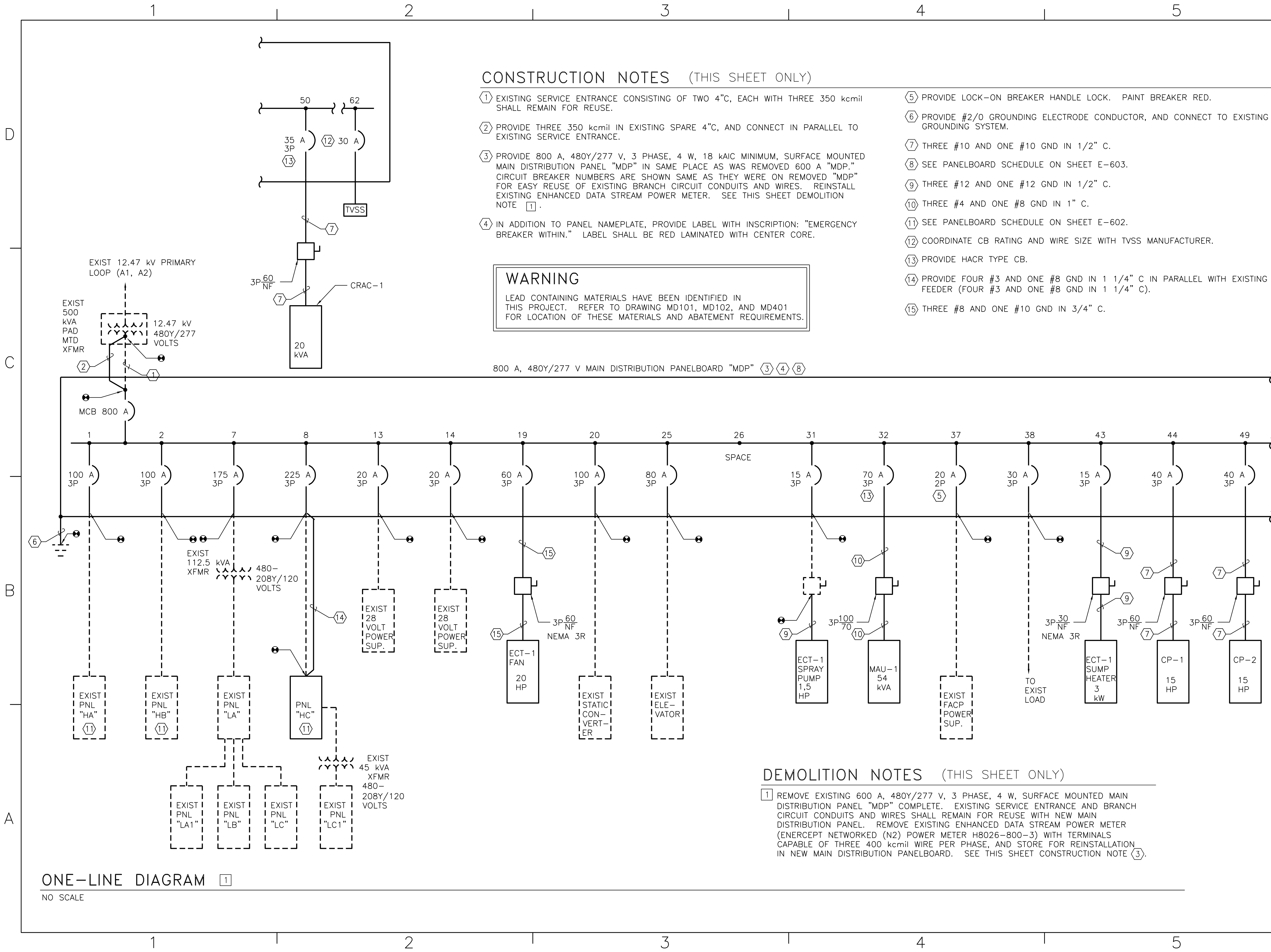
WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

NOTE: IF THIS DRAWING IS A REDUCTION, GRAPHIC SCALE MUST BE USED.



APPROVED		DATE	APPR
ACTIVITY - SATISFACTORY TO			
DATE			
APPROVED			
FOR EFD FOR COMMANDER NAVFAC			
DATE			
A/E	DESIGN	EFD	
BAS	DRAWN		
LAW	REVIEW		
MED	QC		
SGR	CHIEF ARCH/ ENGR.		
SGR	PROJECT MANAGER	JS	
FIRE PROTECTION			
BRANCH MANAGER		RM	
DESIGN DIRECTOR		CC	
NAVAL FACILITIES ENGINEERING COMMAND			
MARINE CORPS AIR STATION, CHERRY POINT, N.C.			
REPLACE HVAC SYSTEM BLDG 4414			
FIRST FLOOR POWER PLAN - NEW WORK			
CODE ID. NO. 80091		SIZE	0
SCALE: AS SHOWN			
FED. NO. 2827328			
STA. PROJ. NO. CP0641M			
SPEC. NO. -			
CONSTR. CONTR. NO. -			
NAVFAC DRAWING NO. 4555288			
SHEET 44 OF 49			
E-101			
DRAWING REVISION JULY 2003			



CONSTRUCTION NOTES (THIS SHEET ONLY)

- 1 EXISTING SERVICE ENTRANCE CONSISTING OF TWO 4"C, EACH WITH THREE 350 kcmil SHALL REMAIN FOR REUSE.
- 2 PROVIDE THREE 350 kcmil IN EXISTING SPARE 4"C, AND CONNECT IN PARALLEL TO EXISTING SERVICE ENTRANCE.
- 3 PROVIDE 800 A, 480Y/277 V, 3 PHASE, 4 W, 18 kAIC MINIMUM, SURFACE MOUNTED MAIN DISTRIBUTION PANEL "MDP" IN SAME PLACE AS WAS REMOVED 600 A "MDP." CIRCUIT BREAKER NUMBERS ARE SHOWN SAME AS THEY WERE ON REMOVED "MDP" FOR EASY REUSE OF EXISTING BRANCH CIRCUIT CONDUITS AND WIRES. REINSTALL EXISTING ENHANCED DATA STREAM POWER METER. SEE THIS SHEET DEMOLITION NOTE 1.
- 4 IN ADDITION TO PANEL NAMEPLATE, PROVIDE LABEL WITH INSCRIPTION: "EMERGENCY BREAKER WITHIN." LABEL SHALL BE RED LAMINATED WITH CENTER CORE.
- 5 PROVIDE LOCK-ON BREAKER HANDLE LOCK. PAINT BREAKER RED.
- 6 PROVIDE #2/0 GROUNDING ELECTRODE CONDUCTOR, AND CONNECT TO EXISTING GROUNDING SYSTEM.
- 7 THREE #10 AND ONE #10 GND IN 1/2" C.
- 8 SEE PANELBOARD SCHEDULE ON SHEET E-603.
- 9 THREE #12 AND ONE #12 GND IN 1/2" C.
- 10 THREE #4 AND ONE #8 GND IN 1" C.
- 11 SEE PANELBOARD SCHEDULE ON SHEET E-602.
- 12 COORDINATE CB RATING AND WIRE SIZE WITH TVSS MANUFACTURER.
- 13 PROVIDE HACR TYPE CB.
- 14 PROVIDE FOUR #3 AND ONE #8 GND IN 1 1/4" C IN PARALLEL WITH EXISTING FEEDER (FOUR #3 AND ONE #8 GND IN 1 1/4" C).
- 15 THREE #8 AND ONE #10 GND IN 3/4" C.

WARNING
LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

DEMOLITION NOTES (THIS SHEET ONLY)

- 1 REMOVE EXISTING 600 A, 480Y/277 V, 3 PHASE, 4 W, SURFACE MOUNTED MAIN DISTRIBUTION PANEL "MDP" COMPLETE. EXISTING SERVICE ENTRANCE AND BRANCH CIRCUIT CONDUITS AND WIRES SHALL REMAIN FOR REUSE WITH NEW MAIN DISTRIBUTION PANEL. REMOVE EXISTING ENHANCED DATA STREAM POWER METER (ENERCEPT NETWORKED (N2) POWER METER H8026-800-3) WITH TERMINALS CAPABLE OF THREE 400 kcmil WIRE PER PHASE, AND STORE FOR REINSTALLATION IN NEW MAIN DISTRIBUTION PANELBOARD. SEE THIS SHEET CONSTRUCTION NOTE 3.

ONE-LINE DIAGRAM 1

NO SCALE

DATE		APPROVED	
DESCRIPTION		SYMBOL	
APPROVED			
ACTIVITY - SATISFACTORY TO			
DATE			
APPROVED			
FOR EFD FOR COMMANDER NAVFAC			
DATE			
A/E	DESIGN	EFD	
BAS	DRAWN		
LAW	REVIEW		
MED	QC		
SGR	CHIEF ARCH/ ENGR.		
SGR	PROJECT MANAGER	JS	
	FIRE PROTECTION		
	BRANCH MANAGER	RM	
	DESIGN DIRECTOR	CC	
NAVAL FACILITIES ENGINEERING COMMAND			
MARINE CORPS AIR STATION, CHERRY POINT, N.C.			
REPLACE HVAC SYSTEM BLDG 4414			
ONE-LINE DIAGRAM			
DEPARTMENT OF THE NAVY			
CODE ID. NO. 80091			
SCALE: AS SHOWN			
FED. NO. 2827328			
STA. PROJ. NO. CP0641M			
SPEC. NO. -			
CONSTR. CONTR. NO. -			
NAVFAC DRAWING NO. 4555291			
SHEET 47 OF 49			
E-601			
DRAWING REVISION JULY 2003			

1

1

D

C

- B

1

A

A

7

(7)

D

C

- B

1

A

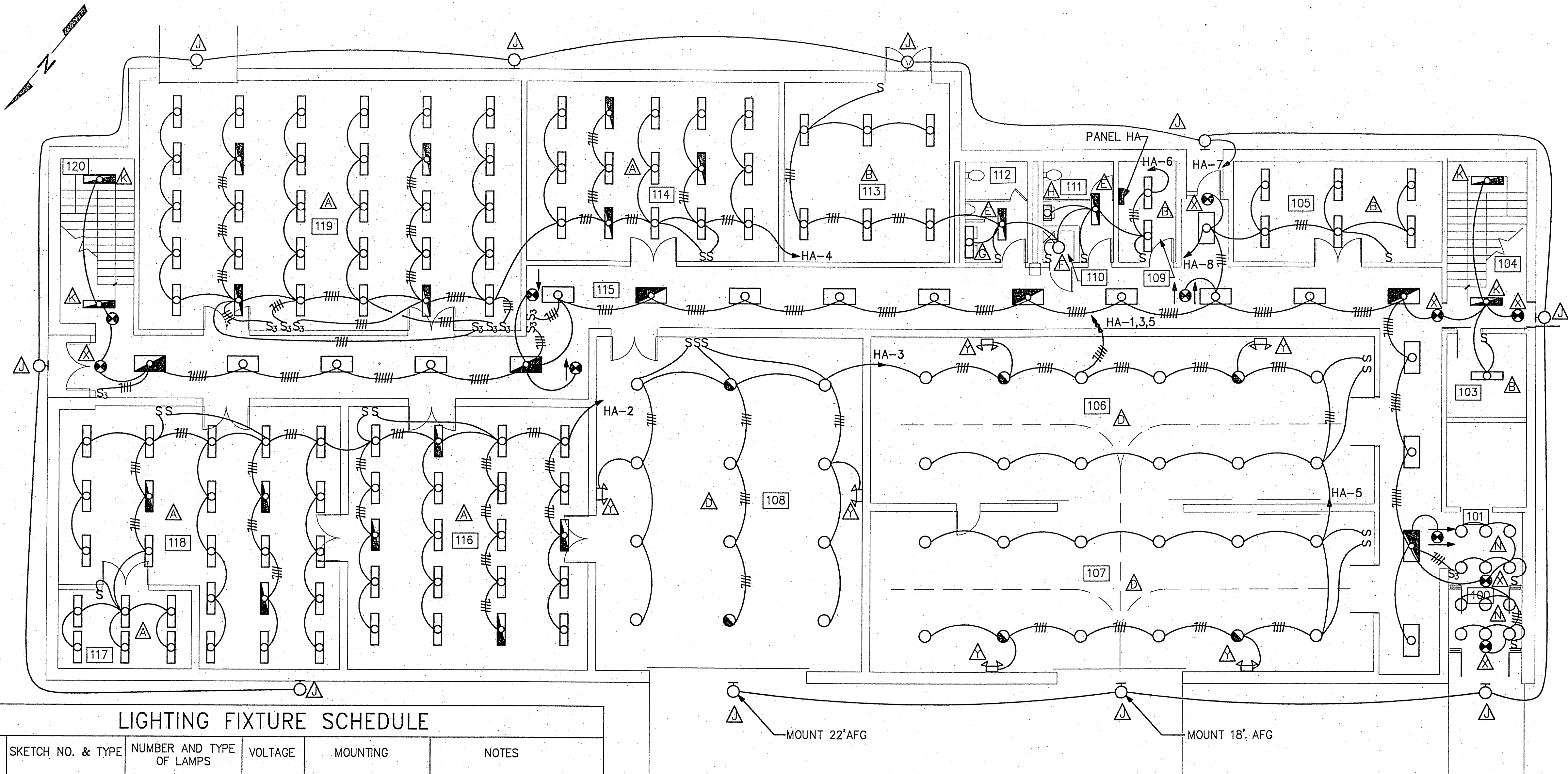
A

WARNING

WARNING

LEAD CONTAINING MATERIALS HAVE BEEN IDENTIFIED IN THIS PROJECT. REFER TO DRAWING MD101, MD102, AND MD401 FOR LOCATION OF THESE MATERIALS AND ABATEMENT REQUIREMENTS.

[illegible]



LIGHTING FIXTURE SCHEDULE

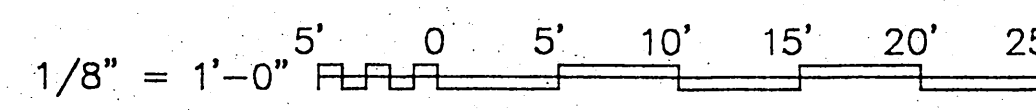
FIXTURE SYMBOL	SKETCH NO. & TYPE	NUMBER AND TYPE OF LAMPS	VOLTAGE	MOUNTING	NOTES
△	NL-2, TYPE B	2-F40CW/T12/RS	277	BOTTOM OF JOIST	SEE NOTE 1
△	NL-6, TYPE B	2-F40CW/T12/RS	277	BOTTOM OF JOIST	
△	NL-3, TYPE C	2-F40CW/T12/RS	277	RECESSED	SEE NOTE 1
△	NL-22, TYPE E	1-250W METAL HALIDE	277	PENDANT	LENS AT LEVEL OF BOTTOM OF JOISTS
△	NL-2, TYPE B	2-F40CW/T12/RS	277	SURFACE, CEILING	SEE NOTE 1
△	NL-13, TYPE D	2-13W COMP FLUOR	277	SURFACE, CEILING	
△	NL-4, TYPE B	2-F40CW/T12/RS	277	WALL 6" ABOVE MIRROR	
△	NL-4, TYPE A	2-F40CW/T12/RS	277	WALL 6" ABOVE MIRROR	
△	NL-56, TYPE B-2	13W DOUBLE TWIN TUBE FLUORESCENT	277	RECESSED	
△	NL-25, TYPE C	150W HPS	277	WALL	MOUNT 12' AFG UON INTEGRAL PHOTOCELL
△	NL-11, TYPE B	2-F40CW/T12/RS	277	SURFACE, CEILING	SEE NOTE 1
△	XL-1	250W HPS	208	XL-23E POLE MOUNT XL-32A MAST ARM MOUNT	INTEGRAL PHOTOCELL TYPE 5 DISTRIBUTION
△	NL-47, TYPE C	2-9W COMP FLUOR	277	WALL 9'-0" AFF	
△	NL-18, TYPE B	2-13W COMP FLUOR	277	RECESSED	
△	NL-61	LED	277	WALL 6" ABOVE DOOR OR CEILING	SEE NOTE 2
△	NL-53	2-8W HALOGEN	277	WALL 7'-0" AFF	SEE NOTE 2

FIRST FLOOR LIGHTING PLAN
SCALE: 1/8" = 1'-0"

- NOTE 1
FIXTURES INDICATED AS EMERGENCY SHALL HAVE BACK-UP BATTERY, AUTOMATIC CHARGER AND SELF DIAGNOSTIC SYSTEM WITH STATUS PANEL MOUNTED ADJACENT TO THE FIXTURE. FIXTURE LAMPS SHALL BE CONTROLLED BY LIGHT SWITCH EXCEPT DURING POWER FAILURE, LAMPS SHALL BE ON
- NOTE 2
FIXTURES SHALL HAVE BACK-UP BATTERY, AUTOMATIC CHARGER AND SELF DIAGNOSTIC SYSTEM WITH STATUS PANEL INTEGRAL TO THE FIXTURE

NOTE
ALL FIXTURES TYPE △ UON.

GRAPHIC SCALE



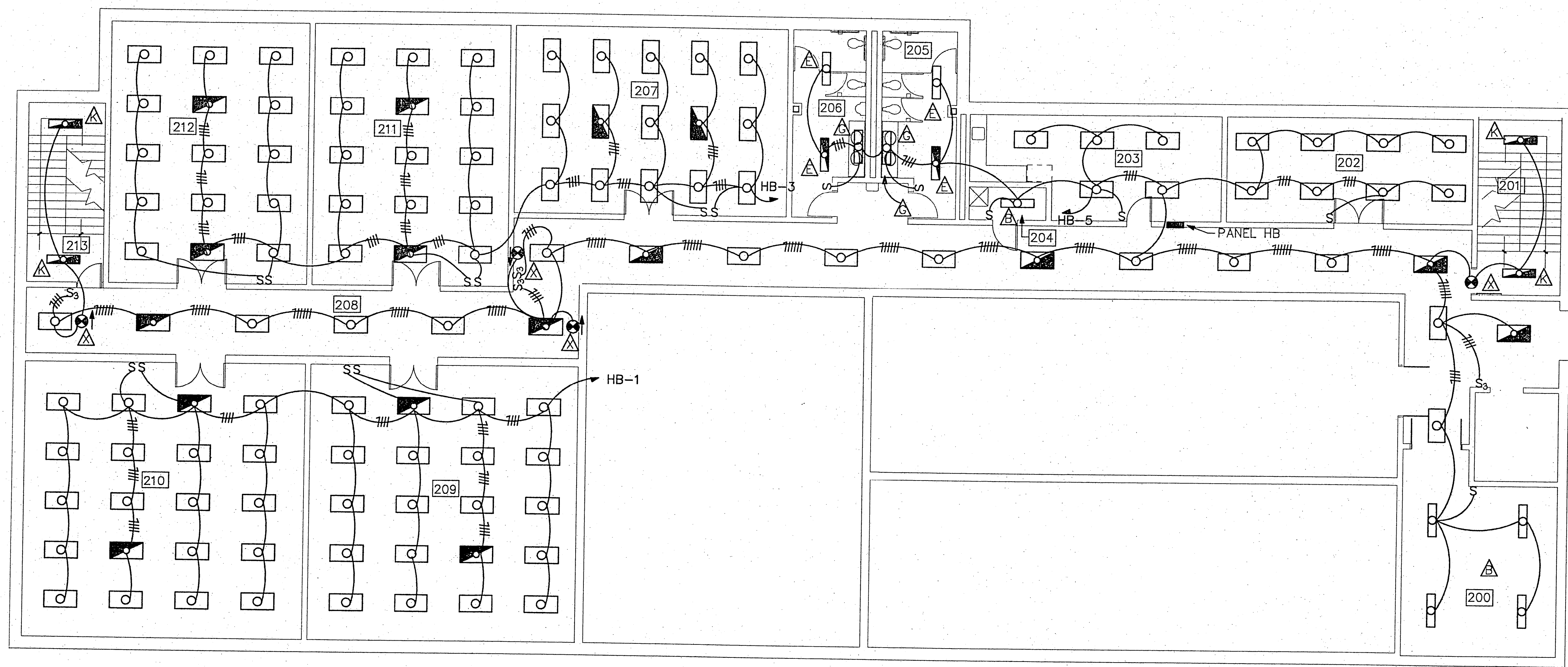
DESIGN	DATE	APPROVED
DRAWN	DATE	APPROVED
CHECK	DATE	APPROVED
PROJECT MANAGER	DATE	APPROVED
QUALITY CONTROL	DATE	APPROVED
BRANCH MANAGER	DATE	APPROVED
DESIGN DIRECTOR	DATE	APPROVED

REVISIONS

NO.	DESCRIPTION	DATE
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

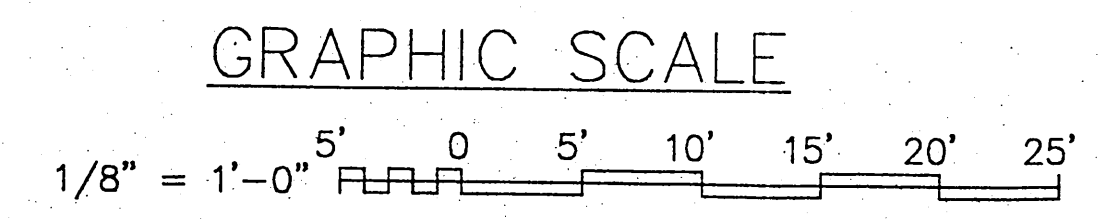
NAVY FACILITIES ENGINEERING COMMAND
ATLANTIC DIVISION
NAVAL STATION
MCAS CHERRY POINT
AIRCRAFT MAINTENANCE TRAINING FACILITY
FIRST FLOOR LIGHTING PLAN

CODE ID. NO. 80091
SCALE AS NOTED
EFD NO. 375636
STA. PROJ. NO. P-043
SPEC. NO. 05-92-2173
CONSTR. CONTR. NO. N62470-92-B-2173
NAVFAC DRAWING NO. 4275636
SHEET 73 OF 83
E-5

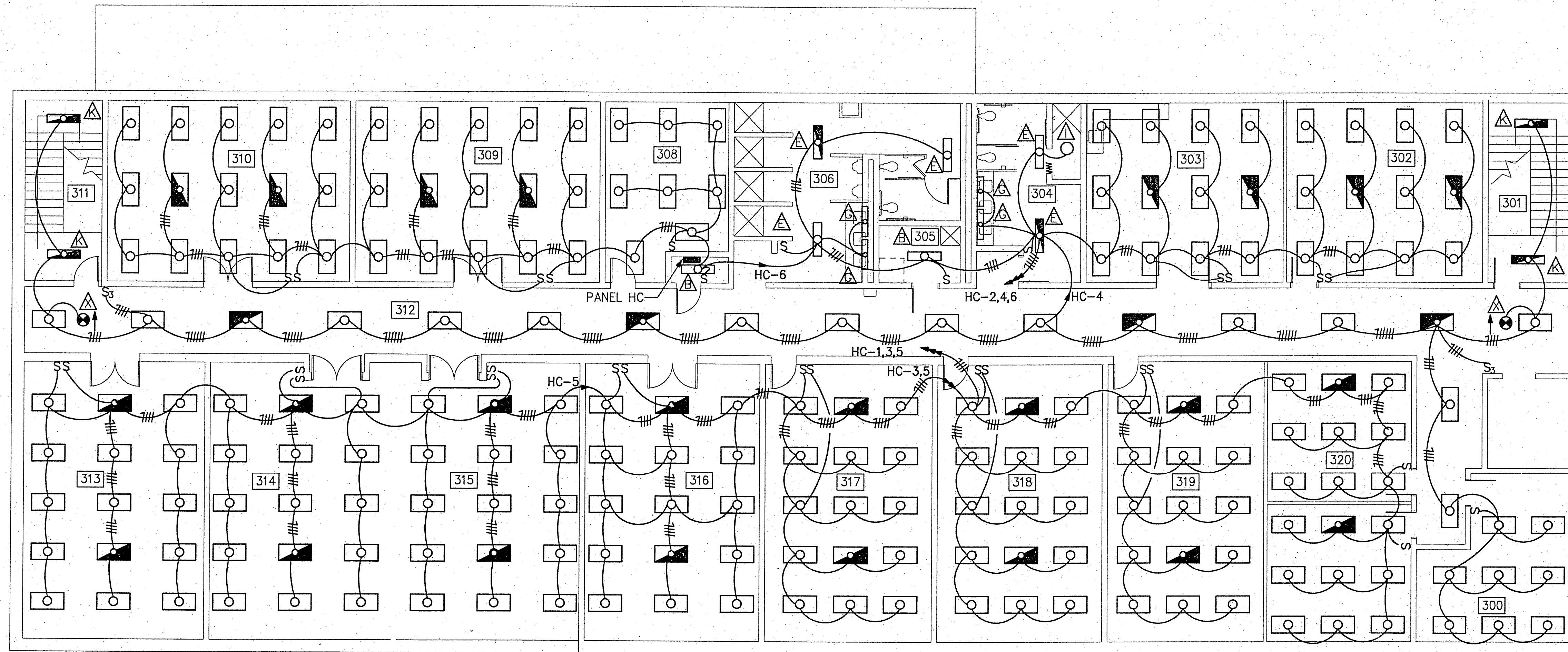
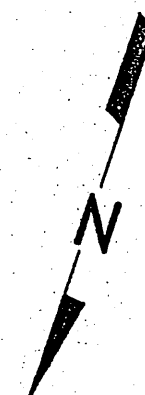


SECOND FLOOR LIGHTING PLAN
SCALE: 1/8" = 1'-0"

NOTE
ALL FIXTURES TYPE Δ UON

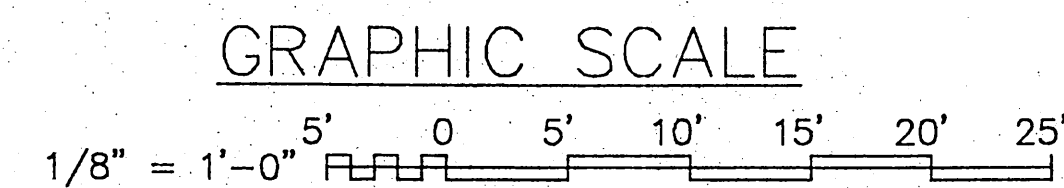


DEPARTMENT OF THE NAVY		ATLANTIC DIVISION		NAVAL FACILITIES ENGINEERING COMMAND	
NAVAL STATION		CHERRY POINT, NC		NORFOLK, VIRGINIA	
MCAS CHERRY POINT		AIRCRAFT MAINTENANCE		TRAINING FACILITY	
SECOND FLOOR LIGHTING PLAN		DATE		1/19/94	
CODE ID. NO. 80091		SIZE		D	
SCALE: AS NOTED		BYD NO. 375637		STA. PROJ. NO. P-043	
SPEC. NO. 05-92-2173		CONSTR. CONTR. NO. NE2470-92-B-2173		NAVJAG DRAWING NO. 4275637	
SHEET 74 OF 83		E-6		REVISIONS	
DESIGN		PKAT/S		GCB	
REVIEW		HBW		HBW	
CHECK		BGA		BGA	
PROJECT MANAGER		EXT		EXT	
TYPE PRODUCTION		C/A		C/A	
QUALITY MANAGER		C/A		C/A	
DESIGN		C/A		C/A	
DATE		1/19/94		APPROVED	
DATE		1/19/94		APPROVED	

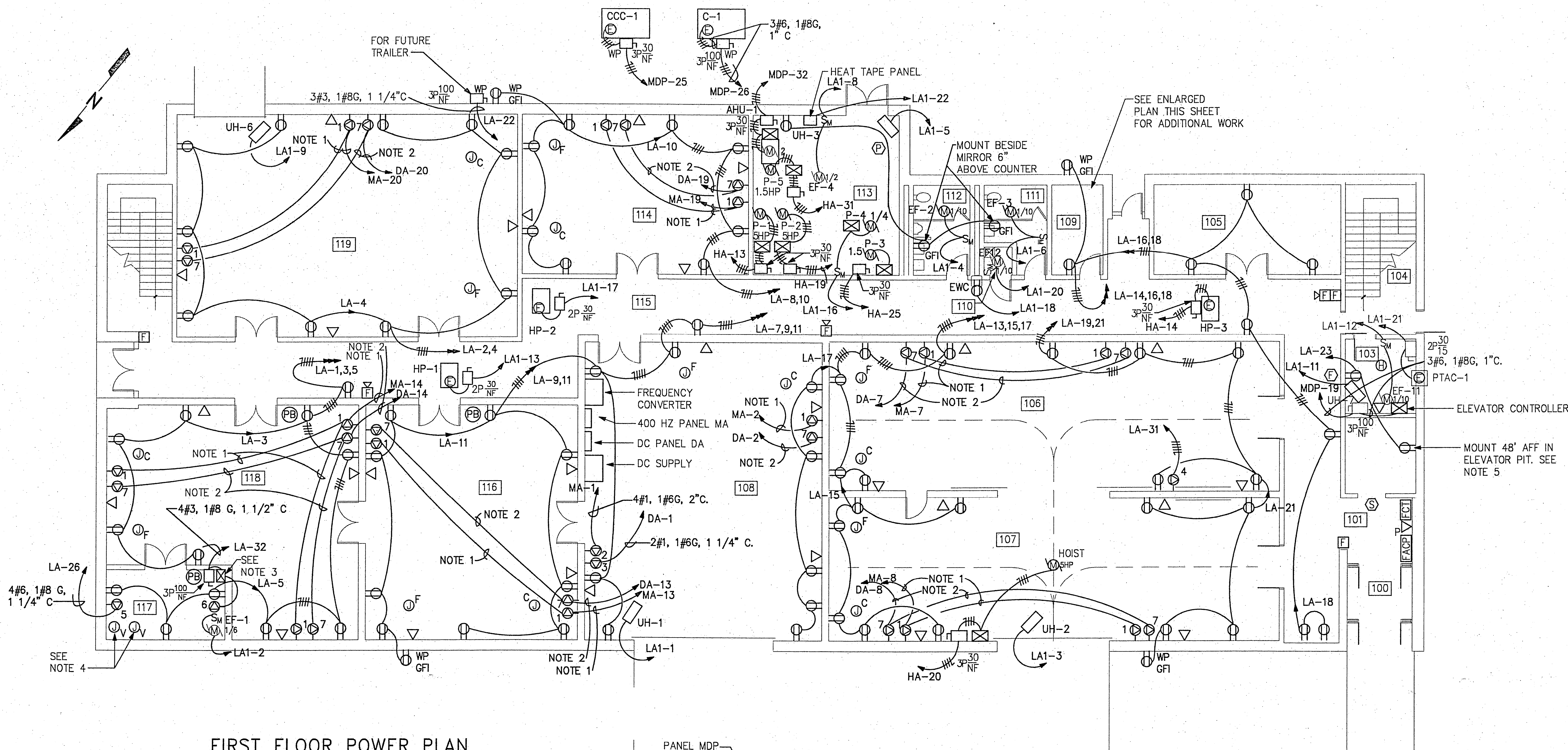


NOTE
ALL FIXTURES TYPE Δ UON.

THIRD FLOOR LIGHTING PLAN
SCALE: 1/8" = 1'-0"



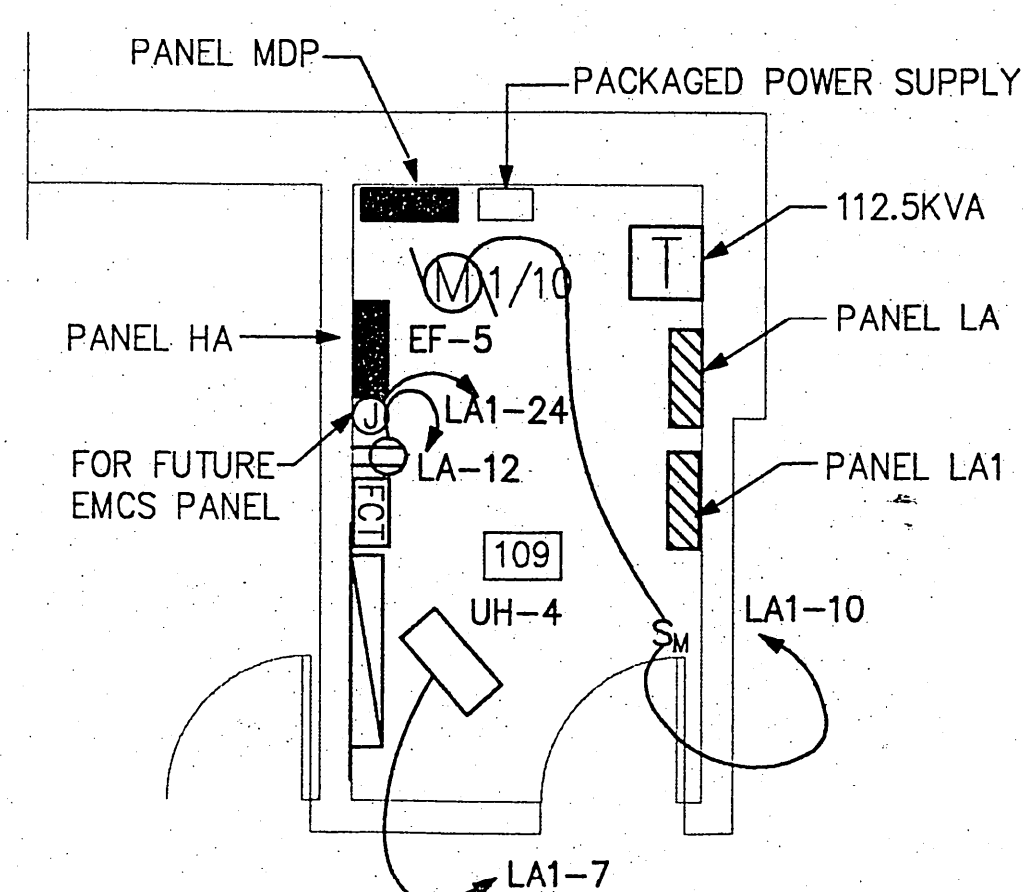
DEPARTMENT OF THE NAVY		NAVAL FACILITIES ENGINEERING COMMAND		ATLANTIC DIVISION		NORFOLK, VIRGINIA	
NAVAL STATION		MCAS CHERRY POINT		AIRCRAFT MAINTENANCE TRAINING FACILITY		CHERRY POINT, NC	
THIRD FLOOR LIGHTING PLAN		DATE		DATE		DATE	
CODE ID. NO. 80091		SIZE D		SCALE: AS NOTED		EFD NO. 375638	
STA. PROJ. NO. P-043		SPEC. NO. 05-94-2173		CONSTR. CONTR. NO. N62470-92-B-2173		NAVFAC DRAWING NO. 4275638	
SHEET 75 OF 83		E-7		REVISIONS		DATE	
DESIGN		GCP		DATE		DATE	
DRAWN		PK/LS		DATE		DATE	
CHECKED		HEWL		DATE		DATE	
PROJECT MANAGER		BOA		DATE		DATE	
FIRE PROTECTION		EPC		DATE		DATE	
QUALITY CONTROL		EPC		DATE		DATE	
BRANCH MANAGER		EPC		DATE		DATE	
DESIGN		EPC		DATE		DATE	



FIRST FLOOR POWER PLAN
SCALE: 1/8" = 1'-0"

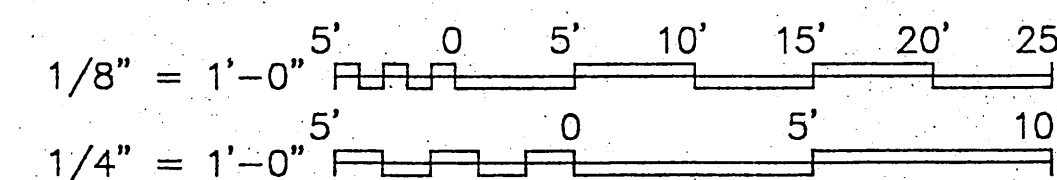
NOTES

- 4#8, 1#10G, 1"C.
- 2#6, 1#10G, 3/4"C
- PROVIDE NEMA SIZE 3 CONTROLLER FOR 208V, 3 PHASE 25 HP GOVERNMENT FURNISHED HYDRAULIC "GENNY". PROVIDE STOP CONTROL CONNECTION TO THE THREE REMOTE KILL SWITCHES (PB) PROVIDE SIGN AT EACH SWITCH READING "HYDRAULIC SYSTEM EMERGENCY FILL SWITCH"
- PROVIDE 2" EMPTY CONDUIT RESPECTIVELY FROM EACH OF TWO VARIDRIVE JUNCTION BOXES TO VARIDRIVE JUNCTION BOXES IN ROOMS 211 & 313.
- PROVIDE HEAVY DUTY PLUG IN MECHANIC'S HOOK LIGHT WITH 20' CORD AND CORD REEL. MOUNT REEL BESIDE RECEPTACLE

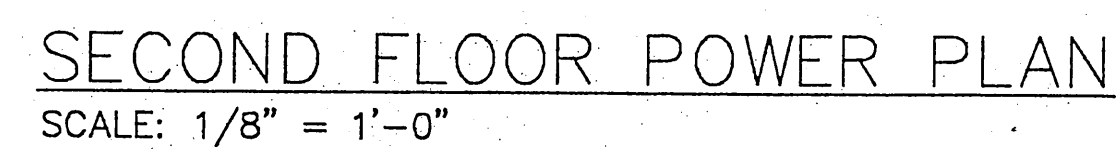


ENLARGED ELECTRICAL ROOM PLAN
SCALE: 1/4" = 1'-0"

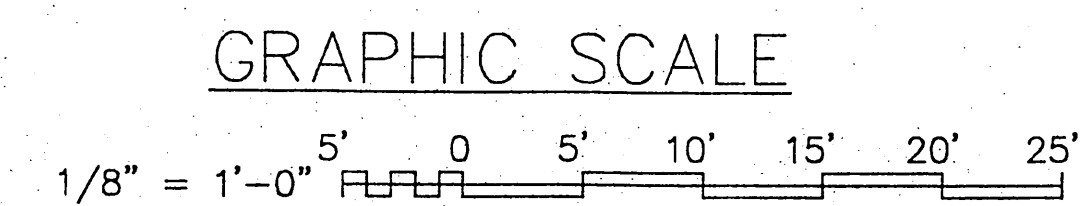
GRAPHIC SCALES



DEPARTMENT OF THE NAVY		NAVAL FACILITIES ENGINEERING COMMAND	
NAVAL STATION		NORFOLK, VIRGINIA	
ATLANTIC DIVISION		CHERRY POINT, NC	
MCAS CHERRY POINT		AIRCRAFT MAINTENANCE	
CONSTR. CONTRL. NO.		TRAINING FACILITY	
NAVFAC DRAWING NO.		FIRST FLOOR POWER PLAN	
4275639			
SCALE: AS NOTED		SIZE	
EFT NO. 375639			
STA. PROJ. NO. P-043			
SPEC. NO. 05-92-217			
NAVFAC DRAWING NO. 4275639			
SHEET 76 OF 83			
F-8			



1. 4#8, 1#10G, 1"C.
2. 2#6, 1#10G, 3/4"C
3. PROVIDE A HEAVY DUTY PLUG-IN MECHANICS HOOK LIGHT WITH 20' CORD AND CORD REEL. MOUNT CORD REEL BESIDE RECEPTACLE.



DEPARTMENT OF THE NAVY		NAVAL FACILITIES ENGINEERING COMMAND	
NAVAL STATION		NORFOLK, VIRGINIA	
ATLANTIC DIVISION		CHERRY POINT, NC	
MCAS CHERRY POINT		AIRCRAFT MAINTENANCE	
TRAINING FACILITY		TRAINING FACILITY	
SECOND FLOOR POWER PLAN			
CODE ID. NO. 80091		SIZE D	
SCALE: AS NOTED			
EFD NO. 375640			
STA. PROJ. NO. P-043			
SPEC. NO. 05-92-2173			
CONSTR. CONTR. NO. N62470-92-B-2173			
NAVYAC DRAWING NO. 4275640			
SHEET 77 OF 83			
F-19			

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND ATLANTIC DIVISION NAVAL STATION NORFOLK, VIRGINIA		CHERRY POINT, NC MCAS CHERRY POINT AIRCRAFT MAINTENANCE TRAINING FACILITY THIRD FLOOR POWER PLAN	
CODE ID. NO. 80091 SCALE: AS NOTED EFD NO. 375641 STA. PROJ. NO. P-043 SPEC. NO. 05-92-2173 CONSTR. CONTR. NO. NAVFAC DRAWING NO. 4275641	DATE 11/9/90 APPROVED <i>[Signature]</i> ACTIVITY - SATISFACTORY TO APPROVED		
M/E DESIGN GCB DRAWN PKT/L REVIEW HBW CHECK MCH/DOR PROJECT MANAGER BGA FIRE PROTECTION RCL QUALITY CONTROL BRANCH MANAGER DE/74		DATE APPROVED	